

Basic Questions (1–10)

- Total number of tickets in the dataset?
- How many unique airlines are present?
- How many unique origin cities are there?
- How many unique destination cities are there?
- What is the average ticket price?
- What is the minimum ticket price?
- What is the maximum ticket price?
- What is the average distance traveled?
- How many tickets belong to each travel class?
- Count of tickets sold by each airline.

Intermediate Questions (11–20)

- What is the average price for each airline?
- What is the average price for each class?
- Which origin city has the most flights?
- Which destination city has the most flights?
- What is the average distance by airline?
- Which airline has the highest average ticket price?
- Which airline has the lowest average ticket price?
- What is the average price for each origin city?
- What is the average price for each destination city?
- How many tickets were booked for each Days_Before_Departure value?

Advanced Questions (21–30)

- Does price increase when distance increases?
- What is the average price by distance range (0–500, 500–1000 etc)?
- What is the average price for each airline and class combination?
- Which route (Origin → Destination) has the highest average price?
- Which route has the longest distance?
- What is the average price when booking early vs late?
- Which airline operates the longest flights?
- What is the average price per km for each airline?
- Which class has the highest average distance?
- Which routes are most frequently traveled?

Expert / Analytical Questions (31–40)

- How does Days_Before_Departure affect price?
- Do early bookings reduce ticket price?
- Which airline has the most expensive business class tickets?
- Which airline offers the cheapest economy tickets?
- What is the price distribution for each class?
- Which Origin-Destination route generates the highest revenue?
- What is the average price per km by class?
- Which airline has the largest number of routes?
- What percentage of tickets belong to each class?
- Which airline dominates each route?

- 41. Which airline generates the highest total revenue? 📌 Airline wise Price_USD sum
- 42. Which class contributes the most to total revenue? 📌 Business vs Economy impact samajhne ke liye
- 43. What is the revenue share (%) of each airline? 📌 Market share analysis
- 44. Which routes are most profitable (high price_per_km)? 📌 Profitability insight
- 45. Which origin city generates the highest revenue?
- 46. How does ticket price vary across different distance ranges? 📌 Visualization (boxplot / lineplot)
- 47. Is there any airline that charges significantly more for the same distance? 📌 Outlier airline detection
- 48. How does price vary with booking type (Online vs Agent)?
- 49. Which days_before_departure range gives the best (lowest) price? 📌 Bucket bana sakte ho (0–5, 5–10, etc.)

50. Is there price volatility in last-minute bookings? 🖱️ Std deviation use karo
51. Which airline has the most consistent pricing? 🖱️ Low standard deviation
52. Which route has the highest price variation?
53. Are longer routes always more expensive per km? 🖱️ Normalize using price_per_km
54. Which airline is best for long-distance travel (lowest price per km)?
55. Can we segment customers based on booking behavior (early vs late)? 🖱️ Clustering idea (future ML use)

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings("ignore")
plt.style.use("ggplot")
```

```
df = pd.read_csv("/content/airline_ticket_prices_dataset.csv")
df.head()
```

	Ticket_ID	Airline	Origin	Destination	Distance_km	Class	Days_Before_Departure	Price_USD	
0	1	Saudia	Mumbai	Paris	2643	Economy	28	555.30	
1	2	Etihad	Istanbul	Jeddah	8266	Business	98	2070.27	
2	3	British Airways	Dubai	Riyadh	2698	First	21	1883.11	
3	4	Emirates	London	Istanbul	7956	First	24	5349.65	
4	5	Qatar Airways	Doha	Dubai	1308	Business	60	516.19	

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

```
df.columns
```

```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',
      'Days_Before_Departure', 'Price_USD'],
      dtype='object')
```

```
for i in df.columns:
    print("columns : -- ",i)
    print("Missing Values:", df[i].isnull().sum())
    print(" ")
```

```
columns : -- Ticket_ID
Missing Values: 0

columns : -- Airline
Missing Values: 0

columns : -- Origin
Missing Values: 0

columns : -- Destination
Missing Values: 0

columns : -- Distance_km
Missing Values: 0

columns : -- Class
Missing Values: 0

columns : -- Days_Before_Departure
Missing Values: 0

columns : -- Price_USD
Missing Values: 0
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 250 entries, 0 to 249
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Ticket_ID             250 non-null   int64
1   Airline                250 non-null   object
2   Origin                 250 non-null   object
3   Destination            250 non-null   object
4   Distance_km            250 non-null   int64
5   Class                  250 non-null   object
6   Days_Before_Departure  250 non-null   int64
7   Price_USD              250 non-null   float64
dtypes: float64(1), int64(3), object(4)
memory usage: 15.8+ KB
```

```
df.describe()
```

	Ticket_ID	Distance_km	Days_Before_Departure	Price_USD	
count	250.000000	250.000000	250.000000	250.000000	
mean	125.500000	5999.952000	59.024000	1999.262720	
std	72.312977	3401.695261	33.736628	1678.803619	
min	1.000000	336.000000	1.000000	106.070000	
25%	63.250000	2991.750000	31.250000	815.892500	
50%	125.500000	6132.500000	60.500000	1519.105000	
75%	187.750000	8832.750000	86.750000	2590.357500	
max	250.000000	11903.000000	118.000000	8852.690000	

df.columns

Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class', 'Days_Before_Departure', 'Price_USD'], dtype='object')

```
for col in df.select_dtypes(include="object").columns:
    print(f"name of columns is {col} : --")
    print(df[col].value_counts())
    print(" ")
```

```
name of columns is Airline : --
Airline
British Airways    36
Qatar Airways     36
Lufthansa         34
Saudia            32
Emirates          32
Turkish Airlines  30
Flynas            26
Etihad            24
Name: count, dtype: int64

name of columns is Origin : --
Origin
Paris      32
Istanbul   31
Doha       28
London     27
Mumbai     26
Riyadh     23
Jeddah     22
New York   21
Cairo      21
Dubai      19
Name: count, dtype: int64

name of columns is Destination : --
Destination
Cairo      36
Riyadh     32
Istanbul   30
Paris      29
New York   24
Dubai      24
London     20
Doha       19
Mumbai     19
Jeddah     17
Name: count, dtype: int64

name of columns is Class : --
Class
Economy    89
First      86
Business   75
Name: count, dtype: int64
```

Total number of tickets in the dataset?

df.columns

Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class', 'Days_Before_Departure', 'Price_USD'], dtype='object')

df["Ticket_ID"].nunique()

250

How many unique airlines are present?

df['Airline'].unique()

array(['Saudia', 'Etihad', 'British Airways', 'Emirates', 'Qatar Airways', 'Turkish Airlines', 'Flynas', 'Lufthansa'], dtype=object)

df['Airline'].nunique()

8

How many unique origin cities are there?

```
df['Origin'].unique()

array(['Mumbai', 'Istanbul', 'Dubai', 'London', 'Doha', 'Paris',
       'New York', 'Jeddah', 'Riyadh', 'Cairo'], dtype=object)
```

```
df['Origin'].nunique()

10
```

How many unique destination cities are there?

```
df['Destination'].unique()

array(['Paris', 'Jeddah', 'Riyadh', 'Istanbul', 'Dubai', 'Doha',
       'New York', 'Cairo', 'London', 'Mumbai'], dtype=object)
```

```
df['Destination'].nunique()

10
```

What is the average ticket price?

```
df.columns

Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',
       'Days_Before_Departure', 'Price_USD'],
      dtype='object')
```

```
print(f"avarage ticket price is ${df["Price_USD"].mean()}")

avarage ticket price is $1999.26272
```

What is the minimum ticket price?

```
print(f"minimum ticket price is ${df["Price_USD"].min()}")

minimum ticket price is $106.07
```

What is the maximum ticket price?

```
print(f"maximum ticket price is ${df["Price_USD"].min()}")

maximum ticket price is $106.07
```

What is the average distance traveled?

```
print(f"average distance traveled is {df['Distance_km'].mean()}")

average distance traveled is 5999.952
```

How many tickets belong to each travel class?

```
df1 = df["Class"].value_counts().reset_index()
df1 = df1.rename(columns={"count": "number_of_ticket"})
df1
```

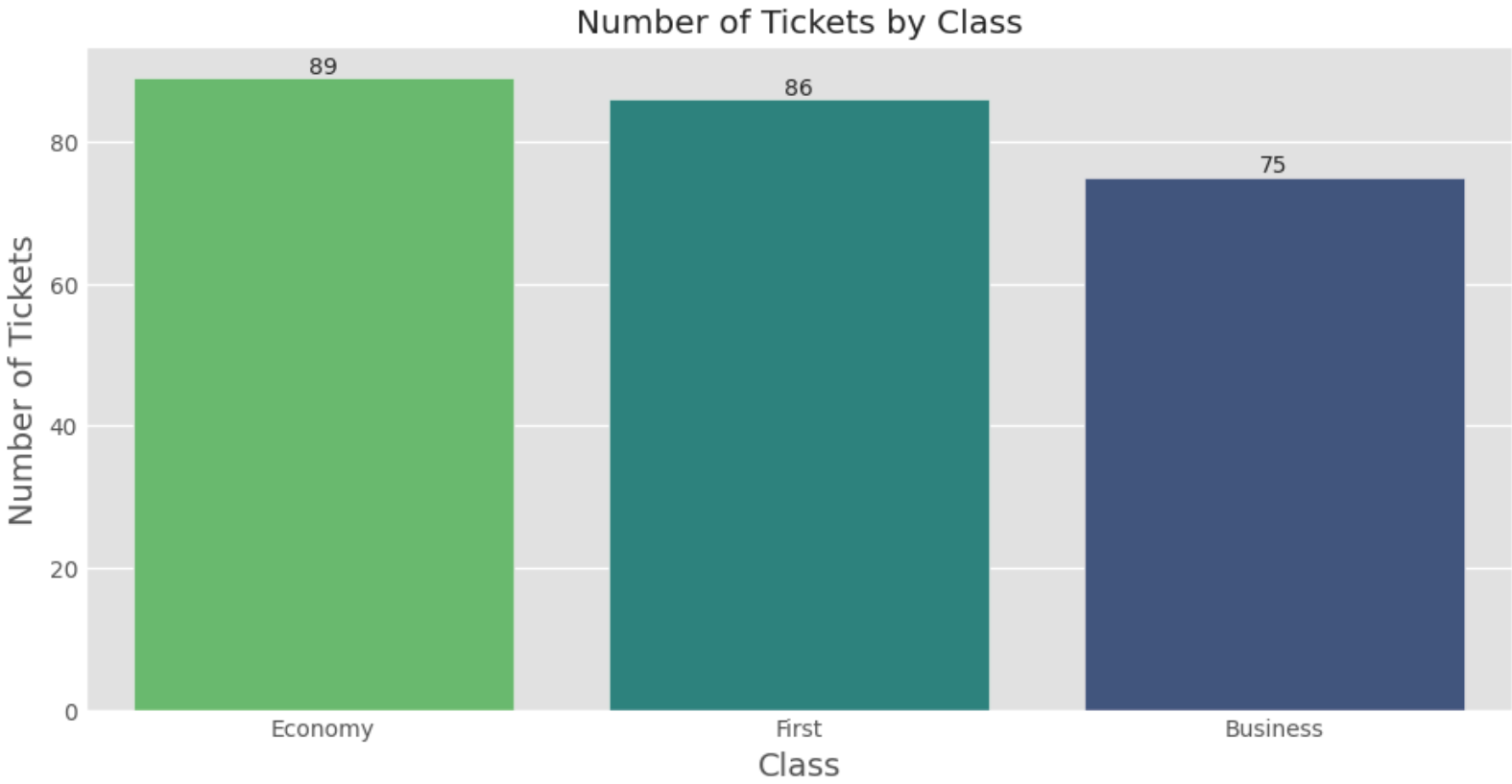
	Class	number_of_ticket	
0	Economy	89	 
1	First	86	
2	Business	75	

Next steps:

[Generate code with df1](#)

[New interactive sheet](#)

```
plt.figure(figsize=(10,5))
ax = sns.barplot(x="Class", y="number_of_ticket", data=df1,palette="viridis_r")
for i in ax.containers:
    ax.bar_label(i)
plt.title("Number of Tickets by Class")
plt.xlabel("Class",fontsize=14)
plt.ylabel("Number of Tickets",fontsize=14)
plt.show()
```



Count of tickets sold by each airline.

```
df.columns
```

```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',
      'Days_Before_Departure', 'Price_USD'],
      dtype='object')
```

```
df["Airline"].value_counts()
```

count	
Airline	
British Airways	36
Qatar Airways	36
Lufthansa	34
Saudia	32
Emirates	32
Turkish Airlines	30
Flynas	26
Etihad	24

dtype: int64

```
df2 = df.groupby("Airline")["Ticket_ID"].count().sort_values(ascending=False).reset_index(name='Ticket_Count')
df2
```

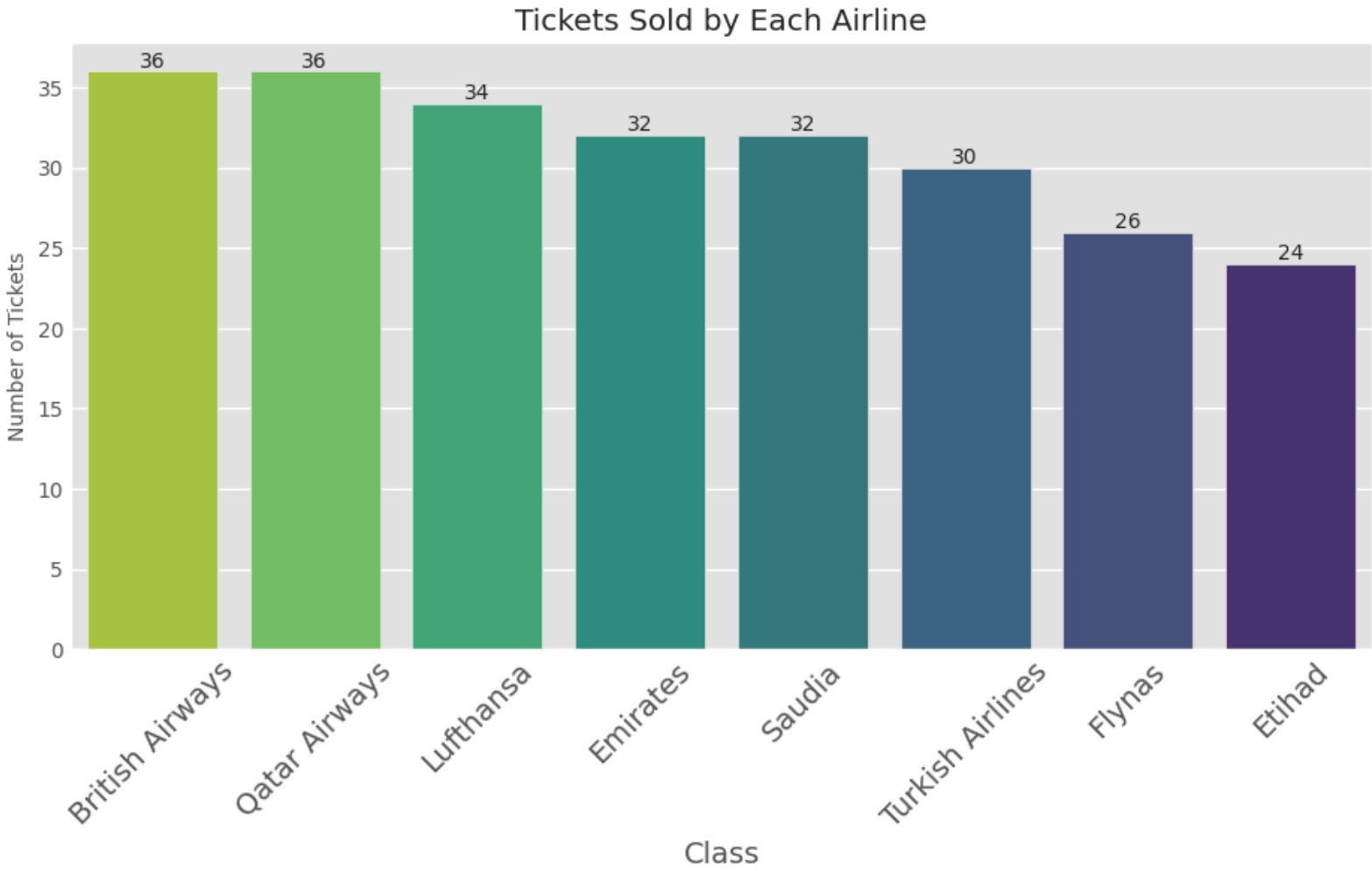
	Airline	Ticket_Count	
0	British Airways	36	
1	Qatar Airways	36	
2	Lufthansa	34	
3	Emirates	32	
4	Saudia	32	
5	Turkish Airlines	30	
6	Flynas	26	
7	Etihad	24	

Next steps:

[Generate code with df2](#)

[New interactive sheet](#)

```
plt.figure(figsize=(10,5))
ax = sns.barplot(x="Airline", y="Ticket_Count", data=df2,palette="viridis_r")
for i in ax.containers:
    ax.bar_label(i)
plt.title("Tickets Sold by Each Airline")
plt.xlabel("Class",fontsize=14)
plt.xticks(rotation=45,fontsize=14)
plt.ylabel("Number of Tickets",fontsize=10)
plt.show()
```



What is the average price for each airline?

```
df.columns
```

```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',  
      'Days_Before_Departure', 'Price_USD'],  
      dtype='object')
```

```
round(df.groupby("Airline")["Price_USD"].mean(),2).reset_index().rename(columns={"Price_USD":"average price"})
```

	Airline	average price	
0	British Airways	1934.80	
1	Emirates	2103.24	
2	Etihad	1486.47	
3	Flynas	1671.06	
4	Lufthansa	2066.63	
5	Qatar Airways	2022.13	
6	Saudia	2389.40	
7	Turkish Airlines	2140.44	

What is the average price for each class?

```
df3 = round(df.groupby("Class")["Price_USD"].mean(),2).reset_index().rename(columns={"Price_USD":"average price"})  
df3
```

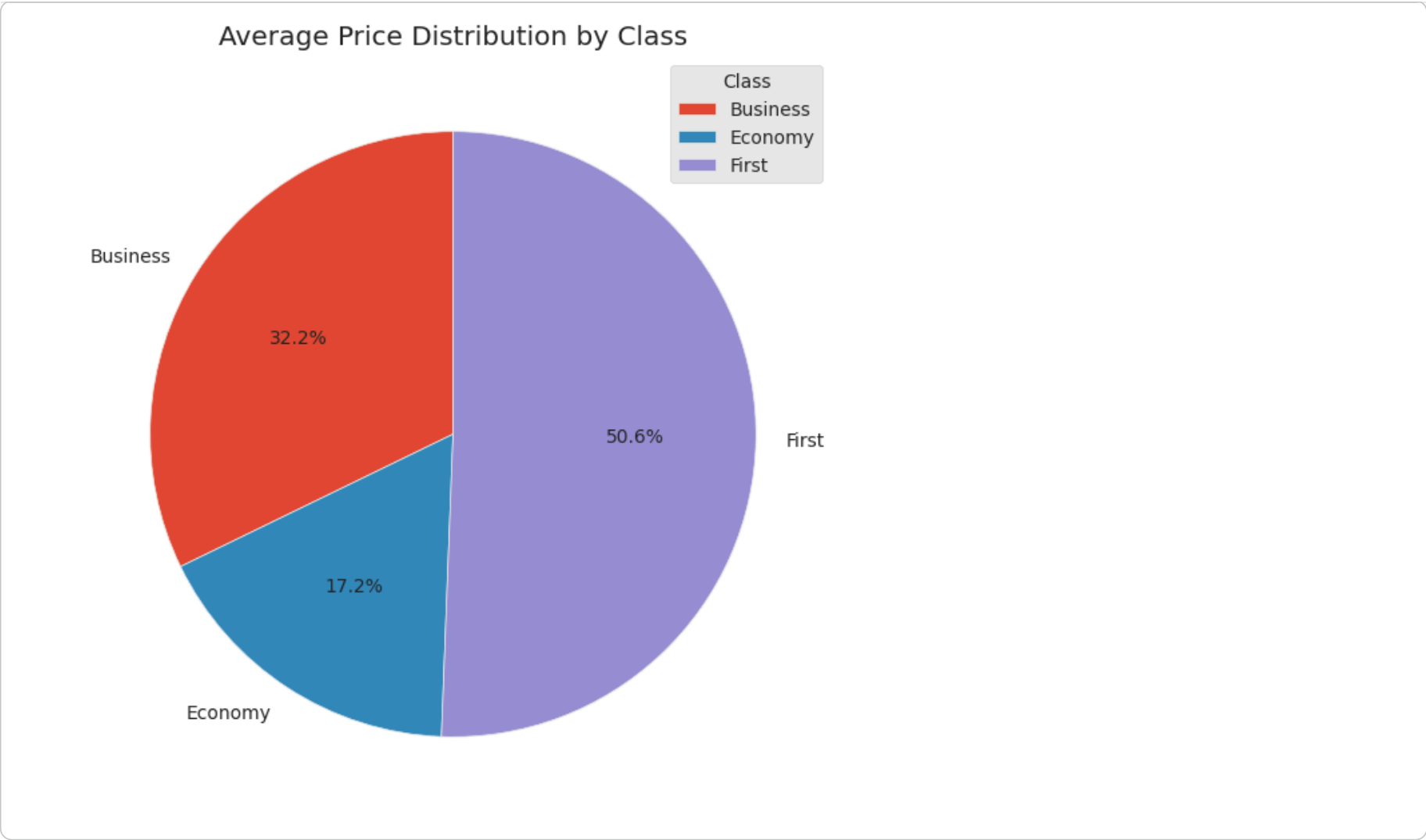
	Class	average price	
0	Business	1939.81	
1	Economy	1035.39	
2	First	3048.61	

Next steps:

[Generate code with df3](#)

[New interactive sheet](#)

```
plt.figure(figsize=(7,7))  
  
plt.pie(df3['average price'],labels=df3['Class'],autopct='%1.1f%%',startangle=90)  
plt.title("Average Price Distribution by Class")  
plt.legend(df3['Class'],title="Class",loc="best")  
plt.show()
```

Which origin city has the most flights?

```
df.columns
```

```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',  
      'Days_Before_Departure', 'Price_USD'],  
      dtype='object')
```

```
df["Origin"].unique()
```

```
array(['Mumbai', 'Istanbul', 'Dubai', 'London', 'Doha', 'Paris',  
      'New York', 'Jeddah', 'Riyadh', 'Cairo'], dtype=object)
```

```
df4 = df.groupby("Origin")["Ticket_ID"].count().sort_values(ascending=False).reset_index().rename(columns={"Ticket_ID": "number of flight", "Origin": "Origin"})
```

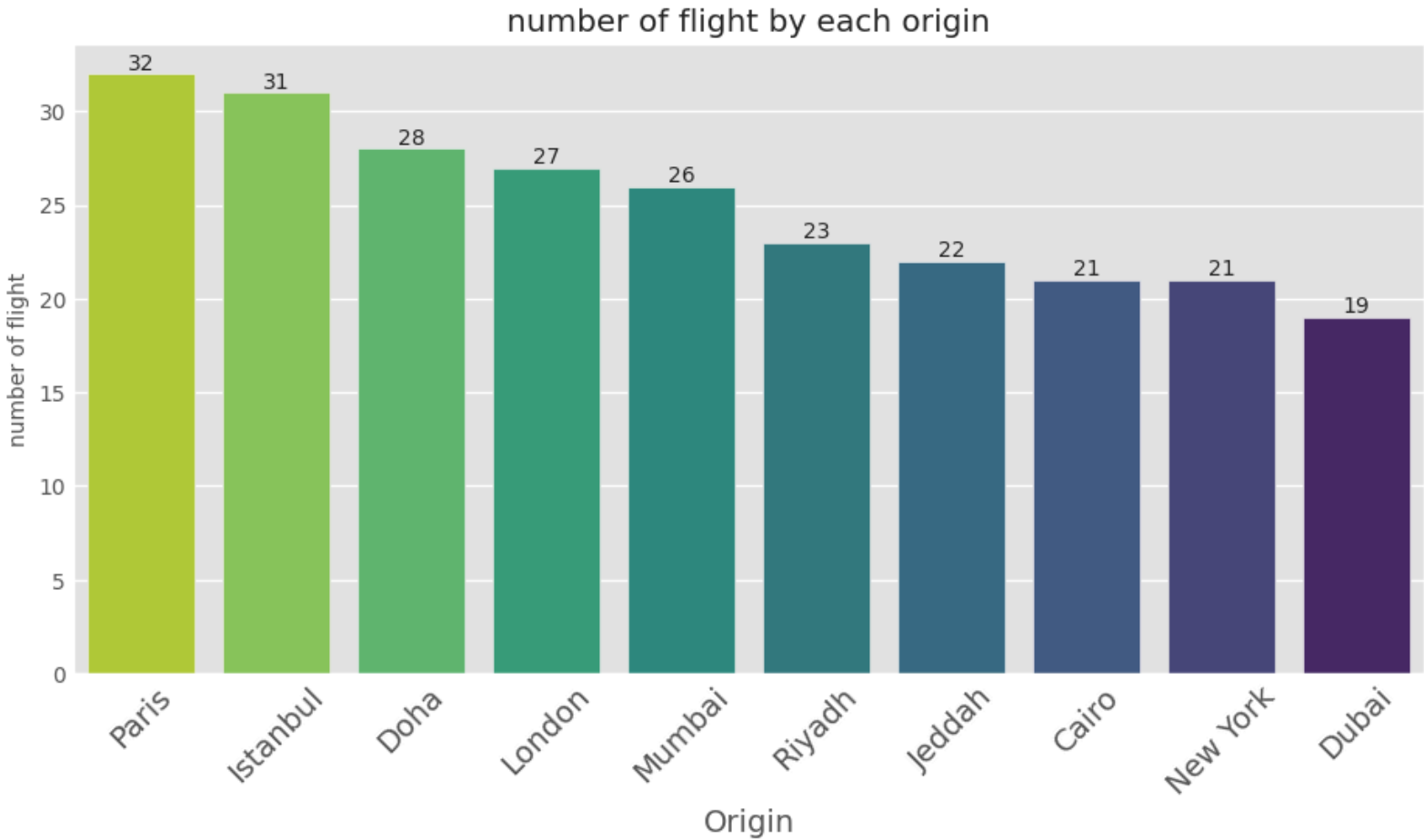
	Origin	number of flight
0	Paris	32
1	Istanbul	31
2	Doha	28
3	London	27
4	Mumbai	26
5	Riyadh	23
6	Jeddah	22
7	Cairo	21
8	New York	21
9	Dubai	19

Next steps:

[Generate code with df4](#)

[New interactive sheet](#)

```
plt.figure(figsize=(10,5))
ax = sns.barplot(x="Origin", y="number of flight", data=df4,palette="viridis_r")
for i in ax.containers:
    ax.bar_label(i)
plt.title("number of flight by each origin")
plt.xlabel("Origin",fontsize=14)
plt.xticks(rotation=45,fontsize=14)
plt.ylabel("number of flight",fontsize=10)
plt.show()
```



```
plt.figure(figsize=(7,7))
plt.pie(df4['number of flight'],labels=df4['Origin'],autopct='%1.1f%%',startangle=90)
plt.title("number of flight by each origin by market share")
plt.legend(df4['Origin'],title="Origin",loc="best")
plt.show()
```



```
df4 = df.groupby("Origin")["Ticket_ID"].count().sort_values(ascending=False).reset_index().rename(columns={"Ticket_ID": "number of flight"})
df4.head(1)
```

	Origin	number of flight	
0	Paris	32	

Next steps:

[Generate code with df4](#)

[New interactive sheet](#)

```
df["Origin"].value_counts()
```


count	
Origin	
Paris	32
Istanbul	31
Doha	28
London	27
Mumbai	26
Riyadh	23
Jeddah	22
New York	21
Cairo	21
Dubai	19

dtype: int64

Which destination city has the most flights?

df.columns

Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',
 'Days_Before_Departure', 'Price_USD'],
 dtype='object')

df5 = df.groupby("Destination")["Ticket_ID"].count().sort_values(ascending=False).reset_index().rename(columns={"Ticket_ID": "number of flight", "Destination": "Destination"})
df5

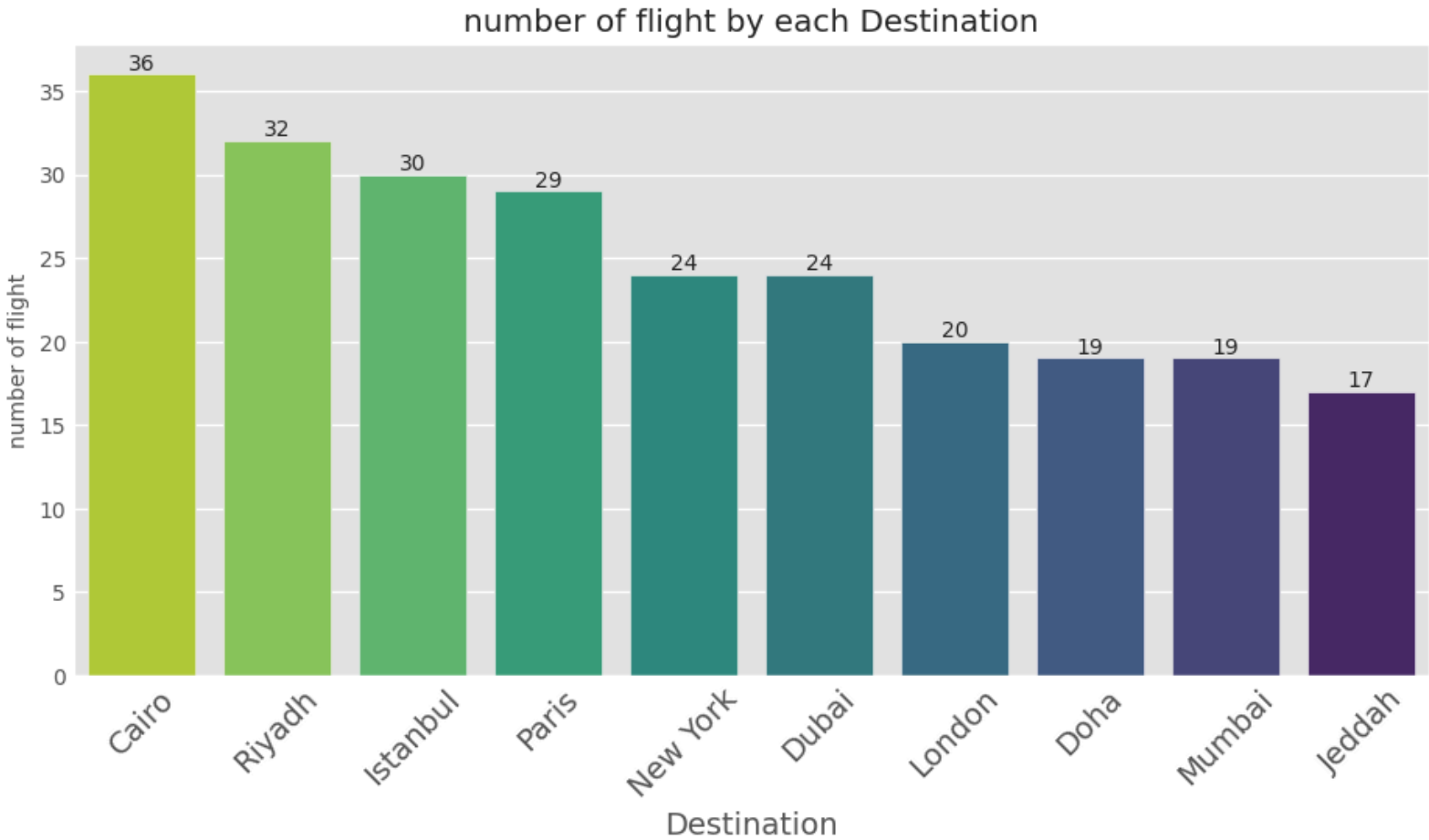
	Destination	number of flight
0	Cairo	36
1	Riyadh	32
2	Istanbul	30
3	Paris	29
4	New York	24
5	Dubai	24
6	London	20
7	Doha	19
8	Mumbai	19
9	Jeddah	17

Next steps:

[Generate code with df5](#)

[New interactive sheet](#)

```
plt.figure(figsize=(10,5))
ax = sns.barplot(x="Destination", y="number of flight", data=df5,palette="viridis_r")
for i in ax.containers:
    ax.bar_label(i)
plt.title("number of flight by each Destination")
plt.xlabel("Destination",fontsize=14)
plt.xticks(rotation=45,fontsize=14)
plt.ylabel("number of flight",fontsize=10)
plt.show()
```

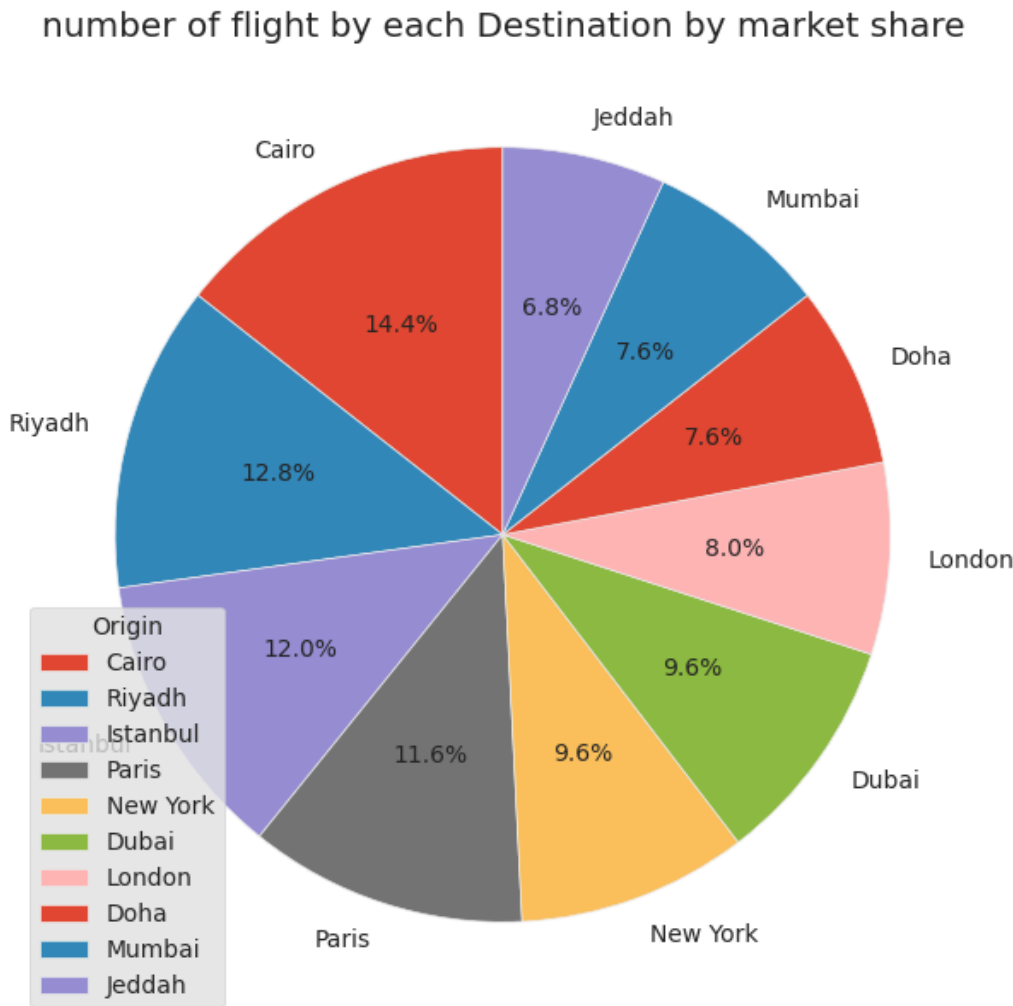


```
df["Destination"].value_counts()
```

	count
Destination	
Cairo	36
Riyadh	32
Istanbul	30
Paris	29
New York	24
Dubai	24
London	20
Doha	19
Mumbai	19
Jeddah	17

dtype: int64

```
plt.figure(figsize=(7,7))
plt.pie(df5['number of flight'],labels=df5['Destination'],autopct='%1.1f%%',startangle=90)
plt.title("number of flight by each Destination by market share")
plt.legend(df5['Destination'],title="Origin",loc="best")
plt.show()
```



What is the average distance by airline

df.columns

Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class', 'Days_Before_Departure', 'Price_USD'], dtype='object')

df6 = df.groupby("Airline")["Distance_km"].mean().round(2).reset_index().rename(columns={"Distance_km":"average distance"}, inplace=True)
df6

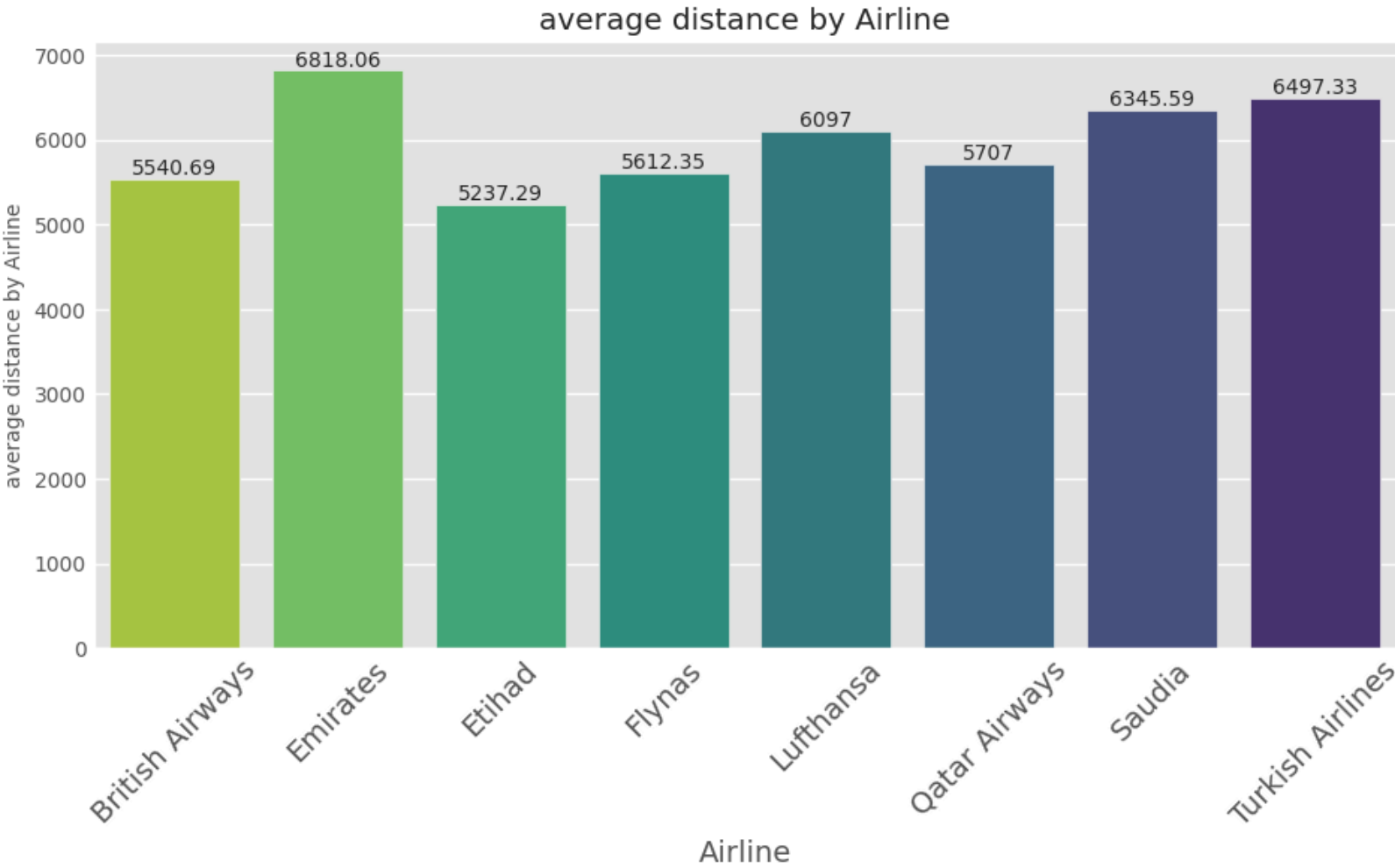
	Airline	average distance
0	British Airways	5540.69
1	Emirates	6818.06
2	Etihad	5237.29
3	Flynas	5612.35
4	Lufthansa	6097.00
5	Qatar Airways	5707.00
6	Saudia	6345.59
7	Turkish Airlines	6497.33

Next steps:

[Generate code with df6](#)

[New interactive sheet](#)

```
plt.figure(figsize=(10,5))
ax = sns.barplot(x="Airline", y="average distance", data=df6,palette="viridis_r")
for i in ax.containers:
    ax.bar_label(i)
plt.title("average distance by Airline")
plt.xlabel("Airline",fontsize=14)
plt.xticks(rotation=45,fontsize=14)
plt.ylabel("average distance by Airline",fontsize=10)
plt.show()
```



Which airline has the highest average ticket price?

df.columns

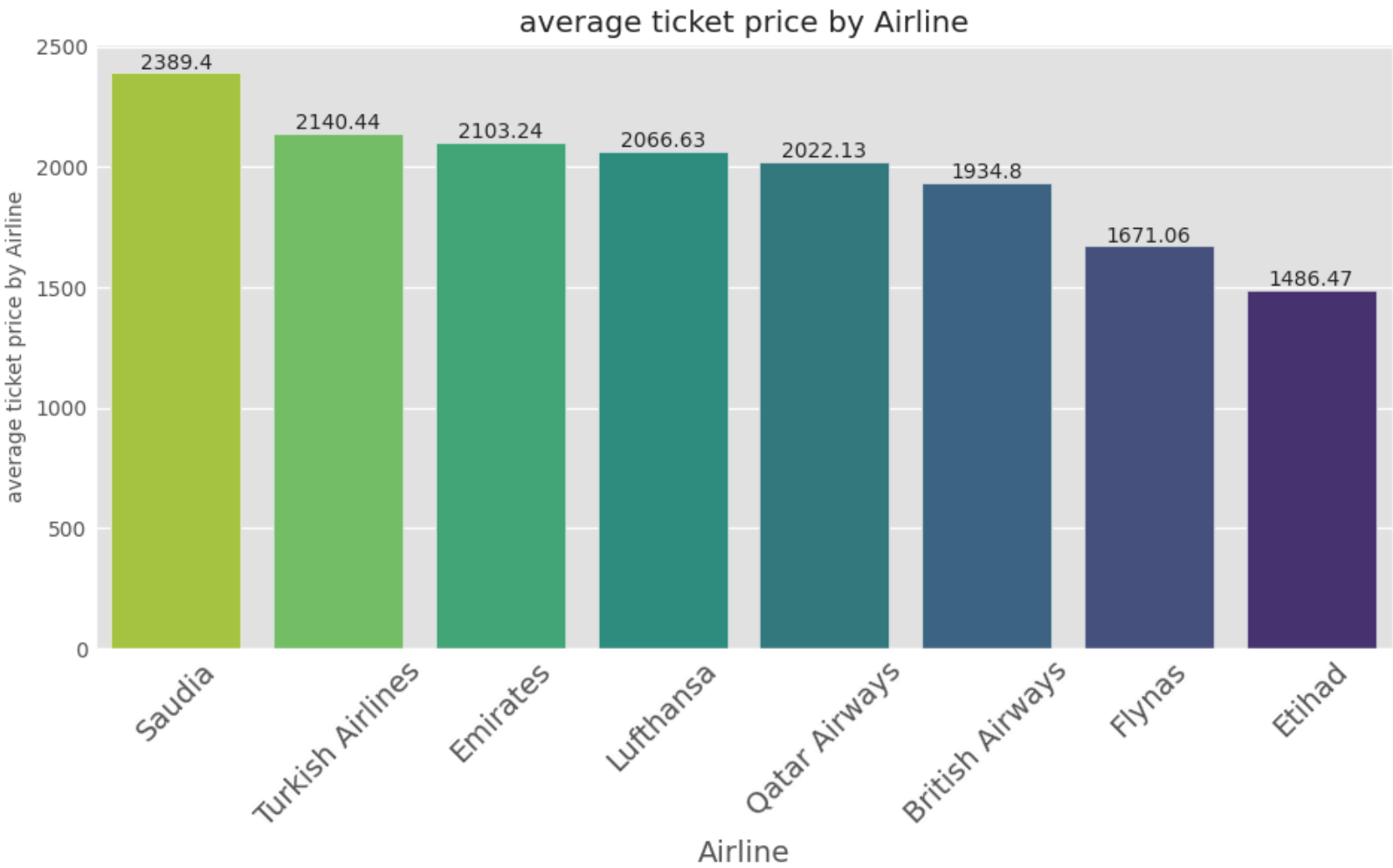
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class', 'Days_Before_Departure', 'Price_USD'], dtype='object')

df7 = df.groupby("Airline")["Price_USD"].mean().round(2).sort_values(ascending=False).reset_index().rename(columns={"Price_USD":"average price"}, inplace=True)
df7

	Airline	average ticket price
0	Saudia	2389.40
1	Turkish Airlines	2140.44
2	Emirates	2103.24
3	Lufthansa	2066.63
4	Qatar Airways	2022.13
5	British Airways	1934.80
6	Flynas	1671.06
7	Etihad	1486.47

Next steps: [Generate code with df7](#) [New interactive sheet](#)

```
plt.figure(figsize=(10,5))
ax = sns.barplot(x="Airline", y="average ticket price", data=df7,palette="viridis_r")
for i in ax.containers:
    ax.bar_label(i)
plt.title("average ticket price by Airline")
plt.xlabel("Airline",fontsize=14)
plt.xticks(rotation=45,fontsize=14)
plt.ylabel("average ticket price by Airline",fontsize=10)
plt.show()
```



Which airline has the lowest average ticket price?

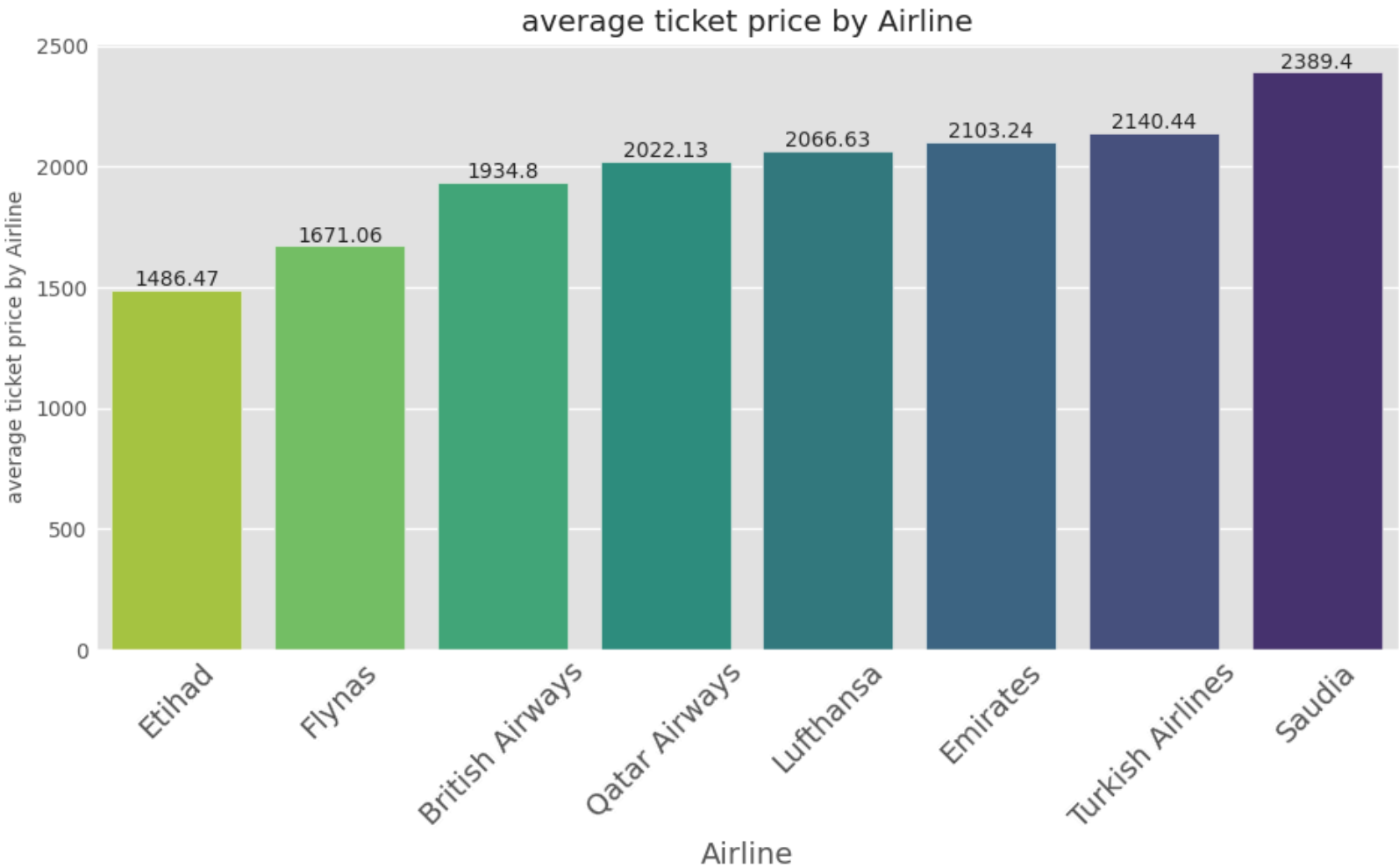
```
df8 = df.groupby("Airline")["Price_USD"].mean().round(2).sort_values(ascending=True).reset_index().rename(columns={'Price_USD': 'average ticket price'})
df8
```

	Airline	average ticket price
0	Etihad	1486.47
1	Flynas	1671.06
2	British Airways	1934.80
3	Qatar Airways	2022.13
4	Lufthansa	2066.63
5	Emirates	2103.24
6	Turkish Airlines	2140.44
7	Saudia	2389.40

Next steps: [Generate code with df8](#) [New interactive sheet](#)

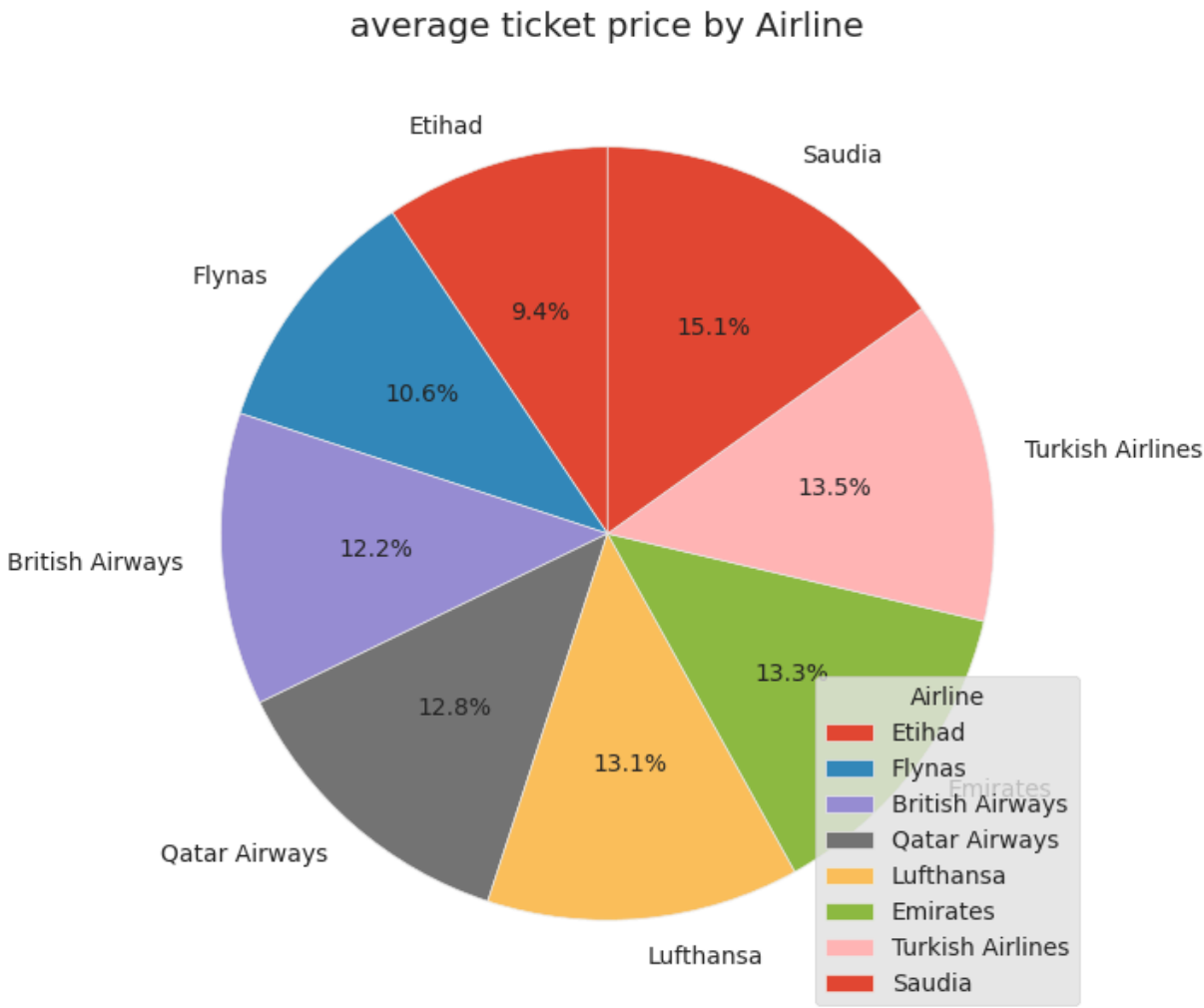
```
plt.figure(figsize=(10,5))
ax = sns.barplot(x="Airline", y="average ticket price", data=df8,palette="viridis_r")
for i in ax.containers:
    ax.bar_label(i)
plt.title("average ticket price by Airline")
plt.xlabel("Airline",fontsize=14)
```

```
plt.xticks(rotation=45,fontsize=14)
plt.ylabel("average ticket price by Airline",fontsize=10)
plt.show()
```



Market Share

```
plt.figure(figsize=(7,7))
plt.pie(df8['average ticket price'],labels=df8['Airline'],autopct='%1.1f%%',startangle=90)
plt.title("average ticket price by Airline")
plt.legend(df8['Airline'],title="Airline",loc=0)
plt.show()
```



What is the average price for each origin city?

```
df.columns
```

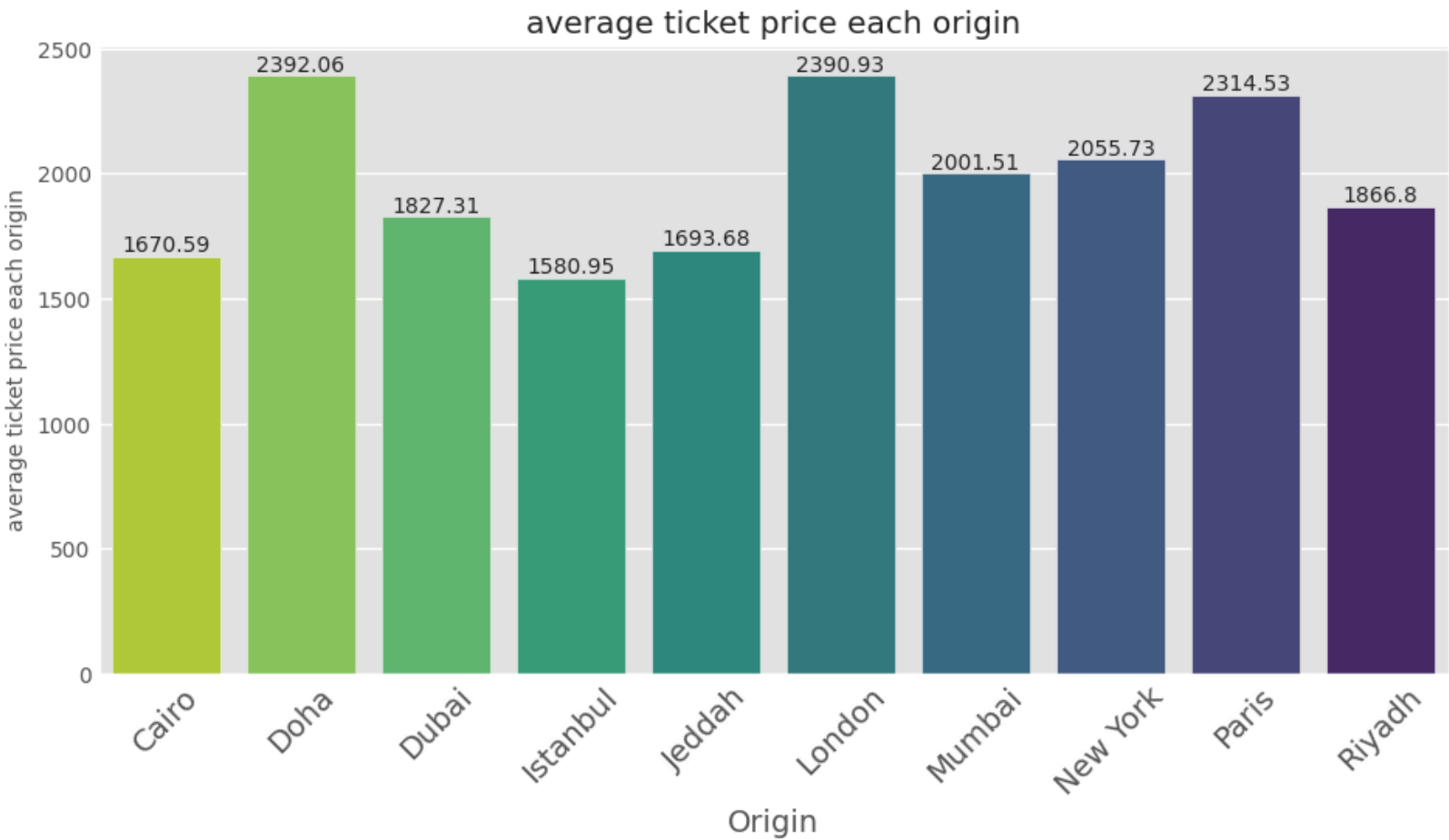
```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',
      'Days_Before_Departure', 'Price_USD'],
      dtype='object')
```

```
df9 = df.groupby("Origin")["Price_USD"].mean().round(2).reset_index().rename(columns={"Price_USD":"average ticket price"})
df9
```

	Origin	average ticket price	
0	Cairo	1670.59	
1	Doha	2392.06	
2	Dubai	1827.31	
3	Istanbul	1580.95	
4	Jeddah	1693.68	
5	London	2390.93	
6	Mumbai	2001.51	
7	New York	2055.73	
8	Paris	2314.53	
9	Riyadh	1866.80	

Next steps: [Generate code with df9](#) [New interactive sheet](#)

```
plt.figure(figsize=(10,5))
ax = sns.barplot(x="Origin", y="average ticket price", data=df9,palette="viridis_r")
for i in ax.containers:
    ax.bar_label(i)
plt.title("average ticket price each origin")
plt.xlabel("Origin",fontsize=14)
plt.xticks(rotation=45,fontsize=14)
plt.ylabel("average ticket price each origin",fontsize=10)
plt.show()
```



What is the average price for each destination city?

```
df10 = df.groupby("Destination")["Price_USD"].mean().round(2).reset_index().rename(columns={"Price_USD":"average ti
df10
```

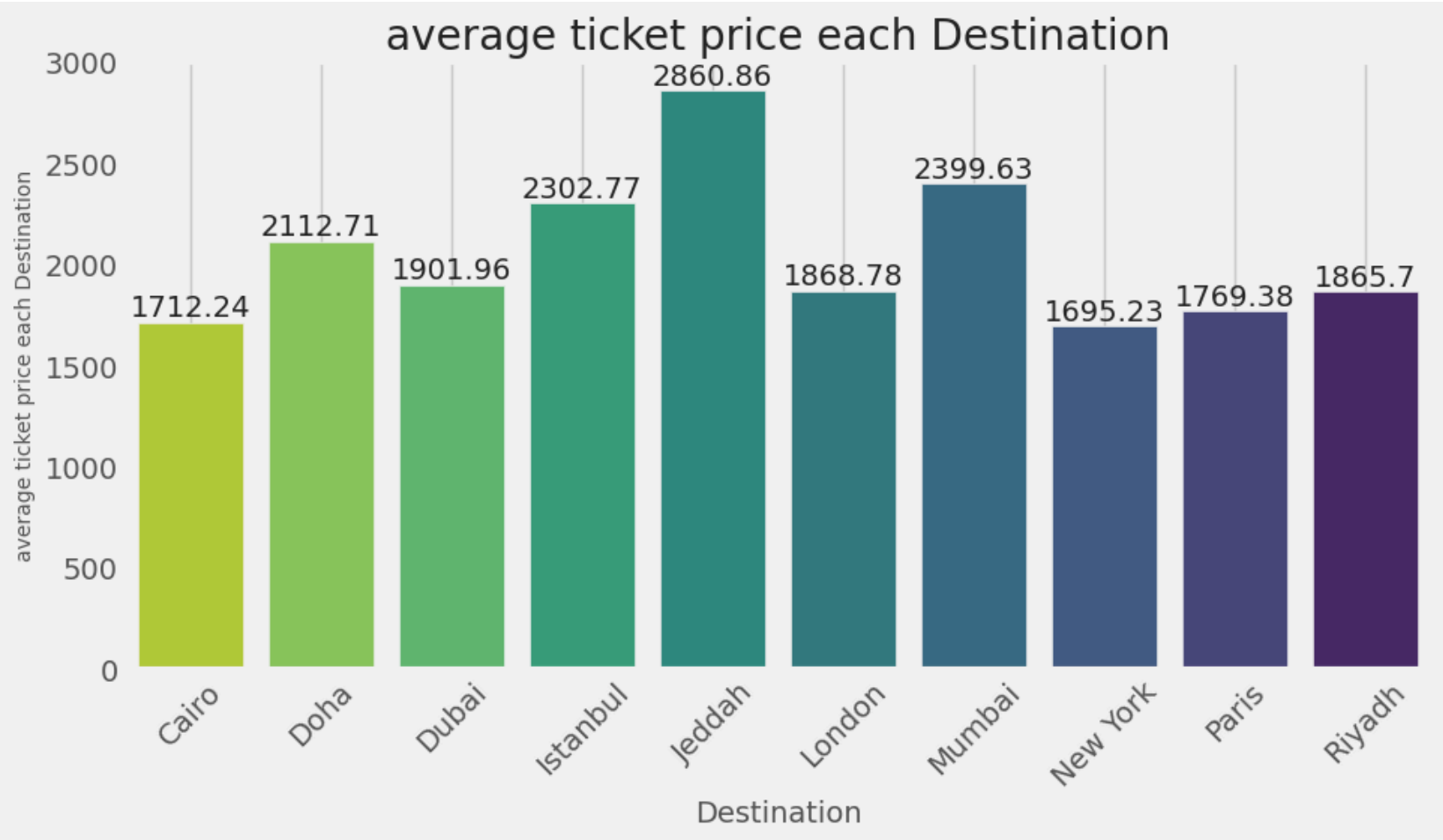
	Destination	average ticket price	
0	Cairo	1712.24	
1	Doha	2112.71	
2	Dubai	1901.96	
3	Istanbul	2302.77	
4	Jeddah	2860.86	
5	London	1868.78	
6	Mumbai	2399.63	
7	New York	1695.23	
8	Paris	1769.38	
9	Riyadh	1865.70	

Next steps: [Generate code with df10](#) [New interactive sheet](#)

```
plt.style.use('fivethirtyeight')
```



```
plt.figure(figsize=(10,5))
ax = sns.barplot(x="Destination", y="average ticket price", data=df10,palette="viridis_r")
for i in ax.containers:
    ax.bar_label(i)
plt.title("average ticket price each Destination")
plt.xlabel("Destination",fontsize=14)
plt.xticks(rotation=45,fontsize=14)
plt.ylabel("average ticket price each Destination",fontsize=10)
plt.grid()
plt.show()
```



How many tickets were booked for each Days_Before_Departure value?

```
df.columns
```

```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',
      'Days_Before_Departure', 'Price_USD'],
      dtype='object')
```

```
df["Days_Before_Departure"].unique()
```

```
array([ 28,  98,  21,  24,  60,  19,  85,  88,  36,  55, 108,  72,  48,
        38,   9,  82,  37, 109,   4,   3,  23,  86,  54,  71,  93,  99,
        34,  83,  44,  53,  45,  40, 111,  43,  95,  33,  39,   6,  15,
        90,  20,  76,  80,  50,  66,  26,  30, 102,  81,  18,  87,   8,
        92,  64,  32,  74,  61,  13,  16, 110, 112, 107,  14,  35,  75,
        73,  27,  96,  25,  62,  68,  47, 105,  91,  97, 114,  41, 117,
        69,   7, 103, 106,  63,   5,  70,  49,  12,  94, 101,  78,  56,
        89,  67,  11,  52, 116, 118, 115,  31, 100, 104,  58,  84,   1,
        65,  79])
```

```
df.groupby("Days_Before_Departure")["Ticket_ID"].count().reset_index().rename(columns={"Ticket_ID": "Ticket_Count"})
```

Days_Before_Departure	Ticket_Count	
15	19	6
4	6	4
29	35	1
42	50	4
44	53	1

```
df_days = df.groupby("Days_Before_Departure")["Ticket_ID"].count().reset_index()
df_days = df_days.rename(columns={"Ticket_ID": "Ticket_Count"})
df_days = df_days.sort_values("Days_Before_Departure")
df_days
```

Days_Before_Departure	Ticket_Count	
0	1	1
1	3	5
2	4	4
3	5	2
4	6	4
...	...	...
101	114	2
102	115	2
103	116	1
104	117	3
105	118	2

106 rows × 2 columns

Next steps: [Generate code with df_days](#) [New interactive sheet](#)

```
# 2. Data preparation
df_days = df.groupby("Days_Before_Departure")["Ticket_ID"].count().reset_index()
df_days = df_days.rename(columns={"Ticket_ID": "Ticket_Count"})

# Sorting important for line plot
df_days = df_days.sort_values("Days_Before_Departure")

# 3. Style (optional but good)
sns.set_style("whitegrid")

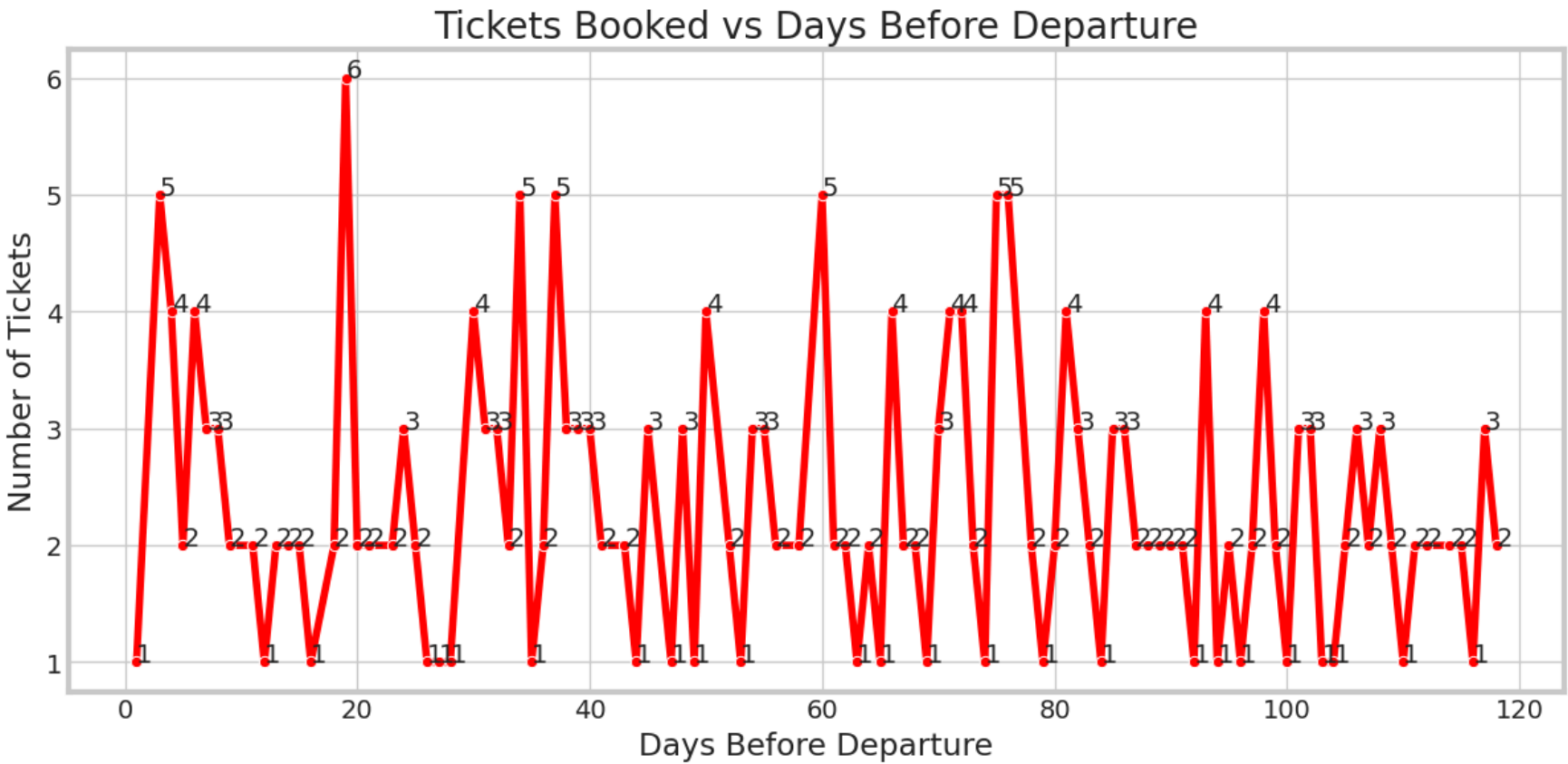
# 4. Line plot
plt.figure(figsize=(13,6))

sns.lineplot(
    data=df_days,
    x="Days_Before_Departure",
    y="Ticket_Count",
    marker='o',
    color = "r"
)

for x, y in zip(df_days["Days_Before_Departure"], df_days["Ticket_Count"]):
    plt.text(x, y, str(y))

# 5. Labels & title
plt.xlabel("Days Before Departure")
plt.ylabel("Number of Tickets")
plt.title("Tickets Booked vs Days Before Departure")

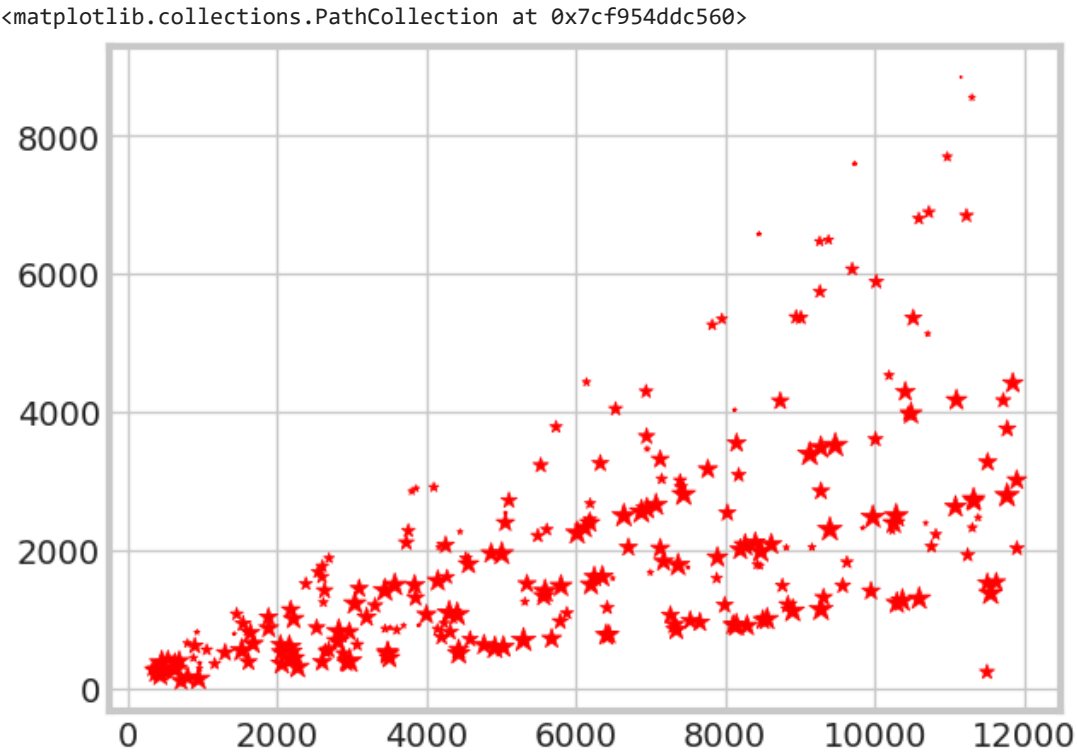
# 6. Show plot
plt.show()
```




```
df.columns
```

```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',  
      'Days_Before_Departure', 'Price_USD'],  
      dtype='object')
```

```
plt.scatter(df['Distance_km'],df['Price_USD'],s=df["Days_Before_Departure"],color ="r",marker="*")
```



```
plt.style.available
```

```
['Solarize_Light2',  
 '_classic_test_patch',  
 '_mpl-gallery',  
 '_mpl-gallery-nogrid',  
 'bmh',  
 'classic',  
 'dark_background',  
 'fast',  
 'fivethirtyeight',  
 'ggplot',  
 'grayscale',  
 'petroff10',  
 'seaborn-v0_8',  
 'seaborn-v0_8-bright',  
 'seaborn-v0_8-colorblind',  
 'seaborn-v0_8-dark',  
 'seaborn-v0_8-dark-palette',  
 'seaborn-v0_8-darkgrid',  
 'seaborn-v0_8-deep',  
 'seaborn-v0_8-muted',  
 'seaborn-v0_8-notebook',  
 'seaborn-v0_8-paper',  
 'seaborn-v0_8-pastel',  
 'seaborn-v0_8-poster',  
 'seaborn-v0_8-talk',  
 'seaborn-v0_8-ticks',  
 'seaborn-v0_8-white',  
 'seaborn-v0_8-whitegrid',  
 'tableau-colorblind10']
```

Does price increase when distance increases?

Yes, usually price increases with distance

```
df.head(1)
```

	Ticket_ID	Airline	Origin	Destination	Distance_km	Class	Days_Before_Departure	Price_USD	
0	1	Saudia	Mumbai	Paris	2643	Economy	28	555.3	

Next steps: [Generate code with df](#) [New interactive sheet](#)

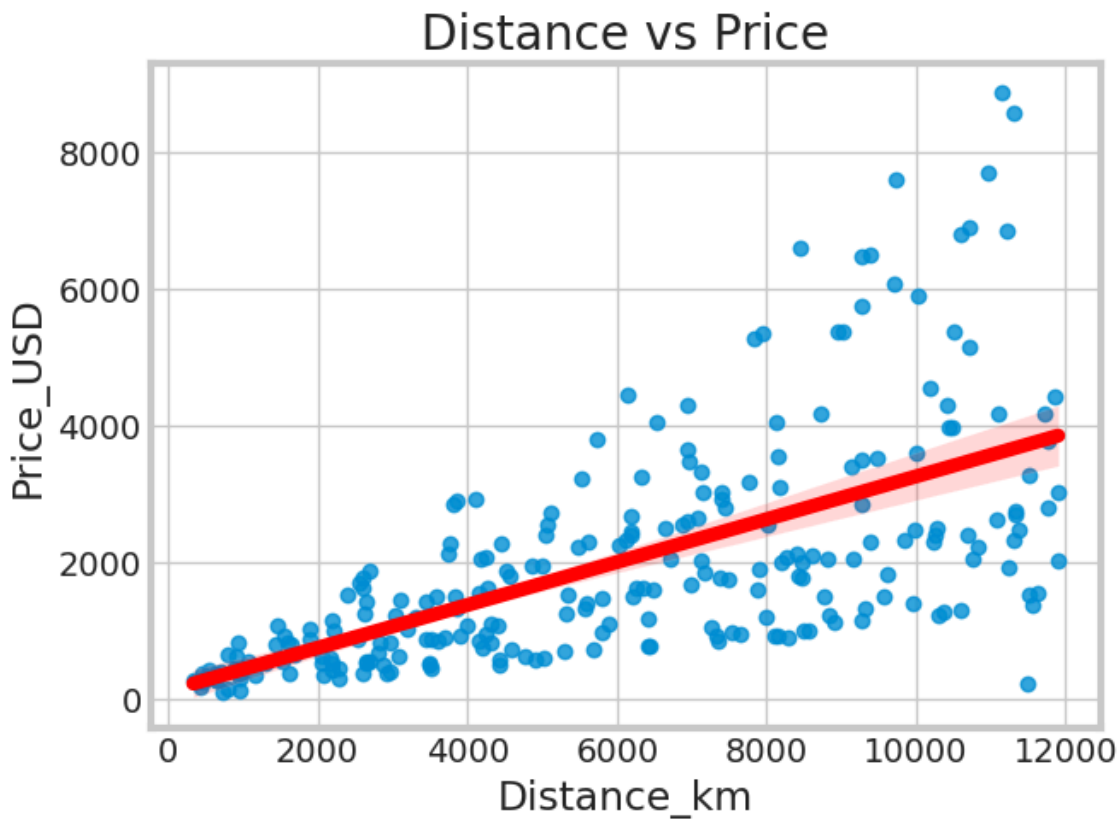
```
df.columns
```

```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',  
      'Days_Before_Departure', 'Price_USD'],  
      dtype='object')
```

```
df["Distance_km"].corr(df["Price_USD"])
```

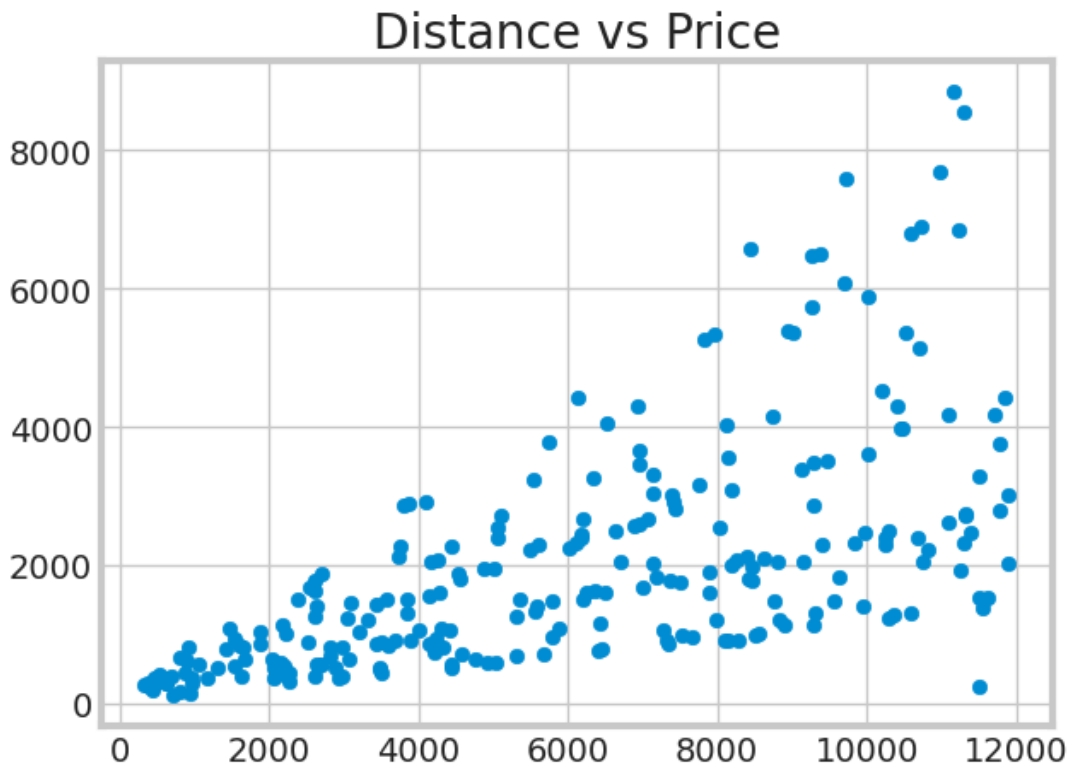
```
np.float64(0.6345413223912628)
```

```
sns.regplot(x="Distance_km", y="Price_USD", data=df,line_kws={"color": "red"})  
plt.title("Distance vs Price")  
plt.show()
```



```
plt.scatter(df['Distance_km'],df['Price_USD'])      # If points go upward → price increases with distance
plt.title("Distance vs Price")                    # If scattered randomly → weak relationship
```

```
Text(0.5, 1.0, 'Distance vs Price')
```



What is the average price by distance range (0–500, 500–1000 etc)?

```
df["Distance_Range"] = pd.cut(df["Distance_km"], bins=5)
```

```
df["Distance_Range"]
```

	Distance_Range
0	(324.433, 2649.4]
1	(7276.2, 9589.6]
2	(2649.4, 4962.8]
3	(7276.2, 9589.6]
4	(324.433, 2649.4]
...	...
245	(324.433, 2649.4]
246	(4962.8, 7276.2]
247	(7276.2, 9589.6]
248	(7276.2, 9589.6]
249	(9589.6, 11903.0]

250 rows × 1 columns

dtype: category

```
bins = [0, 500, 1000, 2000, 5000, 10000, 12000]
```

```
labels = ["0-500", "500-1000", "1000-2000",
          "2000-5000", "5000-10000", "10000+"]
```

```
df["Distance_Range"] = pd.cut(df["Distance_km"], bins=bins, labels=labels)
```

```
df11 = df.groupby("Distance_Range")["Price_USD"].mean()  
df11 = df11.reset_index().rename(columns={"Price_USD":"average price by distance range"})  
df11
```

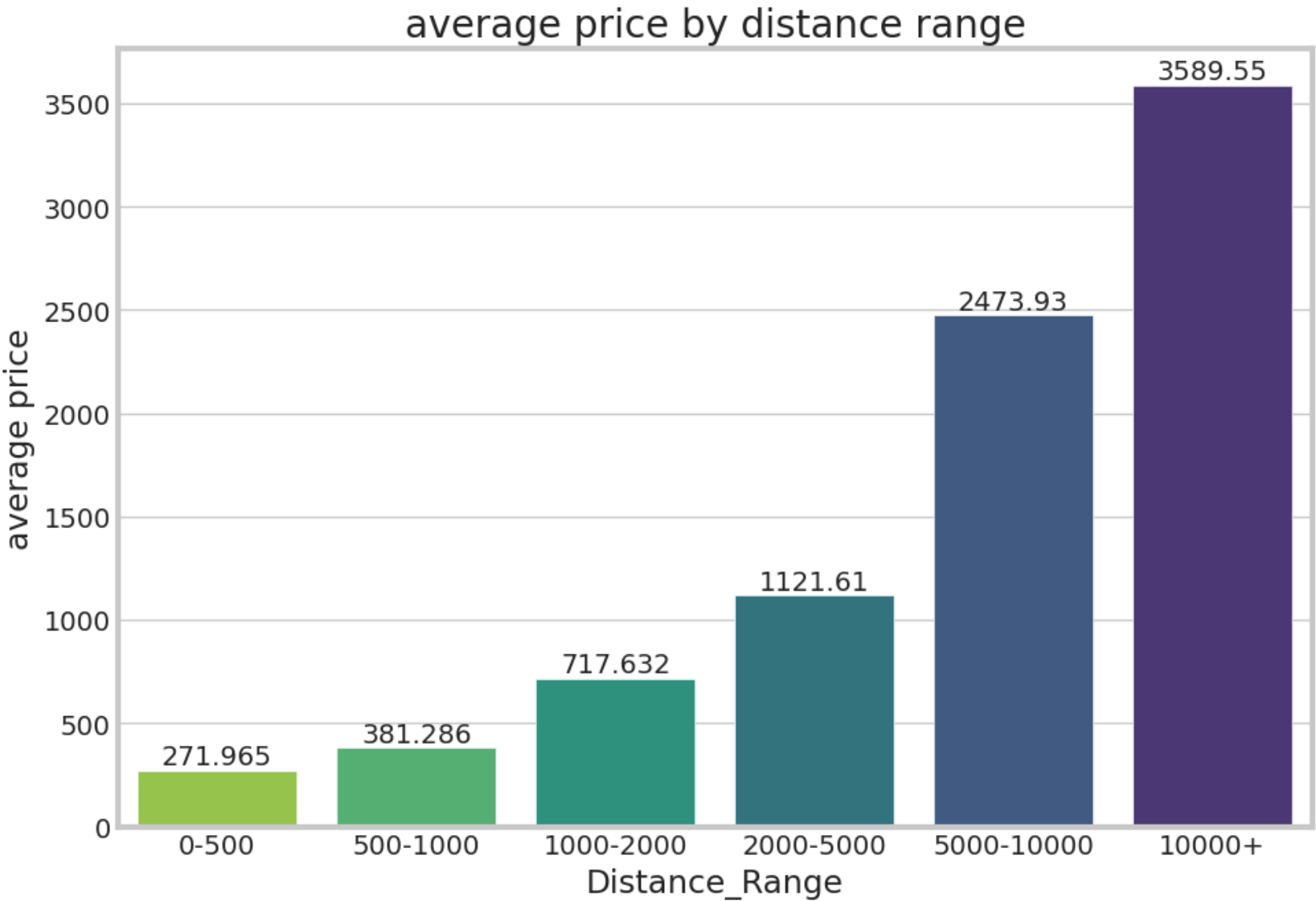
	Distance_Range	average price by distance range	
0	0-500	271.965000	
1	500-1000	381.285714	
2	1000-2000	717.632308	
3	2000-5000	1121.610429	
4	5000-10000	2473.929327	
5	10000+	3589.545366	

Next steps:

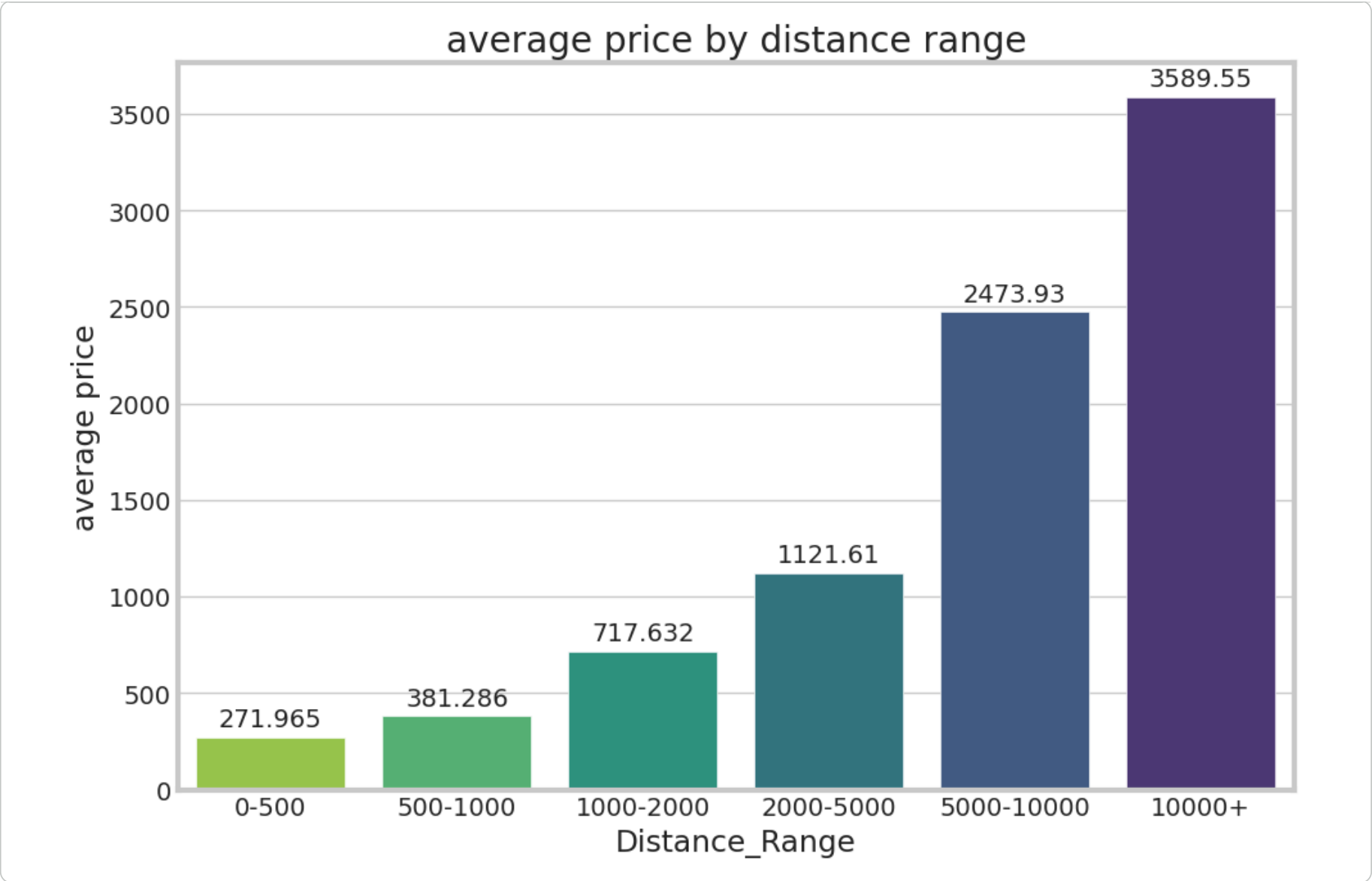
[Generate code with df11](#)

[New interactive sheet](#)

```
plt.figure(figsize=(10,7))  
ax = sns.barplot(x = "Distance_Range", y = "average price by distance range",data = df11,palette="viridis_r")  
for i in ax.containers:  
    ax.bar_label(i)  
plt.xlabel("Distance_Range")  
plt.ylabel("average price")  
plt.title("average price by distance range")  
plt.show()
```



```
plt.figure(figsize=(10,7))  
sns.barplot(x = "Distance_Range", y = "average price by distance range",data = df11,palette="viridis_r")  
for c in plt.gca().containers:  
    plt.gca().bar_label(c,padding=3)  
plt.xlabel("Distance_Range")  
plt.ylabel("average price")  
plt.title("average price by distance range")  
plt.show()
```



What is the average price for each airline and class combination?

```
df.columns

Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',
      'Days_Before_Departure', 'Price_USD', 'Distance_Range'],
      dtype='object')
```

```
df12 = df.groupby(["Airline","Class"])["Price_USD"].mean().reset_index().rename(columns={"Price_USD":"average price'})
df12
```

	Airline	Class	average price	
0	British Airways	Business	1927.000000	
1	British Airways	Economy	991.235714	
2	British Airways	First	3042.127500	
3	Emirates	Business	2175.815714	
4	Emirates	Economy	1109.684667	
5	Emirates	First	3542.779000	
6	Etihad	Business	1477.442727	
7	Etihad	Economy	1120.945714	
8	Etihad	First	1929.470000	
9	Flynas	Business	1450.385556	
10	Flynas	Economy	955.005556	
11	Flynas	First	2724.895000	
12	Lufthansa	Business	2092.140000	
13	Lufthansa	Economy	1079.788462	
14	Lufthansa	First	3321.450000	
15	Qatar Airways	Business	1546.215455	
16	Qatar Airways	Economy	893.142000	
17	Qatar Airways	First	3123.800000	
18	Saudia	Business	3019.644286	
19	Saudia	Economy	920.746667	
20	Saudia	First	3405.711538	
21	Turkish Airlines	Business	2280.015556	
22	Turkish Airlines	Economy	1240.905556	
23	Turkish Airlines	First	2710.416667	

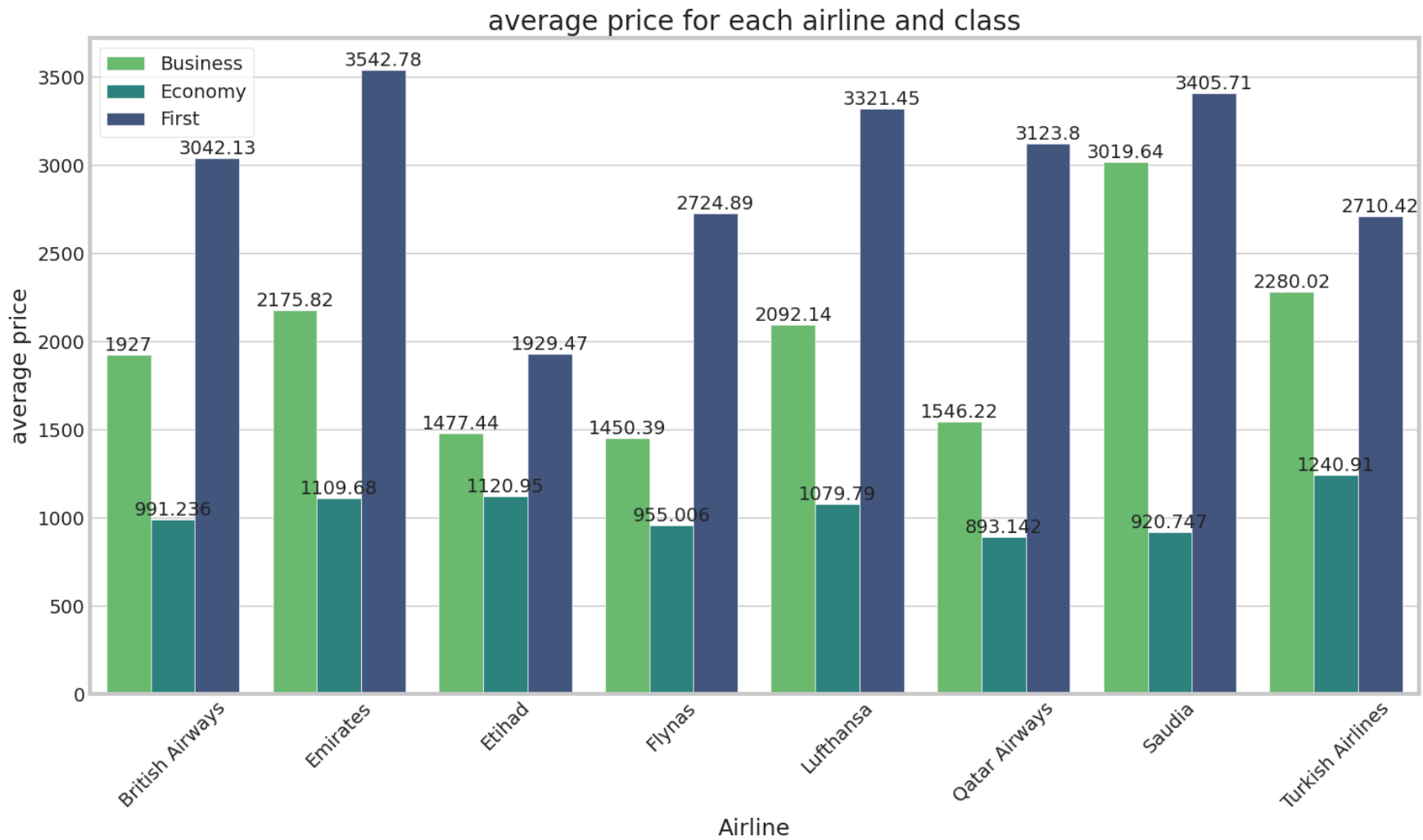
Next steps:

Generate code with df12

New interactive sheet

```
plt.figure(figsize=(15,9))
sns.barplot(x = "Airline", y = "average price",data = df12,palette="viridis_r",hue = "Class")
for c in plt.gca().containers:
    plt.gca().bar_label(c)

plt.xlabel("Airline")
plt.ylabel("average price")
plt.xticks(rotation=45,fontsize=14)
plt.title("average price for each airline and class")
plt.legend(loc=2)
plt.tight_layout()
plt.show()
```



Which route (Origin → Destination) has the highest average price?

```
df['Route'] = df["Origin"] + "→" +df["Destination"]
df.head()
```

	Ticket_ID	Airline	Origin	Destination	Distance_km	Class	Days_Before_Departure	Price_USD	Distance_Range	Route
0	1	Saudia	Mumbai	Paris	2643	Economy	28	555.30	2000-5000	Mumbai→Paris
1	2	Etihad	Istanbul	Jeddah	8266	Business	98	2070.27	5000-10000	Istanbul→Jeddah
2	3	British Airways	Dubai	Riyadh	2698	First	21	1883.11	2000-5000	Dubai→Riyadh
3	4	Emirates	London	Istanbul	7956	First	24	5349.65	5000-10000	London→Istanbul

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
top_route = df.groupby("Route")["Price_USD"].mean().sort_values(ascending=False).reset_index().rename(columns={"Pri
```

Which route has the longest distance?

```
top_route.head(1)
```

	Route	highest average price	
0	London→Mumbai	5576.315	

Next steps: [Generate code with top_route](#) [New interactive sheet](#)

```
top_route1 = df.groupby("Route")["Distance_km"].max().sort_values(ascending=False).reset_index()
top_route1.head(1)
```

Route

Distance_km

0

Istanbul→Mumbai

11903

Next steps: [Generate code with top_route1](#) [New interactive sheet](#)

Which airline operates the longest flights?

df.head()

	Ticket_ID	Airline	Origin	Destination	Distance_km	Class	Days_Before_Departure	Price_USD	Distance_Range	Route
0	1	Saudia	Mumbai	Paris	2643	Economy	28	555.30	2000-5000	Mumbai→Paris
1	2	Etihad	Istanbul	Jeddah	8266	Business	98	2070.27	5000-10000	Istanbul→Jeddah
2	3	British Airways	Dubai	Riyadh	2698	First	21	1883.11	2000-5000	Dubai→Riyadh
3	4	Emirates	London	Istanbul	7956	First	24	5349.65	5000-10000	London→Istanbul

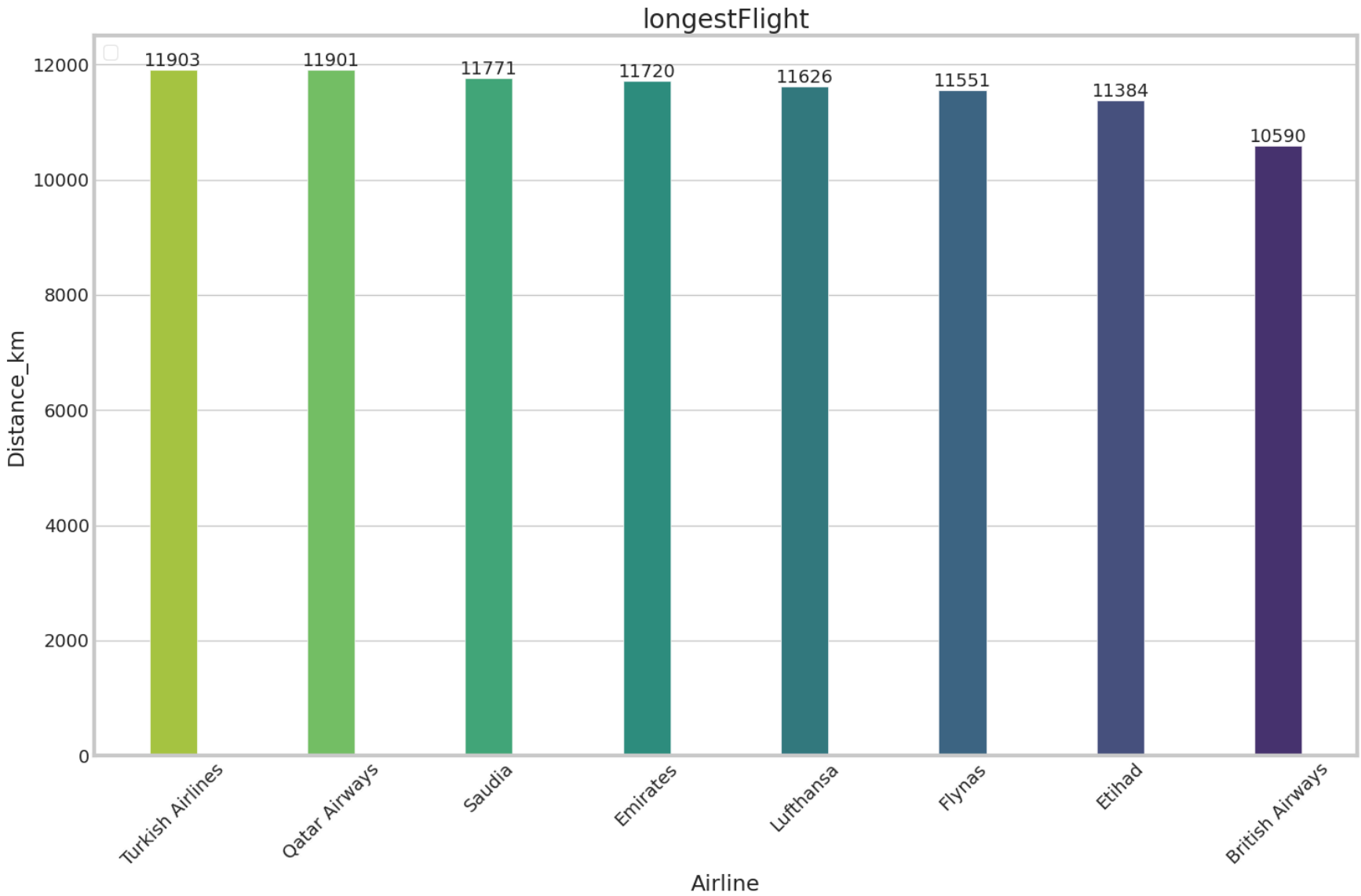
Next steps: [Generate code with df](#) [New interactive sheet](#)

longestFlight = df.groupby("Airline")["Distance_km"].max().sort_values(ascending=False).reset_index()
longestFlight

	Airline	Distance_km
0	Turkish Airlines	11903
1	Qatar Airways	11901
2	Saudia	11771
3	Emirates	11720
4	Lufthansa	11626
5	Flynas	11551
6	Etihad	11384
7	British Airways	10590

Next steps: [Generate code with longestFlight](#) [New interactive sheet](#)

```
plt.figure(figsize=(15,10))
sns.barplot(x = "Airline", y = "Distance_km",data = longestFlight,palette="viridis_r",width=.3)
for c in plt.gca().containers:
    plt.gca().bar_label(c)
plt.xlabel("Airline")
plt.ylabel("Distance_km")
plt.xticks(rotation=45,fontsize=14)
plt.title("longestFlight")
plt.legend(loc=2)
plt.tight_layout()
plt.show()
```

```
longestFlight1 = df.groupby(["Airline","Route"])["Distance_km"].max().sort_values(ascending=False).reset_index()
longestFlight1.head(10)
```

	Airline	Route	Distance_km
0	Turkish Airlines	Istanbul→Mumbai	11903
1	Qatar Airways	Istanbul→Jeddah	11901
2	Qatar Airways	Dubai→Jeddah	11848
3	Turkish Airlines	Doha→Jeddah	11773
4	Saudia	Cairo→Riyadh	11771
5	Emirates	Paris→Riyadh	11720
6	Lufthansa	Riyadh→Dubai	11626
7	Flynas	Riyadh→New York	11551
8	Qatar Airways	Paris→London	11511
9	Emirates	Paris→New York	11507

Next steps: [Generate code with longestFlight1](#) [New interactive sheet](#)

What is the average price when booking early vs late?

```
bins = [0, 10, 30, df["Days_Before_Departure"].max()]
labels = ["Late", "Medium", "Early"]
df["Booking_type"] = pd.cut(df["Days_Before_Departure"],bins=bins,labels=labels,include_lowest=True)
```

df.head()

	Ticket_ID	Airline	Origin	Destination	Distance_km	Class	Days_Before_Departure	Price_USD	Distance_Range	Route
0	1	Saudia	Mumbai	Paris	2643	Economy	28	555.30	2000-5000	Mumbai→Paris
1	2	Etihad	Istanbul	Jeddah	8266	Business	98	2070.27	5000-10000	Istanbul→Jeddah
2	3	British Airways	Dubai	Riyadh	2698	First	21	1883.11	2000-5000	Dubai→Riyadh
3	4	Emirates	London	Istanbul	7956	First	24	5349.65	5000-10000	London→Istanbul

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df13 = df.groupby("Booking_type")["Price_USD"].mean().reset_index().round(2).rename(columns={"Price_USD":"average price"}, inplace=True)
df13
```

	Booking_type	average price
0	Late	2945.86
1	Medium	2618.99
2	Early	1762.27

Next steps:

[Generate code with df13](#)

[New interactive sheet](#)

```
plt.figure(figsize=(15,10))
sns.barplot(x = "Booking_type", y = "average price",data = df13,palette="viridis_r")
for c in plt.gca().containers:
    plt.gca().bar_label(c)
plt.xlabel("Booking_type")
plt.ylabel("average price")
plt.xticks(rotation=45,fontsize=14)
plt.title("Average Price by Booking Type")
plt.legend(loc=2)
plt.tight_layout()
plt.show()
```



What is the average price per km for each airline?

```
df.head()
```

	Ticket_ID	Airline	Origin	Destination	Distance_km	Class	Days_Before_Departure	Price_USD	Distance_Range	Route
0	1	Saudia	Mumbai	Paris	2643	Economy	28	555.30	2000-5000	Mumbai→Paris
1	2	Etihad	Istanbul	Jeddah	8266	Business	98	2070.27	5000-10000	Istanbul→Jeddah
2	3	British Airways	Dubai	Riyadh	2698	First	21	1883.11	2000-5000	Dubai→Riyadh
3	4	Emirates	London	Istanbul	7956	First	24	5349.65	5000-10000	London→Istanbul

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

```
df["price_per_km"] = df["Distance_km"]/df["Price_USD"]
```

```
df.head()
```


	Ticket_ID	Airline	Origin	Destination	Distance_km	Class	Days_Before_Departure	Price_USD	Distance_Range	Route
0	1	Saudia	Mumbai	Paris	2643	Economy	28	555.30	2000-5000	Mumbai→Paris
1	2	Etihad	Istanbul	Jeddah	8266	Business	98	2070.27	5000-10000	Istanbul→Jeddah
2	3	British Airways	Dubai	Riyadh	2698	First	21	1883.11	2000-5000	Dubai→Riyadh
3	4	Emirates	London	Istanbul	7956	First	24	5349.65	5000-10000	London→Istanbul
4	5	Qatar Airways	Doha	Dubai	1308	Business	60	516.19	1000-2000	Doha→Dubai

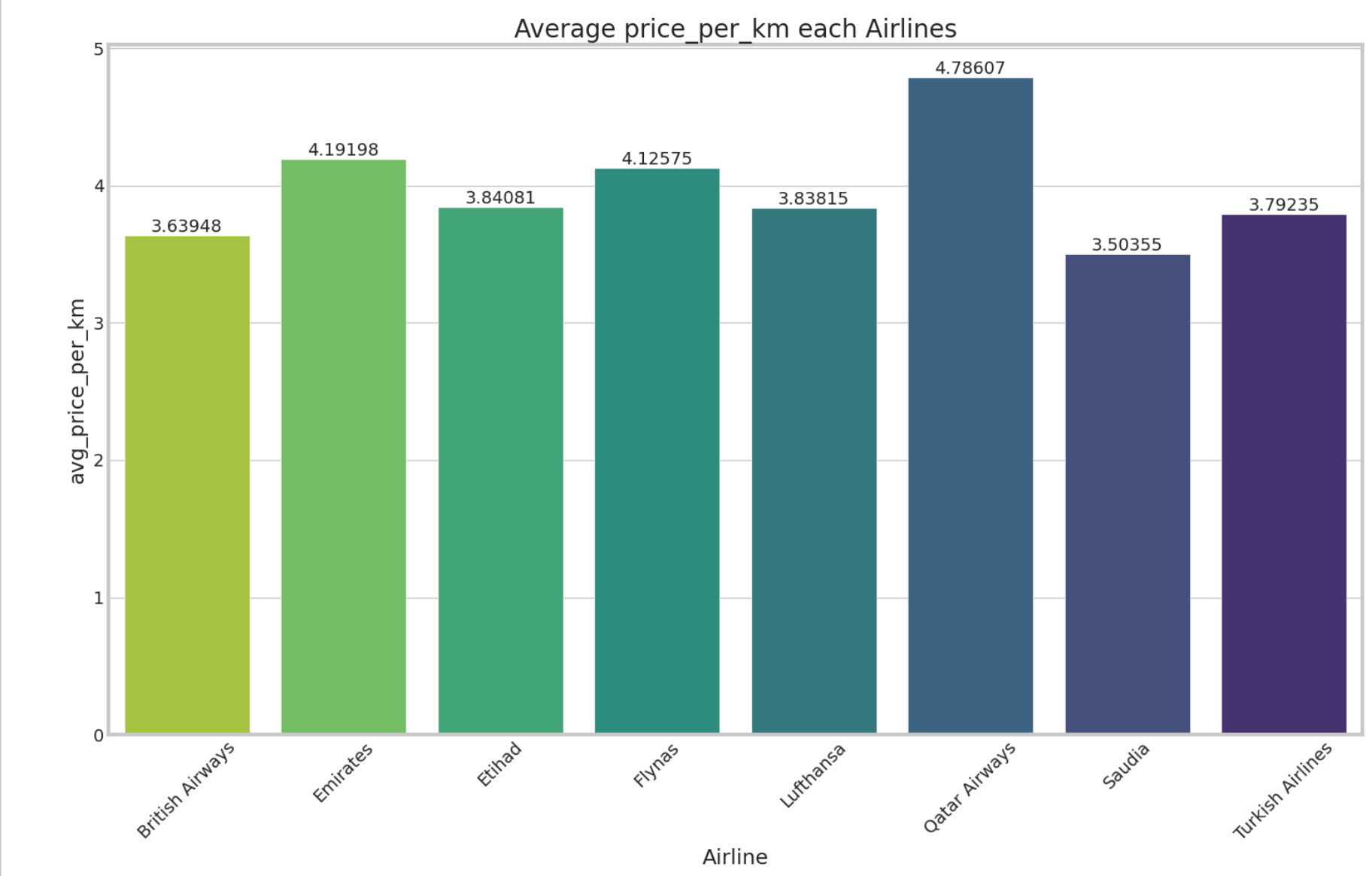
Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df14 = df.groupby("Airline")["price_per_km"].mean().reset_index()
df14 = df14.rename(columns={"price_per_km": "avg_price_per_km"})
df14
```

	Airline	avg_price_per_km	
0	British Airways	3.639479	
1	Emirates	4.191980	
2	Etihad	3.840805	
3	Flynas	4.125753	
4	Lufthansa	3.838152	
5	Qatar Airways	4.786068	
6	Saudia	3.503547	
7	Turkish Airlines	3.792349	

Next steps: [Generate code with df14](#) [New interactive sheet](#)

```
plt.figure(figsize=(15,10))
sns.barplot(x = "Airline", y = "avg_price_per_km",data=df14,palette="viridis_r")
for c in plt.gca().containers:
    plt.gca().bar_label(c)
plt.xlabel("Airline")
plt.ylabel("avg_price_per_km")
plt.xticks(rotation=45,fontsize=14)
plt.title("Average price_per_km each Airlines")
plt.tight_layout()
plt.show()
```



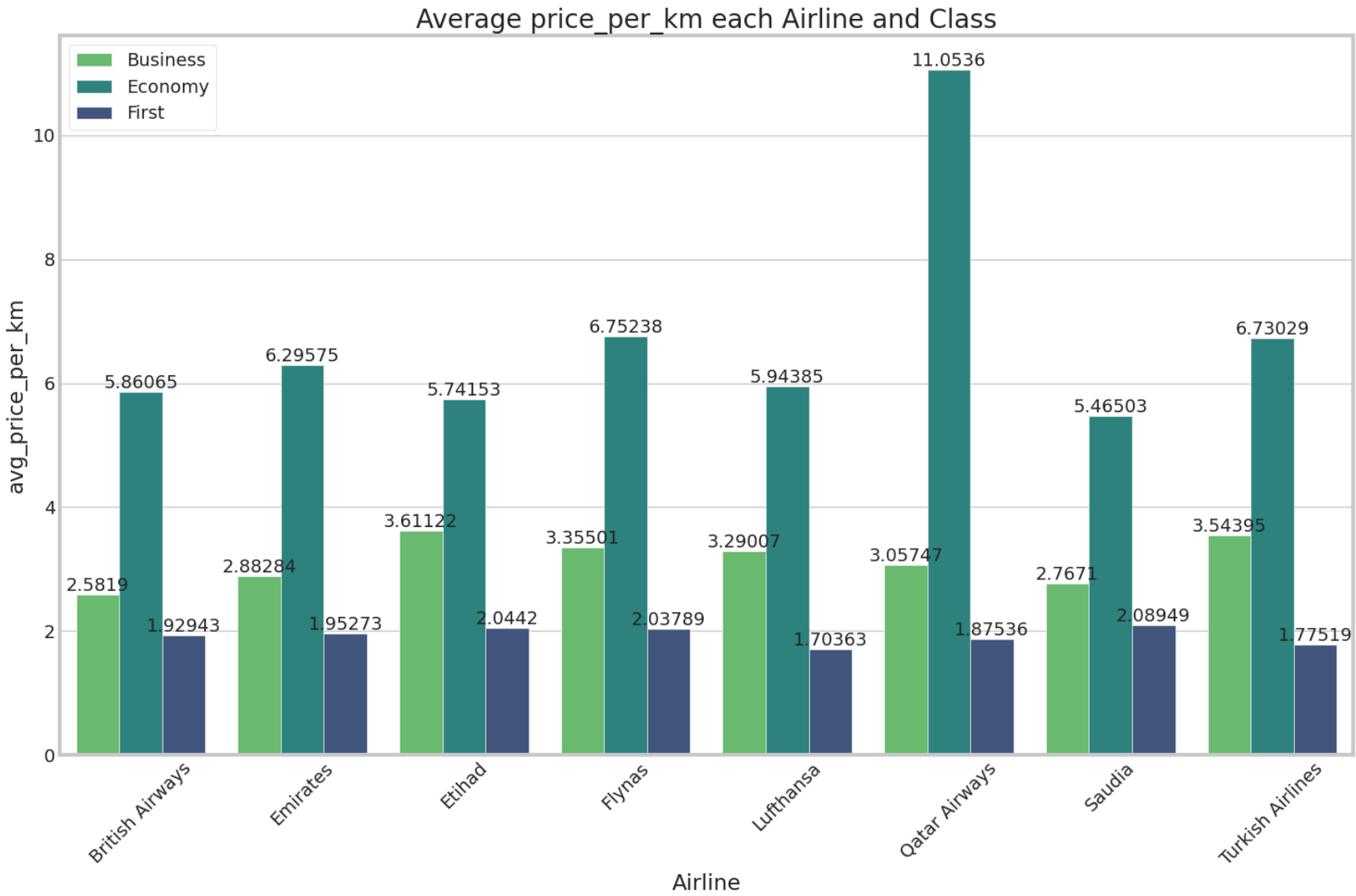
What is the average price per km for each airline and class?

```
df15 = df.groupby(["Airline", "Class"])["price_per_km"].mean().reset_index()
df15 = df15.rename(columns={"price_per_km": "avg_price_per_km"})
df15
```

	Airline	Class	avg_price_per_km	
0	British Airways	Business	2.581904	
1	British Airways	Economy	5.860650	
2	British Airways	First	1.929426	
3	Emirates	Business	2.882843	
4	Emirates	Economy	6.295746	
5	Emirates	First	1.952725	
6	Etihad	Business	3.611216	
7	Etihad	Economy	5.741535	
8	Etihad	First	2.044200	
9	Flynas	Business	3.355007	
10	Flynas	Economy	6.752378	
11	Flynas	First	2.037889	
12	Lufthansa	Business	3.290073	
13	Lufthansa	Economy	5.943851	
14	Lufthansa	First	1.703629	
15	Qatar Airways	Business	3.057475	
16	Qatar Airways	Economy	11.053584	
17	Qatar Airways	First	1.875358	
18	Saudia	Business	2.767104	
19	Saudia	Economy	5.465031	
20	Saudia	First	2.089493	
21	Turkish Airlines	Business	3.543950	
22	Turkish Airlines	Economy	6.730293	
23	Turkish Airlines	First	1.775190	

Next steps: [Generate code with df15](#) [New interactive sheet](#)

```
plt.figure(figsize=(15,10))
sns.barplot(x = "Airline", y = "avg_price_per_km",data = df15,palette="viridis_r",hue="Class")
for c in plt.gca().containers:
    plt.gca().bar_label(c)
plt.xlabel("Airline")
plt.ylabel("avg_price_per_km")
plt.xticks(rotation=45,fontsize=14)
plt.title("Average price_per_km each Airline and Class")
plt.legend(loc=2)
plt.tight_layout()
plt.show()
```



Which class has the highest average distance?

df.columns

```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',
      'Days_Before_Departure', 'Price_USD', 'Distance_Range', 'Route',
      'Booking_type', 'price_per_km'],
      dtype='object')
```

```
df16 = df.groupby("Class")["Distance_km"].mean().reset_index().rename(columns={"Distance_km":"avg_Distance_km"})
df16
```

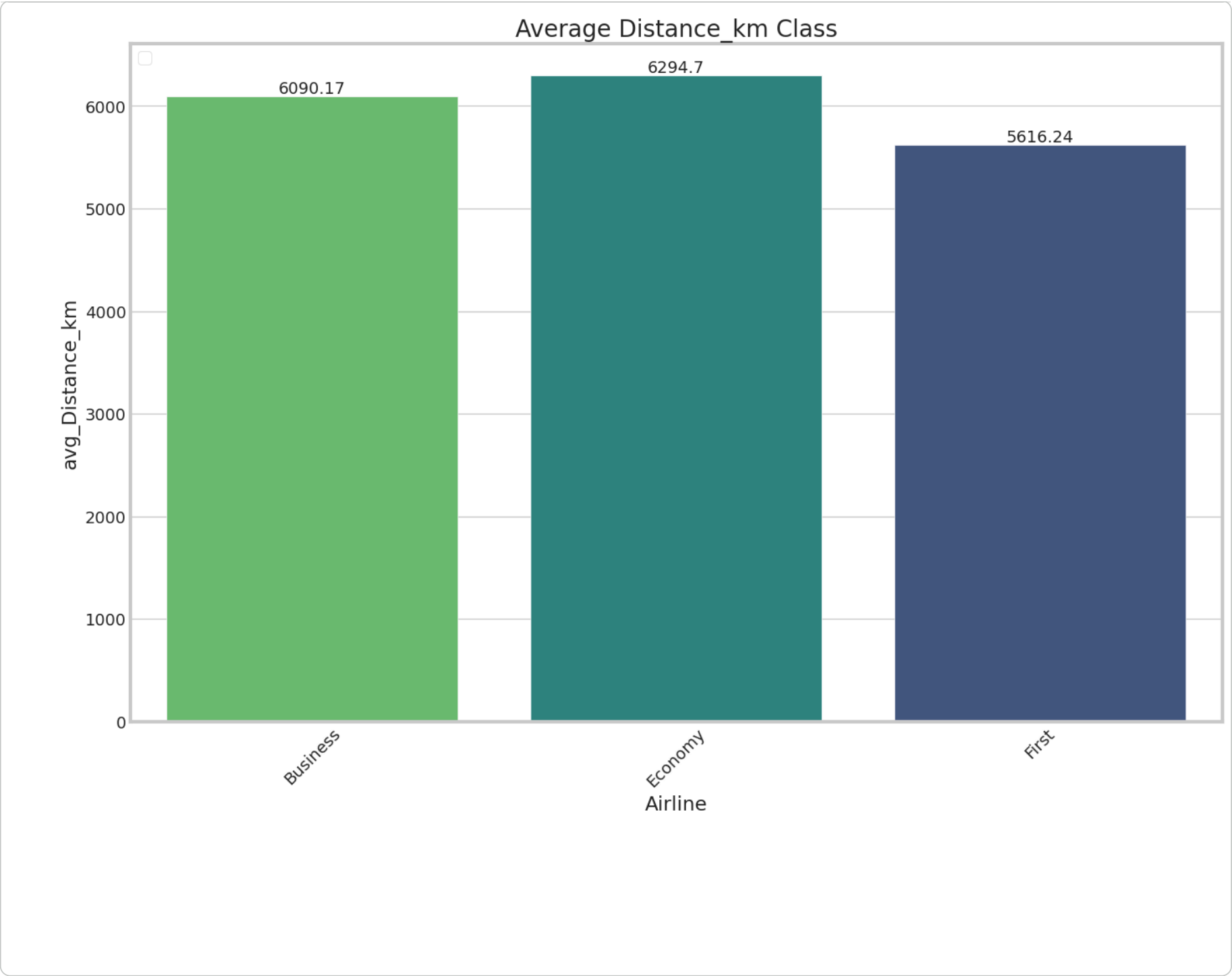
	Class	avg_Distance_km	
0	Business	6090.173333	
1	Economy	6294.696629	
2	First	5616.244186	

Next steps:

[Generate code with df16](#)

[New interactive sheet](#)

```
plt.figure(figsize=(15,10))
sns.barplot(x = "Class", y = "avg_Distance_km",data = df16,palette="viridis_r")
for c in plt.gca().containers:
    plt.gca().bar_label(c)
plt.xlabel("Airline")
plt.ylabel("avg_Distance_km")
plt.xticks(rotation=45,fontsize=14)
plt.title("Average Distance_km Class")
plt.legend(loc=2)
plt.show()
```



Which routes are most frequently traveled?

```
df["Route"].value_counts().reset_index().head(5)
```

	Route	count
0	Mumbai→Paris	7
1	Doha→Cairo	6
2	Jeddah→Cairo	6
3	Mumbai→Istanbul	6
4	Paris→Cairo	6

How does Days_Before_Departure affect price?

```
df.columns
```

```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',  
      'Days_Before_Departure', 'Price_USD', 'Distance_Range', 'Route',  
      'Booking_type', 'price_per_km'],  
      dtype='object')
```

```
df17 = df.groupby("Days_Before_Departure")["Price_USD"].mean().sort_values(ascending=False).reset_index().rename(columns={'Days_Before_Departure': 'Day', 'Price_USD': 'Price'})
df17
```

	Days_Before_Departure	Avg_price
0	1	8852.690000
1	9	5727.360000
2	20	5502.835000
3	56	4561.350000
4	24	4548.723333
...	...	...
101	26	556.440000
102	28	555.300000
103	107	420.225000
104	27	355.780000
105	103	306.500000

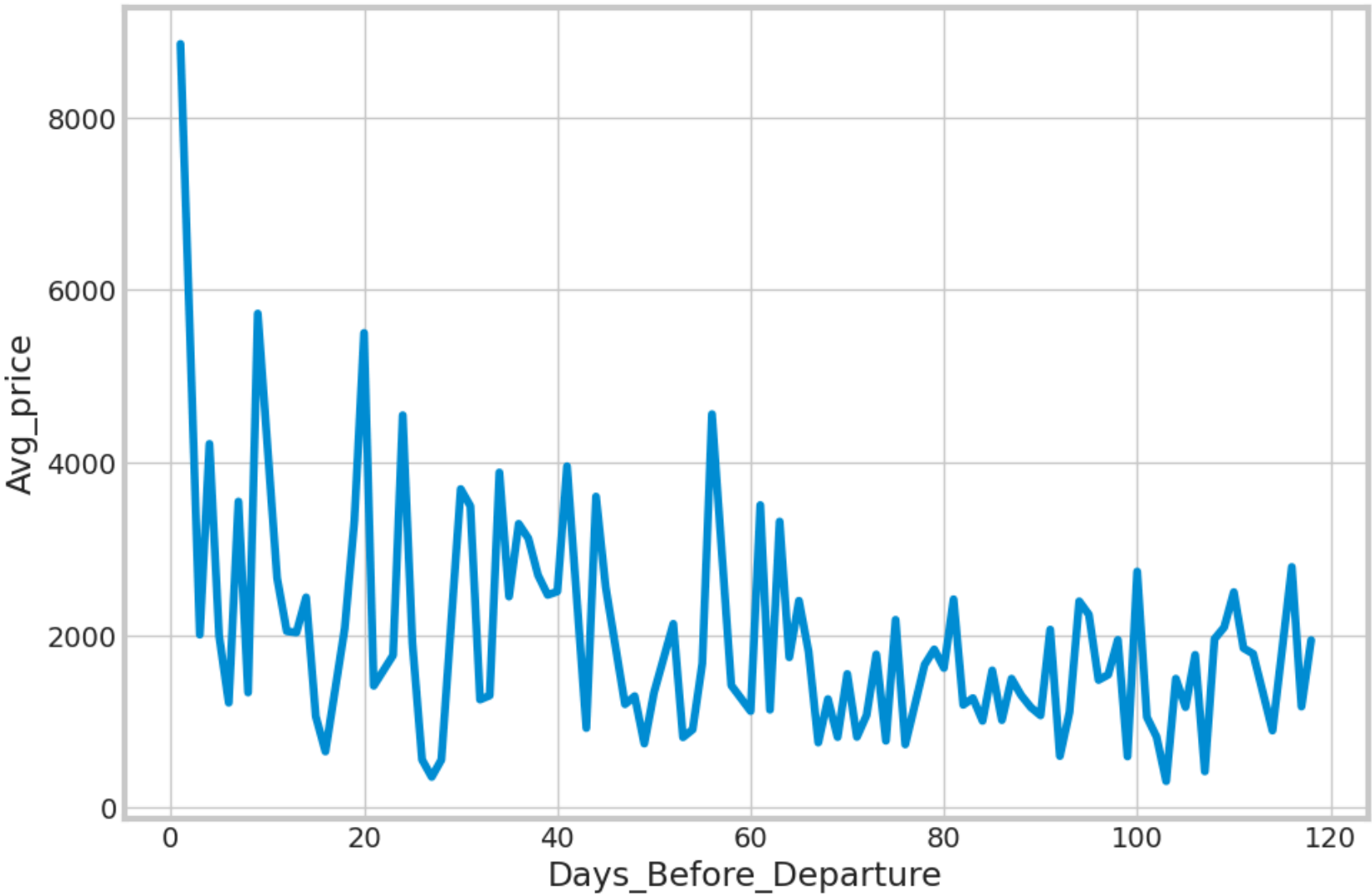
106 rows × 2 columns

Next steps:

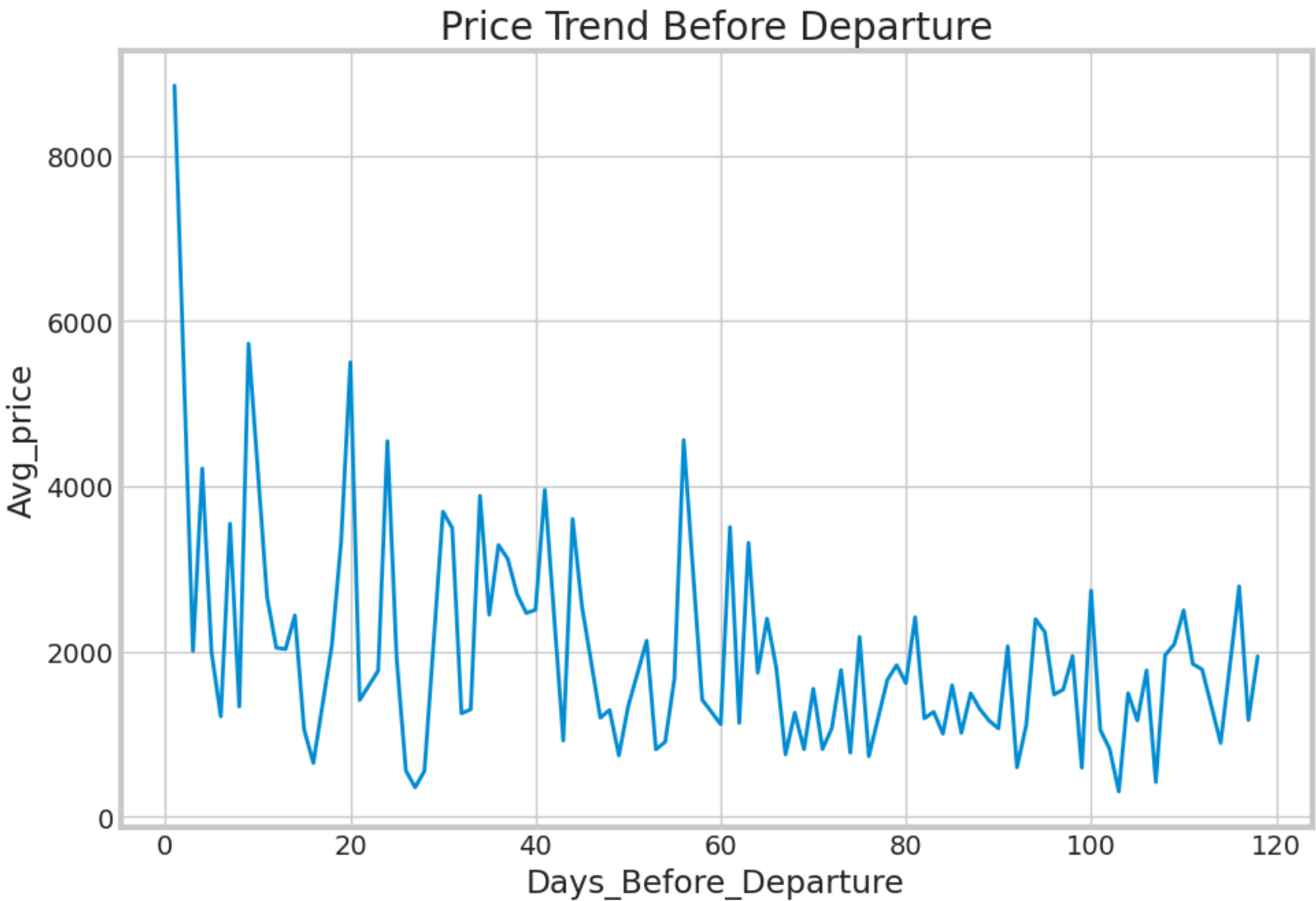
[Generate code with df17](#)

[New interactive sheet](#)

```
plt.figure(figsize=(10,7))
sns.lineplot(x="Days_Before_Departure", y="Avg_price", data=df17)
plt.show()
```



```
plt.figure(figsize=(10,7))
sns.lineplot(x="Days_Before_Departure", y="Avg_price", data=df17,palette="viridis",linewidth=2)
plt.title("Price Trend Before Departure")
plt.show()
```



Do early bookings reduce ticket price?

Yes, early booking generally reduces ticket price.

Which airline has the most expensive business class tickets?

df.columns

```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',
      'Days_Before_Departure', 'Price_USD', 'Distance_Range', 'Route',
      'Booking_type', 'price_per_km'],
      dtype='object')
```

```
df18 = df[df["Class"]=="Business"].groupby("Airline")["Price_USD"].max().reset_index().rename(columns={"Price_USD":'
df18
```

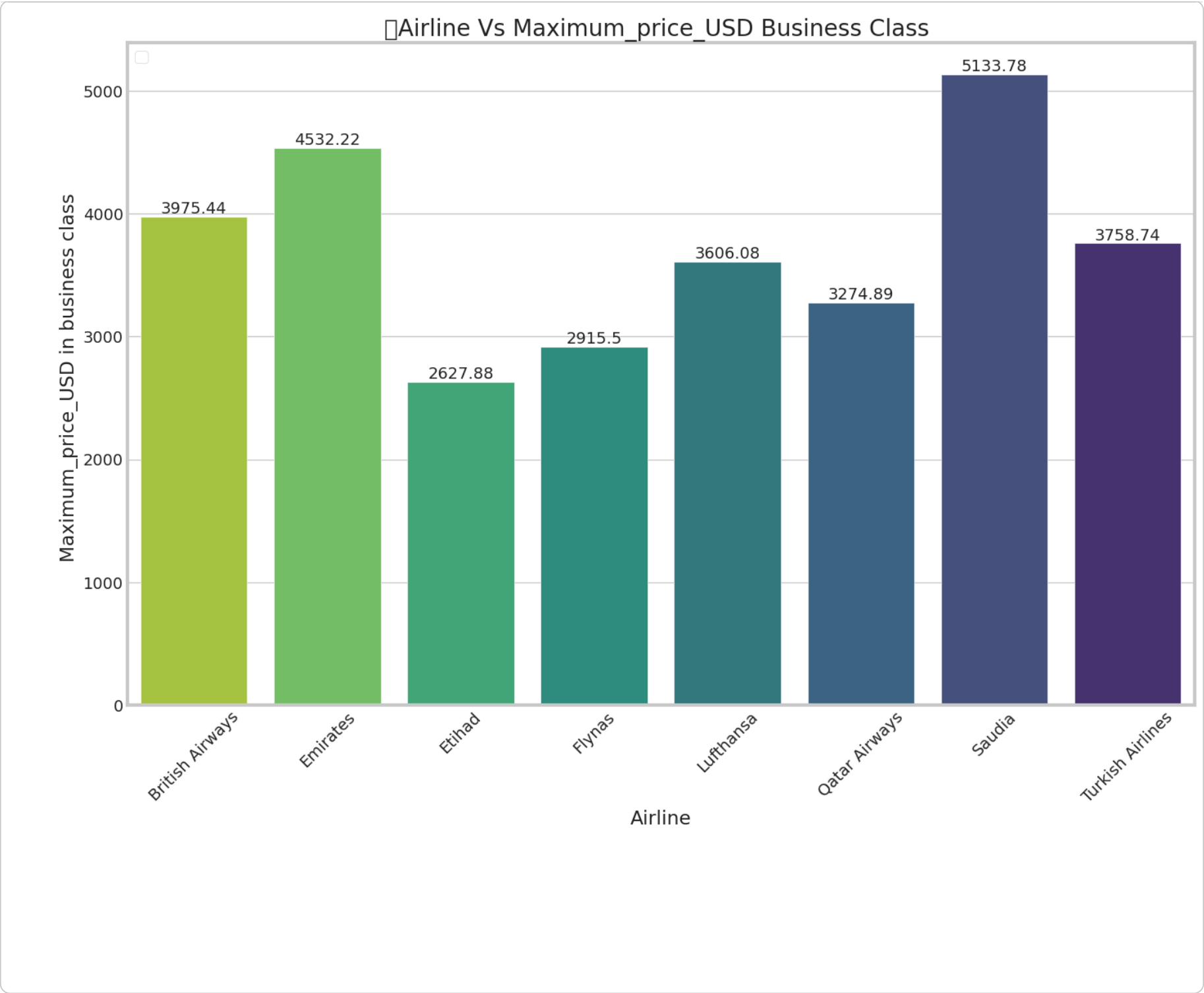
	Airline	Maximum_price_USD in Business Class	
0	British Airways	3975.44	
1	Emirates	4532.22	
2	Etihad	2627.88	
3	Flynas	2915.50	
4	Lufthansa	3606.08	
5	Qatar Airways	3274.89	
6	Saudia	5133.78	
7	Turkish Airlines	3758.74	

Next steps:

[Generate code with df18](#)

[New interactive sheet](#)

```
plt.figure(figsize=(15,10))
sns.barplot(x = "Airline", y = "Maximum_price_USD in Business Class",data = df18,palette="viridis_r")
for c in plt.gca().containers:
    plt.gca().bar_label(c)
plt.xlabel("Airline")
plt.ylabel("Maximum_price_USD in business class ")
plt.xticks(rotation=45,fontsize=14)
plt.title(" Airline Vs Maximum_price_USD Business Class ")
plt.legend(loc=2)
plt.show()
```

Which airline offers the cheapest economy tickets?

```
df.columns

Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',
      'Days_Before_Departure', 'Price_USD', 'Distance_Range', 'Route',
      'Booking_type', 'price_per_km'],
      dtype='object')
```

```
economy = df[df["Class"]=="Economy"]
economy.head()
```

	Ticket_ID	Airline	Origin	Destination	Distance_km	Class	Days_Before_Departure	Price_USD	Distance_Range	Route
0	1	Saudia	Mumbai	Paris	2643	Economy	28	555.30	2000-5000	Mumbai→Paris
6	7	Emirates	Doha	Dubai	3506	Economy	85	439.16	2000-5000	Doha→Dubai
13	14	Qatar Airways	Doha	Cairo	9572	Economy	48	1485.53	5000-10000	Doha→Cairo
17	18	Lufthansa	New York	Riyadh	11244	Economy	37	1929.27	10000+	New York→Riyadh
23	24	Emirates	Istanbul	Jeddah	440	Economy	86	185.70	0-500	Istanbul→Jeddah

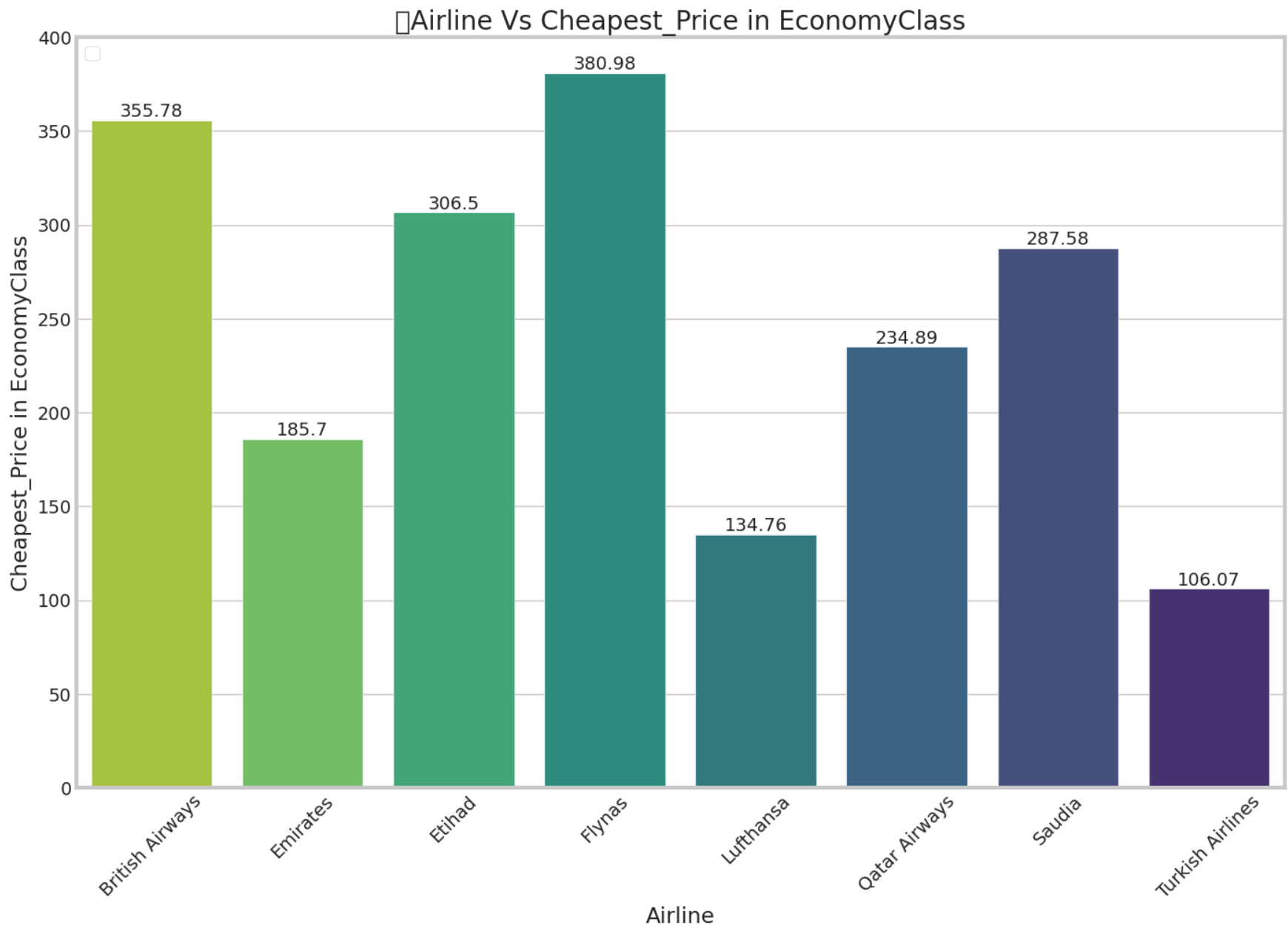
Next steps: [Generate code with economy](#) [New interactive sheet](#)

```
df19 = economy.groupby("Airline")["Price_USD"].min().reset_index().rename(columns={"Price_USD":"Cheapest_Price in Economy Class"})
df19
```

	Airline	Cheapest_Price in EconomyClass	
0	British Airways	355.78	
1	Emirates	185.70	
2	Etihad	306.50	
3	Flynas	380.98	
4	Lufthansa	134.76	
5	Qatar Airways	234.89	
6	Saudia	287.58	
7	Turkish Airlines	106.07	

Next steps: [Generate code with df19](#) [New interactive sheet](#)

```
plt.figure(figsize=(15,10))
sns.barplot(x = "Airline", y = "Cheapest_Price in EconomyClass",data = df19,palette="viridis_r")
for c in plt.gca().containers:
    plt.gca().bar_label(c)
plt.xlabel("Airline")
plt.ylabel("Cheapest_Price in EconomyClass")
plt.xticks(rotation=45,fontsize=14)
plt.title(" Airline Vs Cheapest_Price in EconomyClass")
plt.legend(loc=2)
plt.show()
```



What is the average distance traveled?

```
df.columns
```

```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',
      'Days_Before_Departure', 'Price_USD', 'Distance_Range', 'Route',
      'Booking_type', 'price_per_km'],
      dtype='object')
```

```
df["Distance_km"].mean()
```

```
np.float64(5999.952)
```

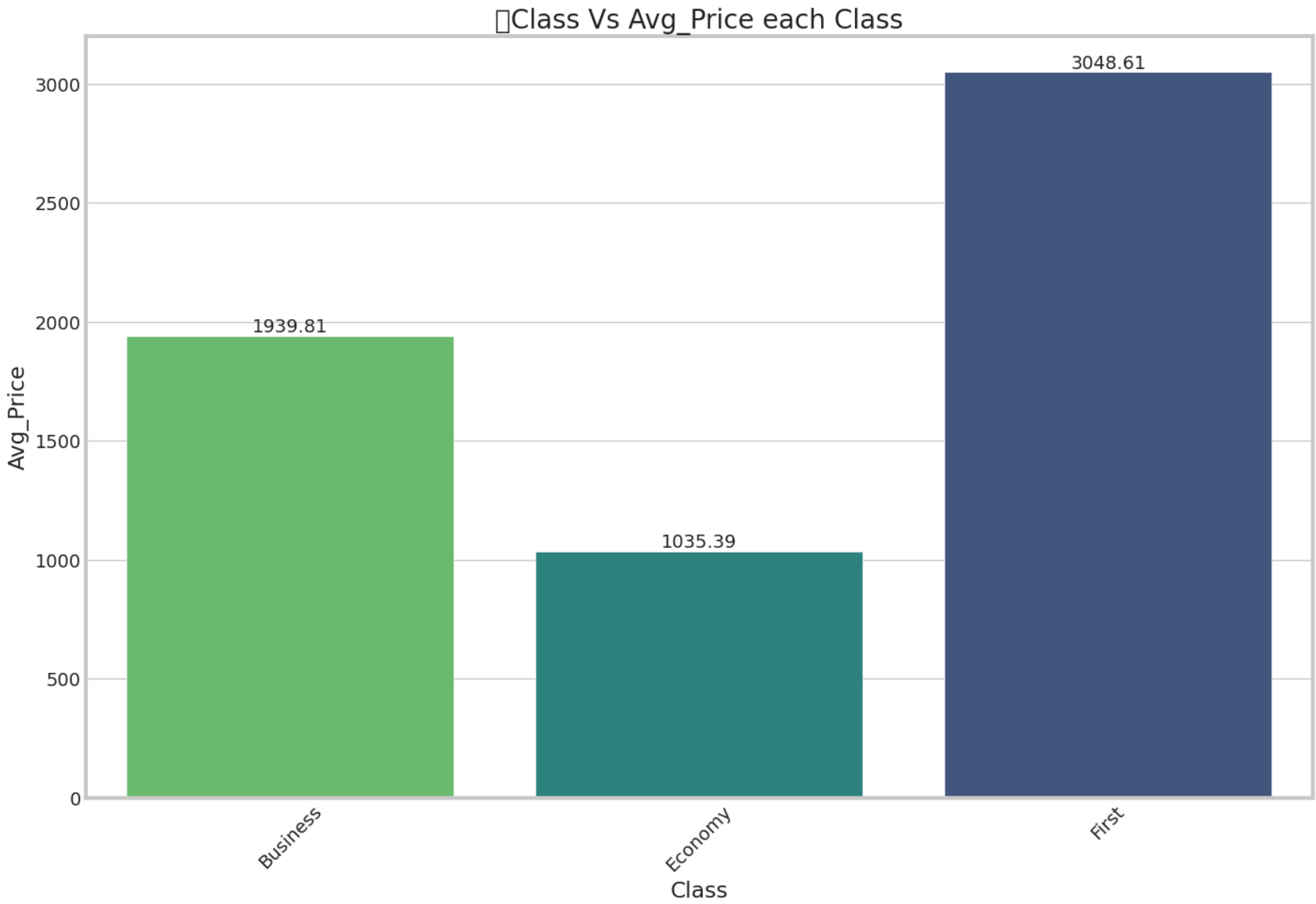
What is the price distribution for each class?


```
df20 = df.groupby("Class")["Price_USD"].mean().round(2).reset_index().rename(columns={"Price_USD": "Avg_Price"})
df20
```

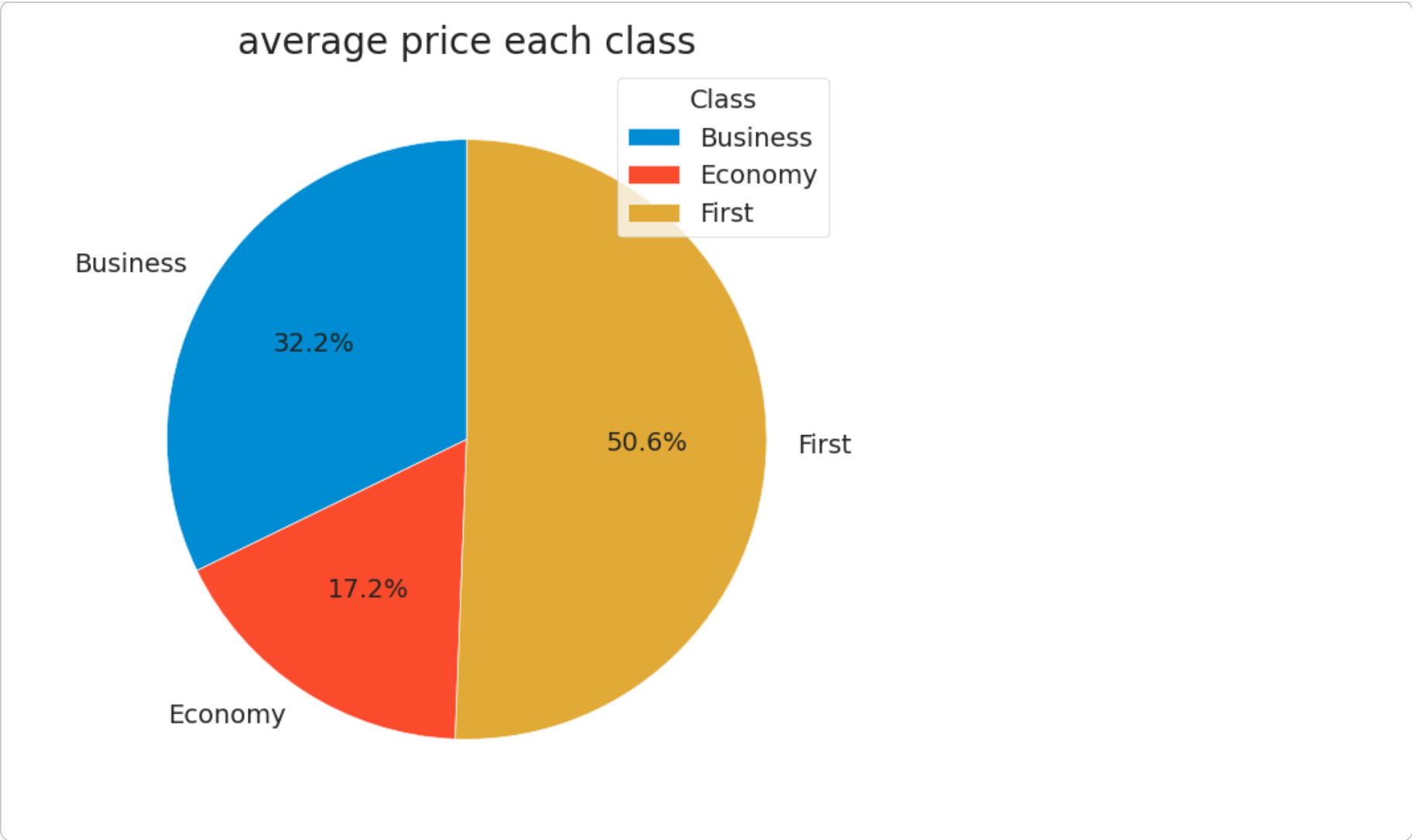
	Class	Avg_Price	
0	Business	1939.81	
1	Economy	1035.39	
2	First	3048.61	

Next steps: [Generate code with df20](#) [New interactive sheet](#)

```
plt.figure(figsize=(15,10))
sns.barplot(x = "Class", y = "Avg_Price",data = df20,palette="viridis_r")
for c in plt.gca().containers:
    plt.gca().bar_label(c)
plt.xlabel("Class")
plt.ylabel("Avg_Price")
plt.xticks(rotation=45,fontsize=14)
plt.title(" Class Vs Avg_Price each Class")
plt.show()
```



```
plt.figure(figsize=(7,7))
plt.pie(df20['Avg_Price'],labels=df20['Class'],autopct='%1.1f%%',startangle=90)
plt.title("average price each class")
plt.legend(df20['Class'],title="Class",loc=1)
plt.show()
```



Which Origin-Destination route generates the highest revenue?

df.head()

	Ticket_ID	Airline	Origin	Destination	Distance_km	Class	Days_Before_Departure	Price_USD	Distance_Range	Route
0	1	Saudia	Mumbai	Paris	2643	Economy	28	555.30	2000-5000	Mumbai→Paris
1	2	Etihad	Istanbul	Jeddah	8266	Business	98	2070.27	5000-10000	Istanbul→Jeddah
2	3	British Airways	Dubai	Riyadh	2698	First	21	1883.11	2000-5000	Dubai→Riyadh
3	4	Emirates	London	Istanbul	7956	First	24	5349.65	5000-10000	London→Istanbul
4	5	Qatar Airways	Doha	Dubai	1308	Business	60	516.19	1000-2000	Doha→Dubai

Next steps: [Generate code with df](#) [New interactive sheet](#)

df.groupby("Route")["Price_USD"].sum().sort_values(ascending=False).reset_index().rename(columns={"Price_USD":"Revenue"})

	Route	Revenue
0	Mumbai→Istanbul	17428.30
1	Doha→Cairo	14726.92
2	Istanbul→Mumbai	14566.42
3	Mumbai→Paris	14239.97
4	London→Dubai	11538.84

What is the average price per km by class?

df.head()

	Ticket_ID	Airline	Origin	Destination	Distance_km	Class	Days_Before_Departure	Price_USD	Distance_Range	Route
0	1	Saudia	Mumbai	Paris	2643	Economy	28	555.30	2000-5000	Mumbai→Paris
1	2	Etihad	Istanbul	Jeddah	8266	Business	98	2070.27	5000-10000	Istanbul→Jeddah
2	3	British Airways	Dubai	Riyadh	2698	First	21	1883.11	2000-5000	Dubai→Riyadh
3	4	Emirates	London	Istanbul	7956	First	24	5349.65	5000-10000	London→Istanbul
4	5	Qatar Airways	Doha	Dubai	1308	Business	60	516.19	1000-2000	Doha→Dubai

Next steps: [Generate code with df](#) [New interactive sheet](#)

df21 = df.groupby("Class")["price_per_km"].mean().reset_index().rename(columns={"price_per_km":"Avg_Price_per_Km"})
df21

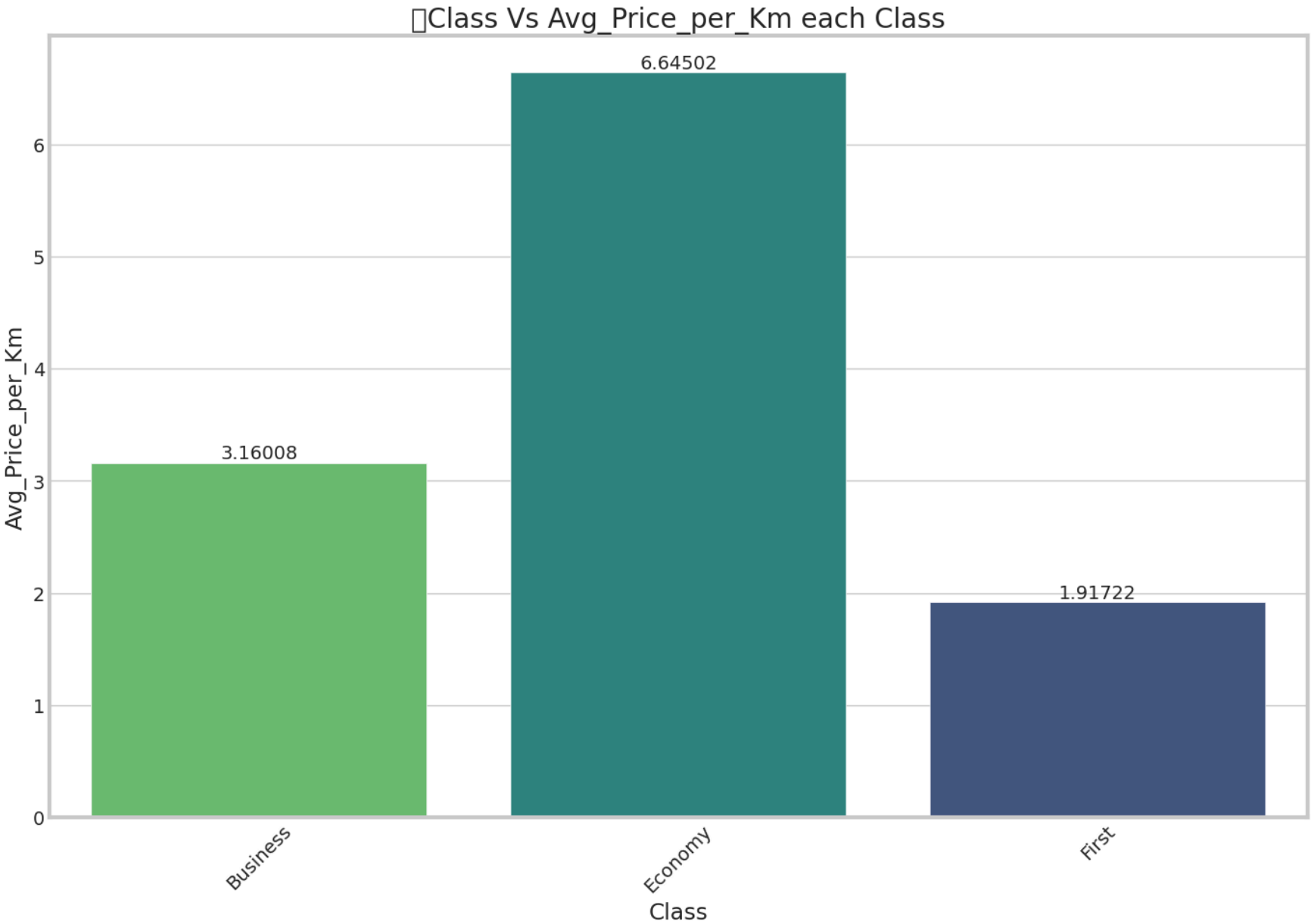
	Class	Avg_Price_per_Km	
0	Business	3.160076	
1	Economy	6.645015	
2	First	1.917221	

Next steps:

[Generate code with df21](#)

[New interactive sheet](#)

```
plt.figure(figsize=(15,10))
sns.barplot(x = "Class", y = "Avg_Price_per_Km",data = df21,palette="viridis_r")
for c in plt.gca().containers:
    plt.gca().bar_label(c)
plt.xlabel("Class")
plt.ylabel("Avg_Price_per_Km")
plt.xticks(rotation=45,fontsize=14)
plt.title(" Class Vs Avg_Price_per_Km each Class")
plt.show()
```



Which airline has the largest number of routes?

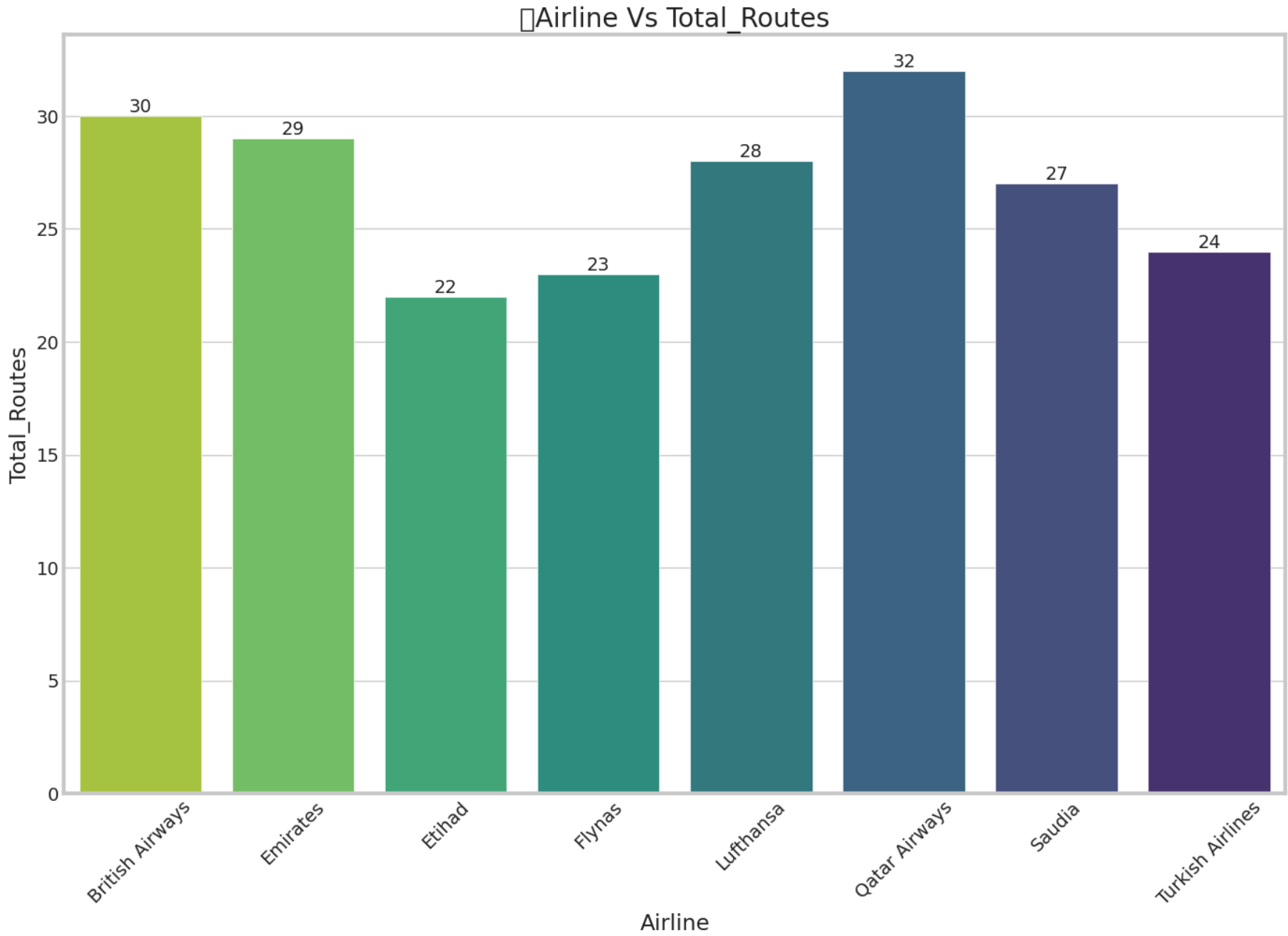
```
df_airline_routes = df.groupby("Airline")["Route"].nunique().reset_index()

df_airline_routes = df_airline_routes.rename(columns={"Route": "Total_Routes"})
```

```
df_airline_routes.sort_values(by = "Total_Routes",ascending=False)
```

	Airline	Total_Routes	
5	Qatar Airways	32	
0	British Airways	30	
1	Emirates	29	
4	Lufthansa	28	
6	Saudia	27	
7	Turkish Airlines	24	
3	Flynas	23	
2	Etihad	22	

```
plt.figure(figsize=(15,10))
sns.barplot(x = "Airline", y = "Total_Routes",data = df_airline_routes,palette="viridis_r")
for c in plt.gca().containers:
    plt.gca().bar_label(c)
plt.xlabel("Airline")
plt.ylabel("Total_Routes")
plt.xticks(rotation=45,fontsize=14)
plt.title(" Airline Vs Total_Routes")
plt.show()
```



What percentage of tickets belong to each class?

```
df.columns
```

```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',
      'Days_Before_Departure', 'Price_USD', 'Distance_Range', 'Route',
      'Booking_type', 'price_per_km'],
      dtype='object')
```

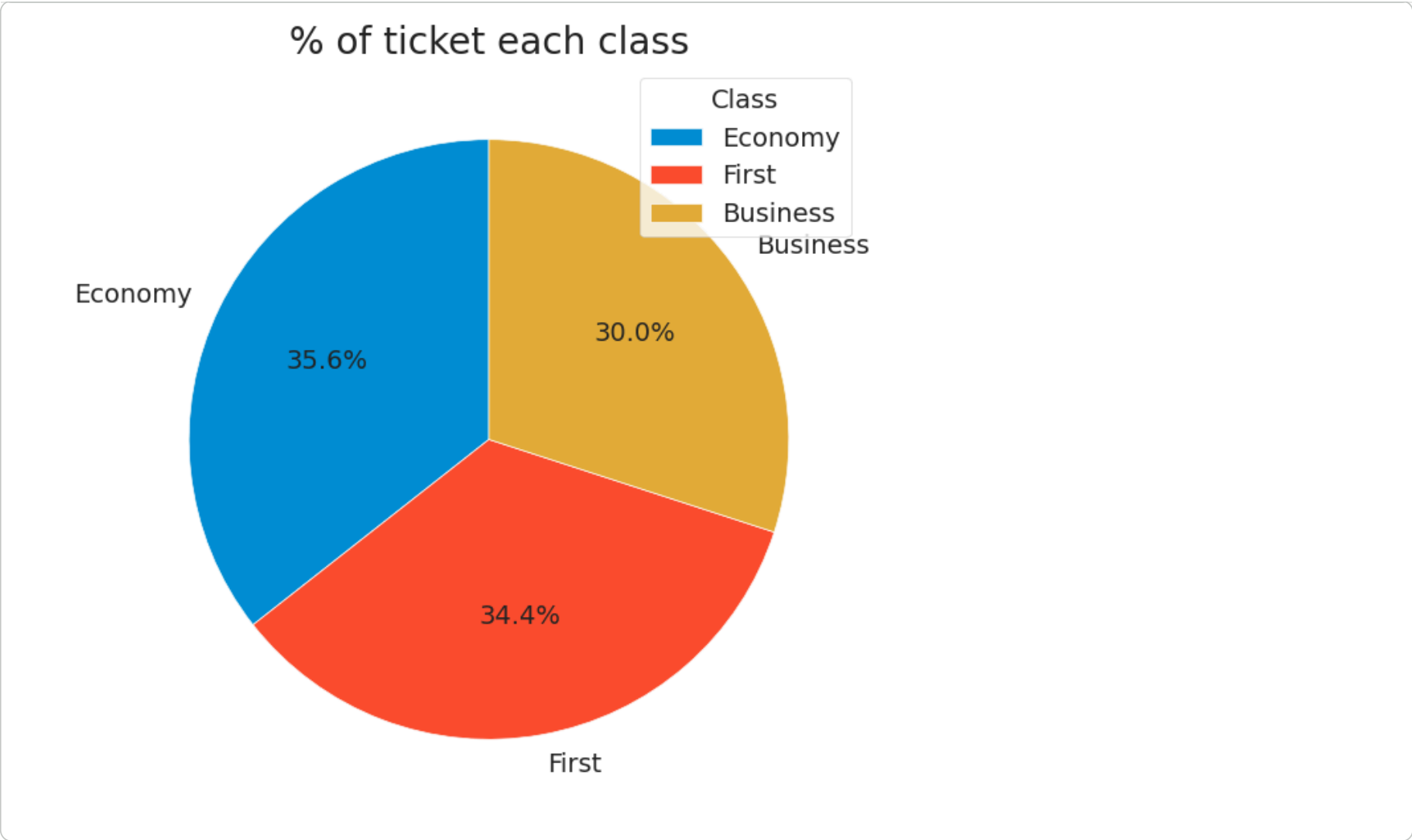
```
df23 = df["Class"].value_counts()*100/len(df)
df23 = df23.reset_index().rename(columns={"count":"percentage"})
```

df23

	Class	percentage	
0	Economy	35.6	
1	First	34.4	
2	Business	30.0	

Next steps: [Generate code with df23](#) [New interactive sheet](#)

```
plt.figure(figsize=(7,7))
plt.pie(df23["percentage"],labels=df23['Class'],autopct='%1.1f%%',startangle=90)
plt.title("% of ticket each class")
plt.legend(df23['Class'],title="Class",loc=1)
plt.show()
```



Which airline dominates each route?

```
df.columns

Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',
      'Days_Before_Departure', 'Price_USD', 'Distance_Range', 'Route',
      'Booking_type', 'price_per_km'],
      dtype='object')
```

```
route_airline_count = df.groupby(["Airline","Route"])["Ticket_ID"].size().reset_index(name='ticket_count')
route_airline_count
```

	Airline	Route	ticket_count	
0	British Airways	Cairo→Doha	1	
1	British Airways	Cairo→Riyadh	1	
2	British Airways	Doha→Istanbul	1	
3	British Airways	Doha→Jeddah	1	
4	British Airways	Doha→Riyadh	1	
...	...	...	...	
210	Turkish Airlines	Paris→Istanbul	1	
211	Turkish Airlines	Paris→New York	1	
212	Turkish Airlines	Riyadh→Cairo	1	
213	Turkish Airlines	Riyadh→Mumbai	1	
214	Turkish Airlines	Riyadh→Paris	2	

215 rows × 3 columns

Next steps:

[Generate code with route_airline_count](#)

[New interactive sheet](#)

```
dominant_airline = (route_airline_count.sort_values(['Route', 'ticket_count'], ascending=[True,False]).drop_duplicates('Route'))
```

```
dominant_airline.sort_values(by = "ticket_count",ascending=False).head(10)
```

	Airline	Route	ticket_count	
106	Lufthansa	Cairo→Riyadh	3	
133	Qatar Airways	Cairo→Paris	3	
10	British Airways	Jeddah→Paris	3	
206	Turkish Airlines	Mumbai→Istanbul	3	
169	Saudia	Doha→Dubai	2	
170	Saudia	Doha→London	2	
180	Saudia	Mumbai→New York	2	
209	Turkish Airlines	Paris→Cairo	2	
11	British Airways	London→Cairo	2	
150	Qatar Airways	London→Mumbai	2	

Which airline generates the highest total revenue?

df.columns

Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',
 'Days_Before_Departure', 'Price_USD', 'Distance_Range', 'Route',
 'Booking_type', 'price_per_km'],
 dtype='object')

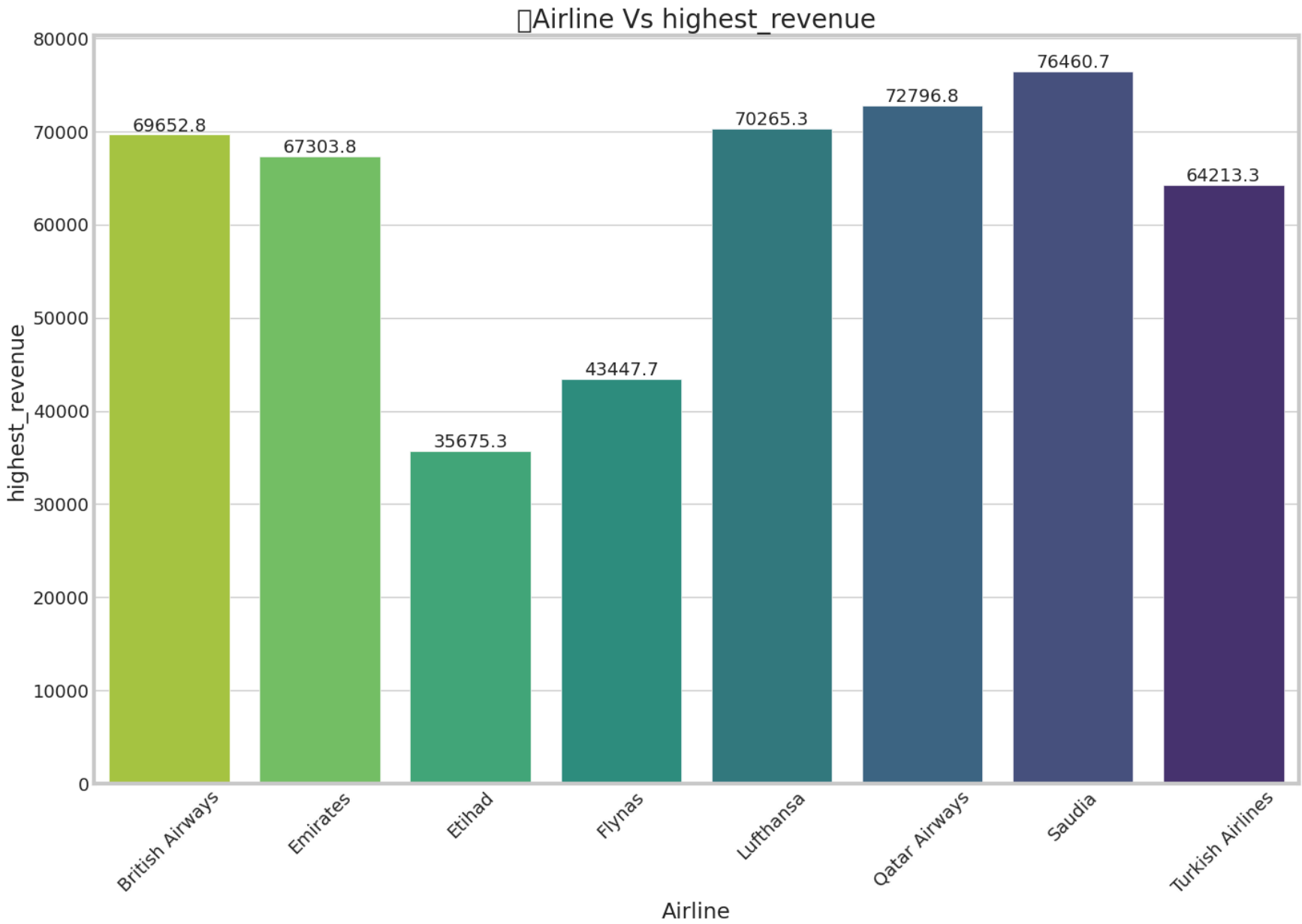
highest_revenue = df.groupby("Airline")["Price_USD"].sum()

highest_revenue = highest_revenue.reset_index().rename(columns={"Price_USD": "highest_revenue"})
highest_revenue

	Airline	highest_revenue	
0	British Airways	69652.83	
1	Emirates	67303.77	
2	Etihad	35675.31	
3	Flynas	43447.68	
4	Lufthansa	70265.29	
5	Qatar Airways	72796.79	
6	Saudia	76460.72	
7	Turkish Airlines	64213.29	

Next steps: [Generate code with highest_revenue](#) [New interactive sheet](#)

```
plt.figure(figsize=(15,10))
sns.barplot(x = "Airline", y = "highest_revenue",data = highest_revenue,palette="viridis_r")
for c in plt.gca().containers:
    plt.gca().bar_label(c)
plt.xlabel("Airline")
plt.ylabel("highest_revenue")
plt.xticks(rotation=45,fontsize=14)
plt.title(" Airline Vs highest_revenue")
plt.show()
```

What is the revenue share (%) of each airline?

```
df.columns
```

```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',  
      'Days_Before_Departure', 'Price_USD', 'Distance_Range', 'Route',  
      'Booking_type', 'price_per_km'],  
      dtype='object')
```

```
total_revenue1 = float(df["Price_USD"].sum())  
total_revenue1
```

499815.68

```
revenue = df.groupby("Airline")["Price_USD"].sum().reset_index().rename(columns={"Price_USD":"revenue"})
```

revenue

	Airline	revenue
0	British Airways	69652.83
1	Emirates	67303.77
2	Etihad	35675.31
3	Flynas	43447.68
4	Lufthansa	70265.29
5	Qatar Airways	72796.79
6	Saudia	76460.72
7	Turkish Airlines	64213.29

Next steps:

[Generate code with revenue](#)

[New interactive sheet](#)

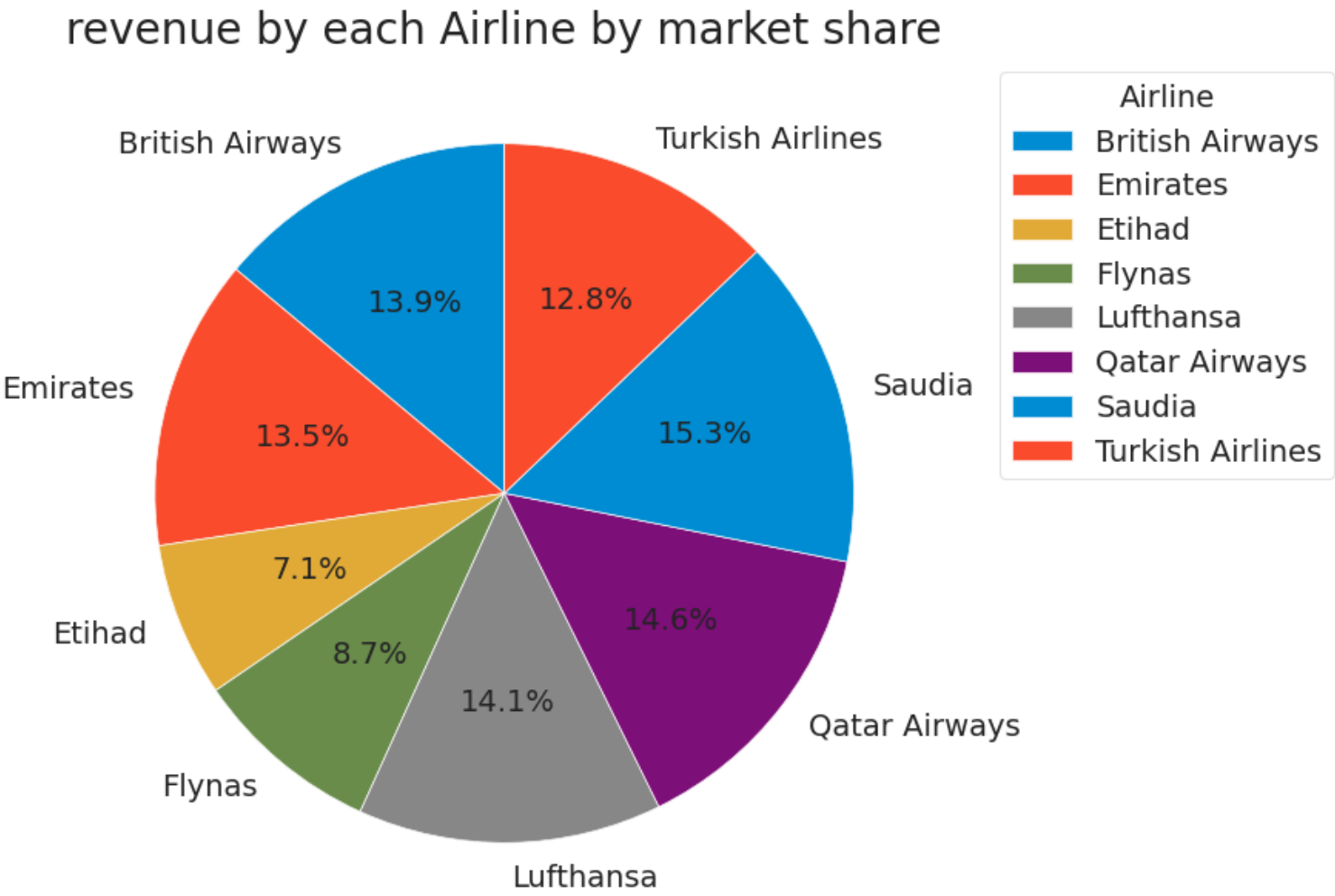
```
revenue["Market_Share %"] = ((revenue["revenue"]/total_revenue1)*100).round(2)
```

revenue

	Airline	revenue	Market_Share %
0	British Airways	69652.83	13.94
1	Emirates	67303.77	13.47
2	Etihad	35675.31	7.14
3	Flynas	43447.68	8.69
4	Lufthansa	70265.29	14.06
5	Qatar Airways	72796.79	14.56
6	Saudia	76460.72	15.30
7	Turkish Airlines	64213.29	12.85

Next steps: [Generate code with revenue](#) [New interactive sheet](#)

```
plt.figure(figsize=(7,7))
plt.pie(revenue['revenue'],labels=revenue['Airline'],autopct='%1.1f%%',startangle=90)
plt.title("revenue by each Airline by market share")
plt.legend(revenue['Airline'],title="Airline",loc=2,bbox_to_anchor=(1.05, 1))
plt.show()
```



Which routes are most profitable (high price_per_km)?

df.columns

```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',
      'Days_Before_Departure', 'Price_USD', 'Distance_Range', 'Route',
      'Booking_type', 'price_per_km'],
      dtype='object')
```

```
profitable_route = df.groupby("Route")["price_per_km"].sum().sort_values(ascending=False).reset_index()
profitable_route
```

	Route	price_per_km
0	Istanbul→Dubai	62.373823
1	Jeddah→Cairo	30.229196
2	Cairo→Riyadh	30.001983
3	Paris→Cairo	23.382705
4	Dubai→Riyadh	22.335051
...	...	...
80	Cairo→Jeddah	1.876550
81	Paris→Mumbai	1.678894
82	Riyadh→Doha	1.486663
83	New York→Dubai	1.235346
84	Dubai→London	1.143636

85 rows × 2 columns

Next steps:

Generate code with profitable_route

New interactive sheet

```
profitable_route = profitable_route.head(10)
profitable_route
```

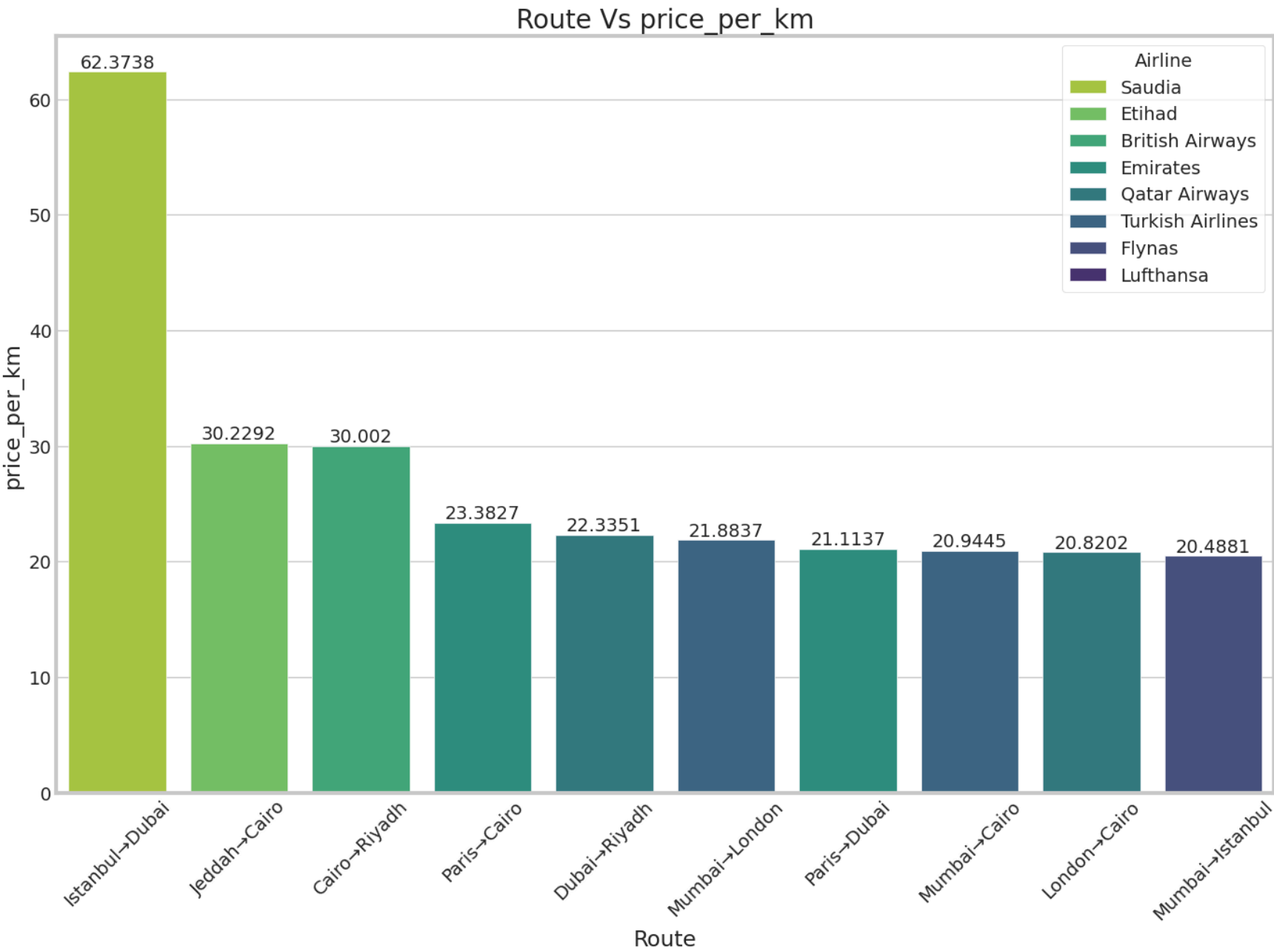
	Route	price_per_km	
0	Istanbul→Dubai	62.373823	
1	Jeddah→Cairo	30.229196	
2	Cairo→Riyadh	30.001983	
3	Paris→Cairo	23.382705	
4	Dubai→Riyadh	22.335051	
5	Mumbai→London	21.883701	
6	Paris→Dubai	21.113653	
7	Mumbai→Cairo	20.944544	
8	London→Cairo	20.820240	
9	Mumbai→Istanbul	20.488089	

Next steps:

Generate code with profitable_route

New interactive sheet

```
plt.figure(figsize=(15,10))
sns.barplot(x = "Route", y = "price_per_km",data = profitable_route,palette="viridis_r",hue=df["Airline"])
for c in plt.gca().containers:
    plt.gca().bar_label(c)
plt.xlabel("Route")
plt.ylabel("price_per_km")
plt.xticks(rotation=45,fontsize=14)
plt.title("Route Vs price_per_km")
plt.show()
```



Which origin city generates the highest revenue?

```
df.columns
```

```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',
      'Days_Before_Departure', 'Price_USD', 'Distance_Range', 'Route',
```

```
'Booking_type', 'price_per_km'],
dtype='object')
```

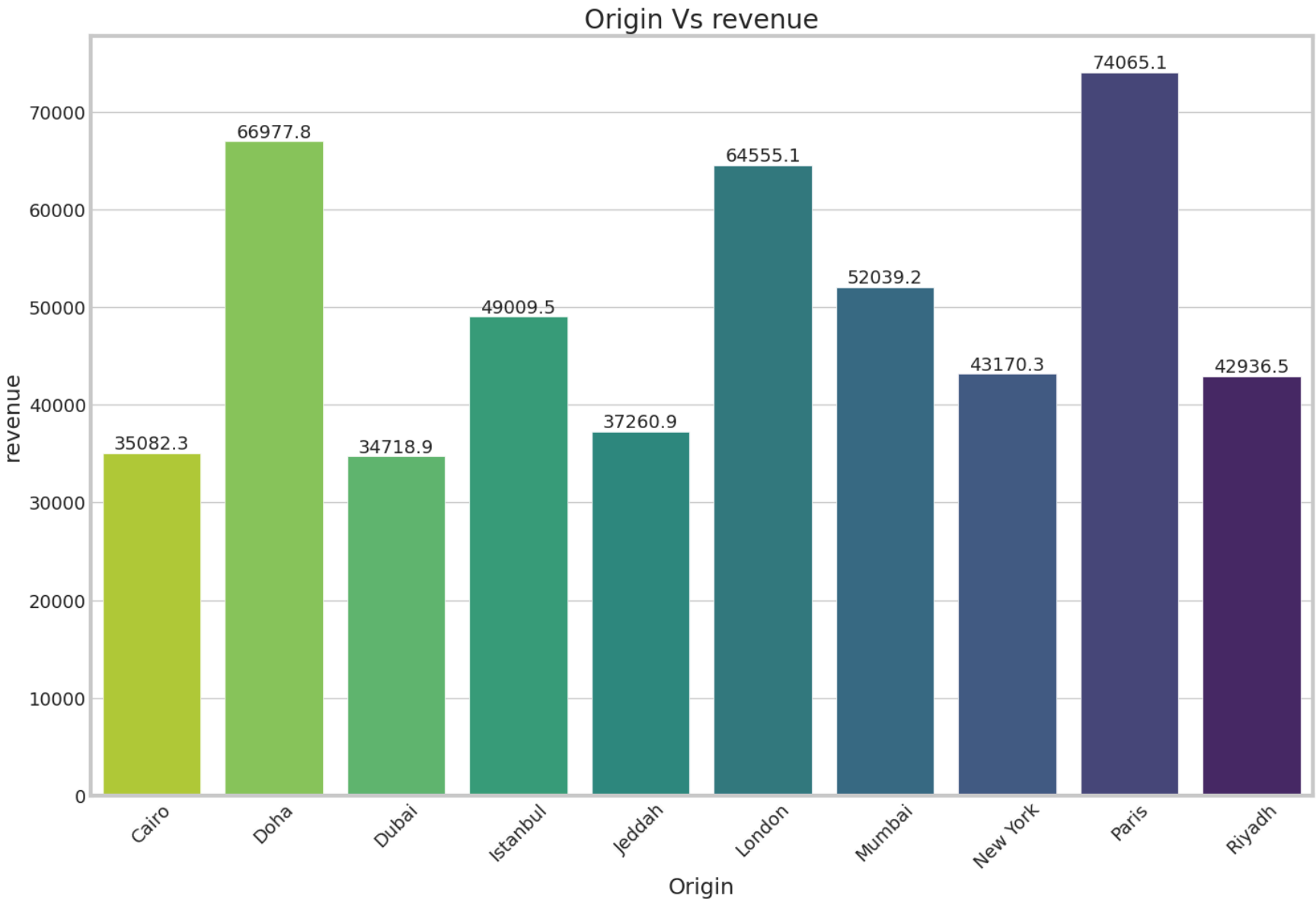
```
revenueOrigin = df.groupby("Origin")["Price_USD"].sum()
revenueOrigin = revenueOrigin.reset_index().rename(columns={"Price_USD":"revenue"})
```

revenueOrigin

	Origin	revenue	
0	Cairo	35082.33	
1	Doha	66977.77	
2	Dubai	34718.95	
3	Istanbul	49009.52	
4	Jeddah	37260.94	
5	London	64555.07	
6	Mumbai	52039.23	
7	New York	43170.26	
8	Paris	74065.10	
9	Riyadh	42936.51	

Next steps: [Generate code with revenueOrigin](#) [New interactive sheet](#)

```
plt.figure(figsize=(15,10))
sns.barplot(x = "Origin", y = "revenue",data = revenueOrigin,palette="viridis_r")
for c in plt.gca().containers:
    plt.gca().bar_label(c)
plt.xlabel("Origin")
plt.ylabel("revenue")
plt.xticks(rotation=45,fontsize=14)
plt.title("Origin Vs revenue")
plt.show()
```



```
total_revenue2 = float(df["Price_USD"].sum())
total_revenue2
```

499815.68

```
revenueOrigin["originShare %"] = ((revenueOrigin['revenue']/total_revenue2)*100).round(2)
```

revenueOrigin

	Origin	revenue	originShare %
0	Cairo	35082.33	7.02
1	Doha	66977.77	13.40
2	Dubai	34718.95	6.95
3	Istanbul	49009.52	9.81
4	Jeddah	37260.94	7.45
5	London	64555.07	12.92
6	Mumbai	52039.23	10.41
7	New York	43170.26	8.64
8	Paris	74065.10	14.82
9	Riyadh	42936.51	8.59

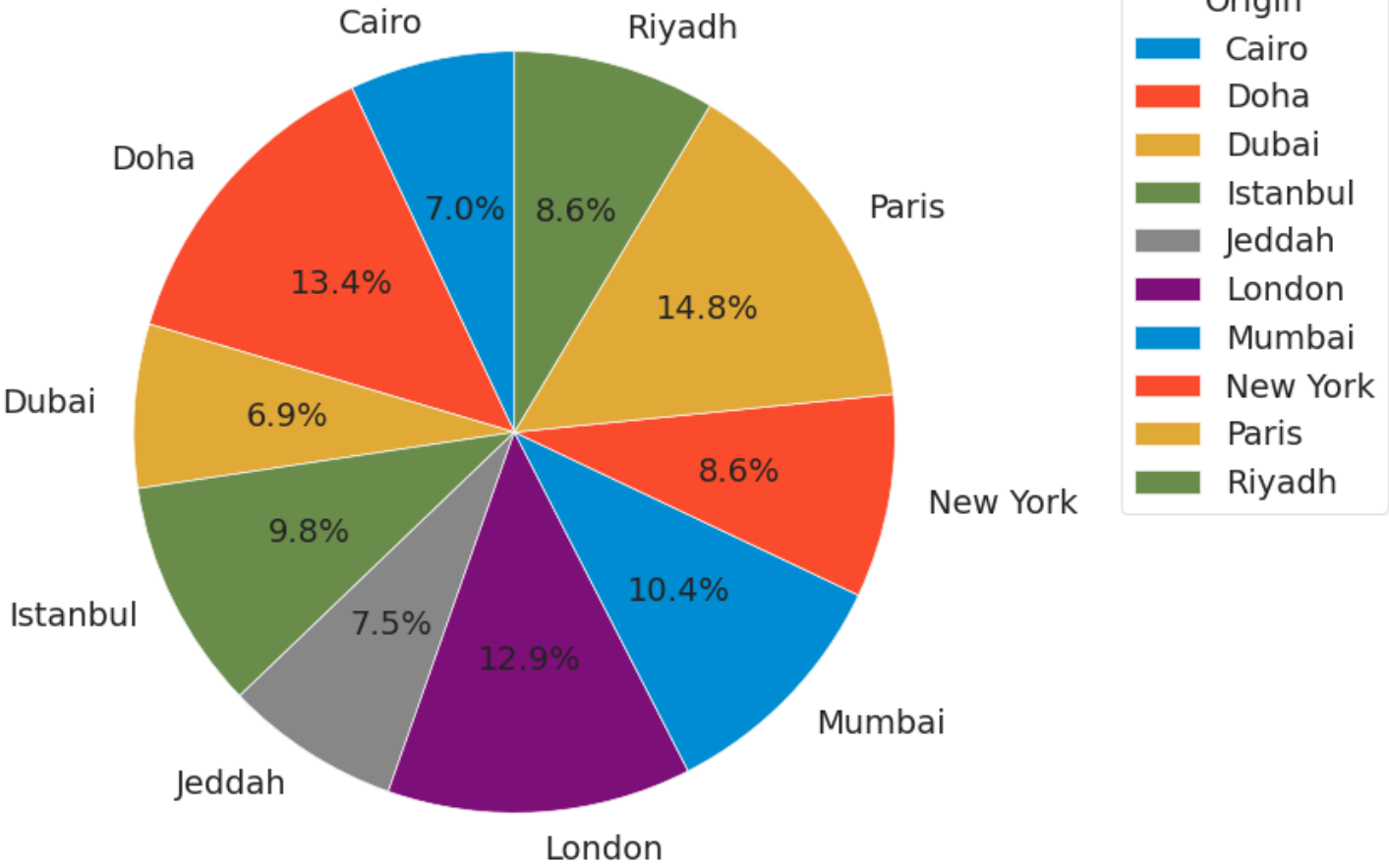
Next steps:

[Generate code with revenueOrigin](#)

[New interactive sheet](#)

```
plt.figure(figsize=(7,7))
plt.pie(revenueOrigin['revenue'],labels=revenueOrigin['Origin'],autopct='%1.1f%%',startangle=90)
plt.title("revenue by each Origin by market share")
plt.legend(revenueOrigin['Origin'],title="Origin",loc=2,bbbox_to_anchor=(1.12, 1))
plt.show()
```

revenue by each Origin by market share



How does ticket price vary across different distance ranges?

df.head()

	Ticket_ID	Airline	Origin	Destination	Distance_km	Class	Days_Before_Departure	Price_USD	Distance_Range	Route
0	1	Saudia	Mumbai	Paris	2643	Economy	28	555.30	2000-5000	Mumbai→Paris
1	2	Etihad	Istanbul	Jeddah	8266	Business	98	2070.27	5000-10000	Istanbul→Jeddah
2	3	British Airways	Dubai	Riyadh	2698	First	21	1883.11	2000-5000	Dubai→Riyadh
3	4	Emirates	London	Istanbul	7956	First	24	5349.65	5000-10000	London→Istanbul
4	5	Qatar Airways	Doha	Dubai	1308	Business	60	516.19	1000-2000	Doha→Dubai

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

df["Distance_Range"].unique()

['2000-5000', '5000-10000', '1000-2000', '10000+', '0-500', '500-1000']
Categories (6, object): ['0-500' < '500-1000' < '1000-2000' < '2000-5000' < '5000-10000' < '10000+']

df24 = df.groupby("Distance_Range")["Price_USD"].mean().reset_index()
df24

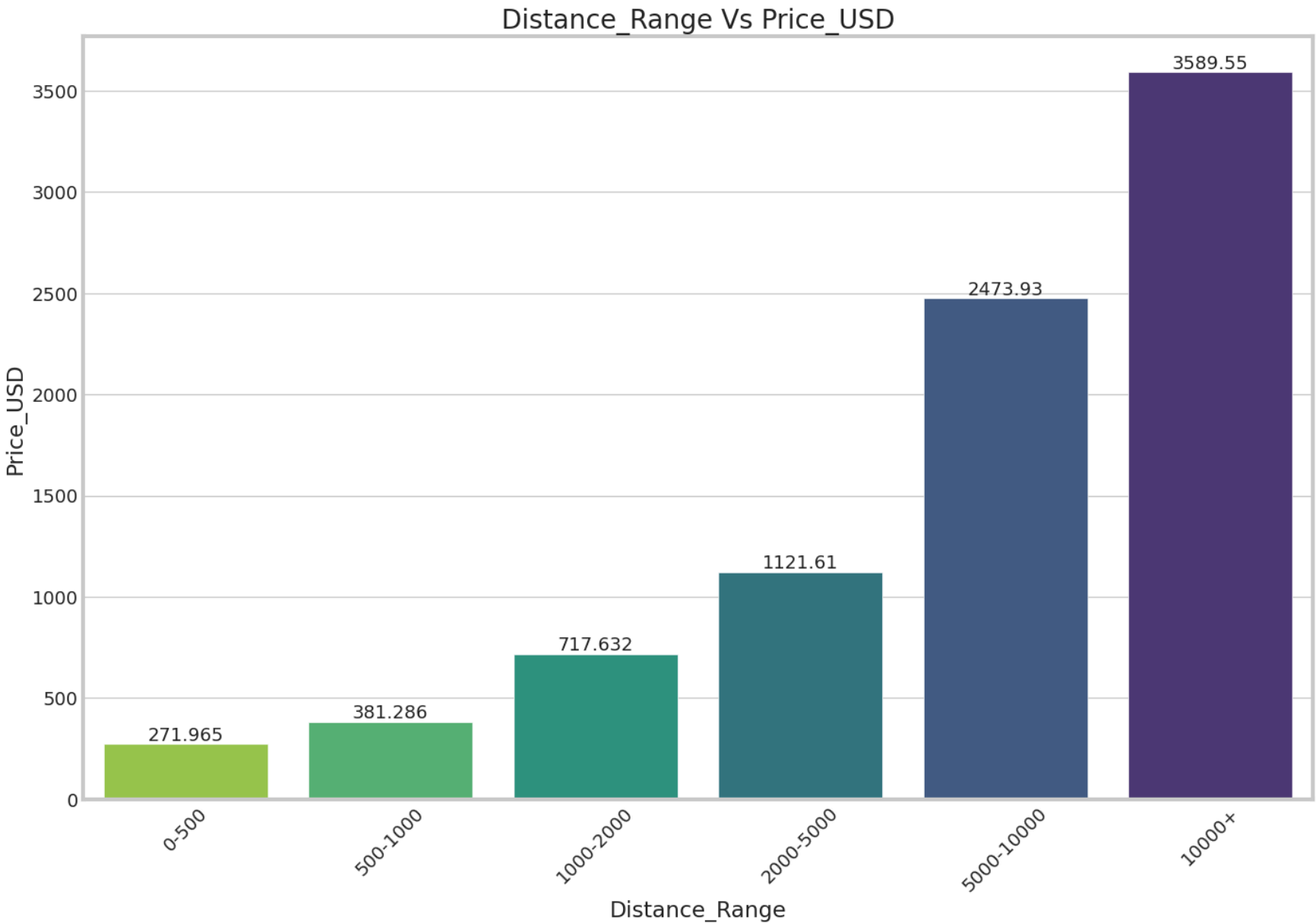
	Distance_Range	Price_USD	
0	0-500	271.965000	
1	500-1000	381.285714	
2	1000-2000	717.632308	
3	2000-5000	1121.610429	
4	5000-10000	2473.929327	
5	10000+	3589.545366	

Next steps:

[Generate code with df24](#)

[New interactive sheet](#)

```
plt.figure(figsize=(15,10))
sns.barplot(x = "Distance_Range", y = "Price_USD",data = df24,palette="viridis_r")
for c in plt.gca().containers:
    plt.gca().bar_label(c)
plt.xlabel("Distance_Range")
plt.ylabel("Price_USD")
plt.xticks(rotation=45,fontsize=14)
plt.title("Distance_Range Vs Price_USD")
plt.show()
```



Is there any airline that charges significantly more for the same distance?

```
df.columns
```

```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',
      'Days_Before_Departure', 'Price_USD', 'Distance_Range', 'Route',
      'Booking_type', 'price_per_km'],
      dtype='object')
```

```
avg_price_km = df.groupby("Airline")["price_per_km"].mean().reset_index()
```

```
avg_price_km = avg_price_km.sort_values(by = "price_per_km",ascending=False)
avg_price_km
```

	Airline	price_per_km	
5	Qatar Airways	4.786068	
1	Emirates	4.191980	
3	Flynas	4.125753	
2	Etihad	3.840805	
4	Lufthansa	3.838152	
7	Turkish Airlines	3.792349	
0	British Airways	3.639479	
6	Saudia	3.503547	

Next steps: [Generate code with avg_price_km](#) [New interactive sheet](#)

```
mean_price = avg_price_km["price_per_km"].mean()
std_price = avg_price_km["price_per_km"].std()
```

mean_price

np.float64(3.9647665551012197)

std_price

0.40209068619707006

```
threshold = mean_price + std_price
threshold
```

np.float64(4.366857241298289)

```
avg_price_km["price_per_km"] > threshold
```

	price_per_km
5	True
1	False
3	False
2	False
4	False
7	False
0	False
6	False

dtype: bool

```
expensive_airlines = avg_price_km[avg_price_km["price_per_km"] > threshold]
```

expensive_airlines

	Airline	price_per_km	
5	Qatar Airways	4.786068	

How does price vary with booking type (Online vs Agent)?

```
df.head()
```

	Ticket_ID	Airline	Origin	Destination	Distance_km	Class	Days_Before_Departure	Price_USD	Distance_Range	Route
0	1	Saudia	Mumbai	Paris	2643	Economy	28	555.30	2000-5000	Mumbai→Paris
1	2	Etihad	Istanbul	Jeddah	8266	Business	98	2070.27	5000-10000	Istanbul→Jeddah
2	3	British Airways	Dubai	Riyadh	2698	First	21	1883.11	2000-5000	Dubai→Riyadh
3	4	Emirates	London	Istanbul	7956	First	24	5349.65	5000-10000	London→Istanbul
4	5	Qatar Airways	Doha	Dubai	1308	Business	60	516.19	1000-2000	Doha→Dubai

Next steps: [Generate code with df](#) [New interactive sheet](#)

df.columns

```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',
      'Days_Before_Departure', 'Price_USD', 'Distance_Range', 'Route',
```

```
'Booking_type', 'price_per_km'],
dtype='object')
```

```
df["Booking_type"].unique()
```

```
['Medium', 'Early', 'Late']
Categories (3, object): ['Late' < 'Medium' < 'Early']
```

```
df["Booking_Channel"] = df["Booking_type"].apply(
    lambda x: "Online" if x == "Early" else "Agent"
)
```

```
df["Booking_Channel"] = df["Booking_type"].map({
    "Early": "Online",
    "Medium": "Agent",
    "Late": "Agent"
})
```

```
df["Booking_Channel"].unique()
```

```
array(['Agent', 'Online'], dtype=object)
```

```
df.head()
```

	Ticket_ID	Airline	Origin	Destination	Distance_km	Class	Days_Before_Departure	Price_USD	Distance_Range	Route
0	1	Saudia	Mumbai	Paris	2643	Economy	28	555.30	2000-5000	Mumbai→Paris
1	2	Etihad	Istanbul	Jeddah	8266	Business	98	2070.27	5000-10000	Istanbul→Jeddah
2	3	British Airways	Dubai	Riyadh	2698	First	21	1883.11	2000-5000	Dubai→Riyadh
3	4	Emirates	London	Istanbul	7956	First	24	5349.65	5000-10000	London→Istanbul
4	5	Qatar Airways	Doha	Dubai	1308	Business	60	516.19	1000-2000	Doha→Dubai

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

```
df.groupby("Booking_Channel")["Price_USD"].mean().reset_index()
```

	Booking_Channel	Price_USD
0	Agent	2749.737167
1	Online	1762.270789

Which days_before_departure range gives the best (lowest) price?

```
df.columns
```

```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',
      'Days_Before_Departure', 'Price_USD', 'Distance_Range', 'Route',
      'Booking_type', 'price_per_km', 'Booking_Channel'],
      dtype='object')
```

```
df.head()
```

	Ticket_ID	Airline	Origin	Destination	Distance_km	Class	Days_Before_Departure	Price_USD	Distance_Range	Route
0	1	Saudia	Mumbai	Paris	2643	Economy	28	555.30	2000-5000	Mumbai→Paris
1	2	Etihad	Istanbul	Jeddah	8266	Business	98	2070.27	5000-10000	Istanbul→Jeddah
2	3	British Airways	Dubai	Riyadh	2698	First	21	1883.11	2000-5000	Dubai→Riyadh
3	4	Emirates	London	Istanbul	7956	First	24	5349.65	5000-10000	London→Istanbul
4	5	Qatar Airways	Doha	Dubai	1308	Business	60	516.19	1000-2000	Doha→Dubai

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

```
df["Days_Before_Departure"].unique()
```

```
array([ 28,  98,  21,  24,  60,  19,  85,  88,  36,  55, 108,  72,  48,
        38,   9,  82,  37, 109,   4,   3,  23,  86,  54,  71,  93,  99,
        34,  83,  44,  53,  45,  40, 111,  43,  95,  33,  39,   6,  15,
        90,  20,  76,  80,  50,  66,  26,  30, 102,  81,  18,  87,   8,
        92,  64,  32,  74,  61,  13,  16, 110, 112, 107,  14,  35,  75,
        73,  27,  96,  25,  62,  68,  47, 105,  91,  97, 114,  41, 117,
        69,   7, 103, 106,  63,   5,  70,  49,  12,  94, 101,  78,  56,
        89,  67,  11,  52, 116, 118, 115,  31, 100, 104,  58,  84,   1,
        65,  79])
```

```
bins = [0, 10, 30, 60, 120]
labels = ["0-10 (Late)", "10-30 (Medium)", "30-60 (Early)", "60+ (Very Early)"]
```



```
df["Departure_Range"] = pd.cut(df["Days_Before_Departure"], bins=bins, labels=labels)
```

df.head()

	Ticket_ID	Airline	Origin	Destination	Distance_km	Class	Days_Before_Departure	Price_USD	Distance_Range	Route
0	1	Saudia	Mumbai	Paris	2643	Economy	28	555.30	2000-5000	Mumbai→Paris
1	2	Etihad	Istanbul	Jeddah	8266	Business	98	2070.27	5000-10000	Istanbul→Jeddah
2	3	British Airways	Dubai	Riyadh	2698	First	21	1883.11	2000-5000	Dubai→Riyadh
3	4	Emirates	London	Istanbul	7956	First	24	5349.65	5000-10000	London→Istanbul
4	5	Qatar Airways	Doha	Dubai	1308	Business	60	516.19	1000-2000	Doha→Dubai

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

df.columns

```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class', 'Days_Before_Departure', 'Price_USD', 'Distance_Range', 'Route', 'Booking_type', 'price_per_km', 'Booking_Channel', 'Departure_Range'], dtype='object')
```

```
lowest_price = df.groupby("Departure_Range")["Price_USD"].mean().reset_index().rename(columns={"Price_USD": "Avg_Price", "Departure_Range": "lowest_price"})
```

	Departure_Range	Avg_Price	
0	0-10 (Late)	2945.855833	
1	10-30 (Medium)	2618.991389	
2	30-60 (Early)	2253.249077	
3	60+ (Very Early)	1506.962080	

Next steps:

[Generate code with lowest_price](#)

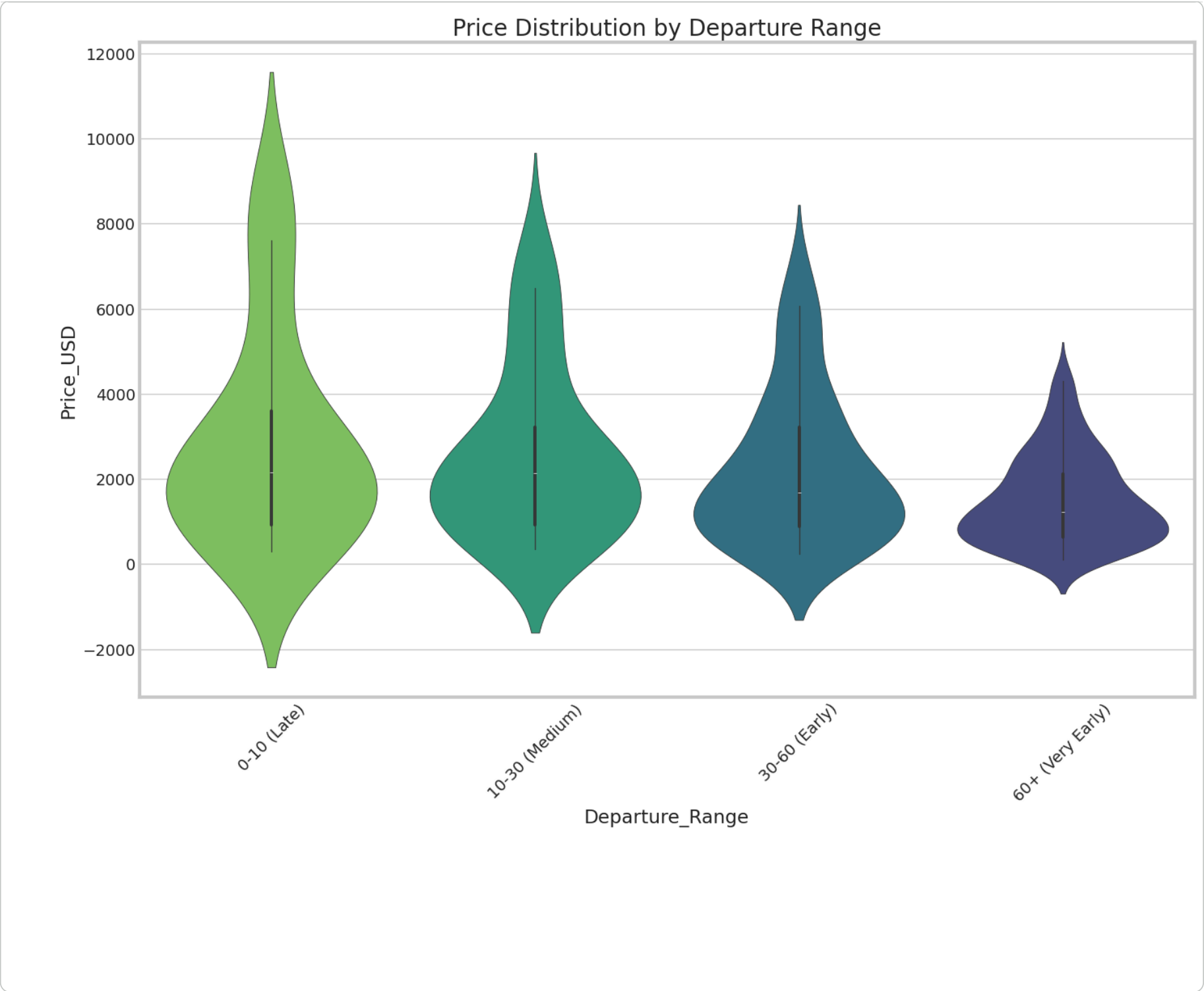
[New interactive sheet](#)

```
plt.figure(figsize=(15,10))

sns.violinplot(x="Departure_Range", y="Price_USD", data=df, palette="viridis_r")

plt.xlabel("Departure_Range")
plt.ylabel("Price_USD")
plt.xticks(rotation=45, fontsize=14)
plt.title("Price Distribution by Departure Range")

plt.show()
```



Is there price volatility in last-minute bookings? 📌 Std deviation use karo

df.columns

Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class', 'Days_Before_Departure', 'Price_USD', 'Distance_Range', 'Route', 'Booking_type', 'price_per_km', 'Booking_Channel', 'Departure_Range'], dtype='object')

df["Booking_type"].unique()

['Medium', 'Early', 'Late']
Categories (3, object): ['Late' < 'Medium' < 'Early']

mean = df.groupby("Booking_type")["Price_USD"].mean().reset_index()
mean

	Booking_type	Price_USD
0	Late	2945.855833
1	Medium	2618.991389
2	Early	1762.270789

Next steps:

Generate code with mean

New interactive sheet

STD = df.groupby("Booking_type")["Price_USD"].std().reset_index()
STD

	Booking_type	Price_USD
0	Late	2560.955064
1	Medium	2014.131627
2	Early	1385.762280

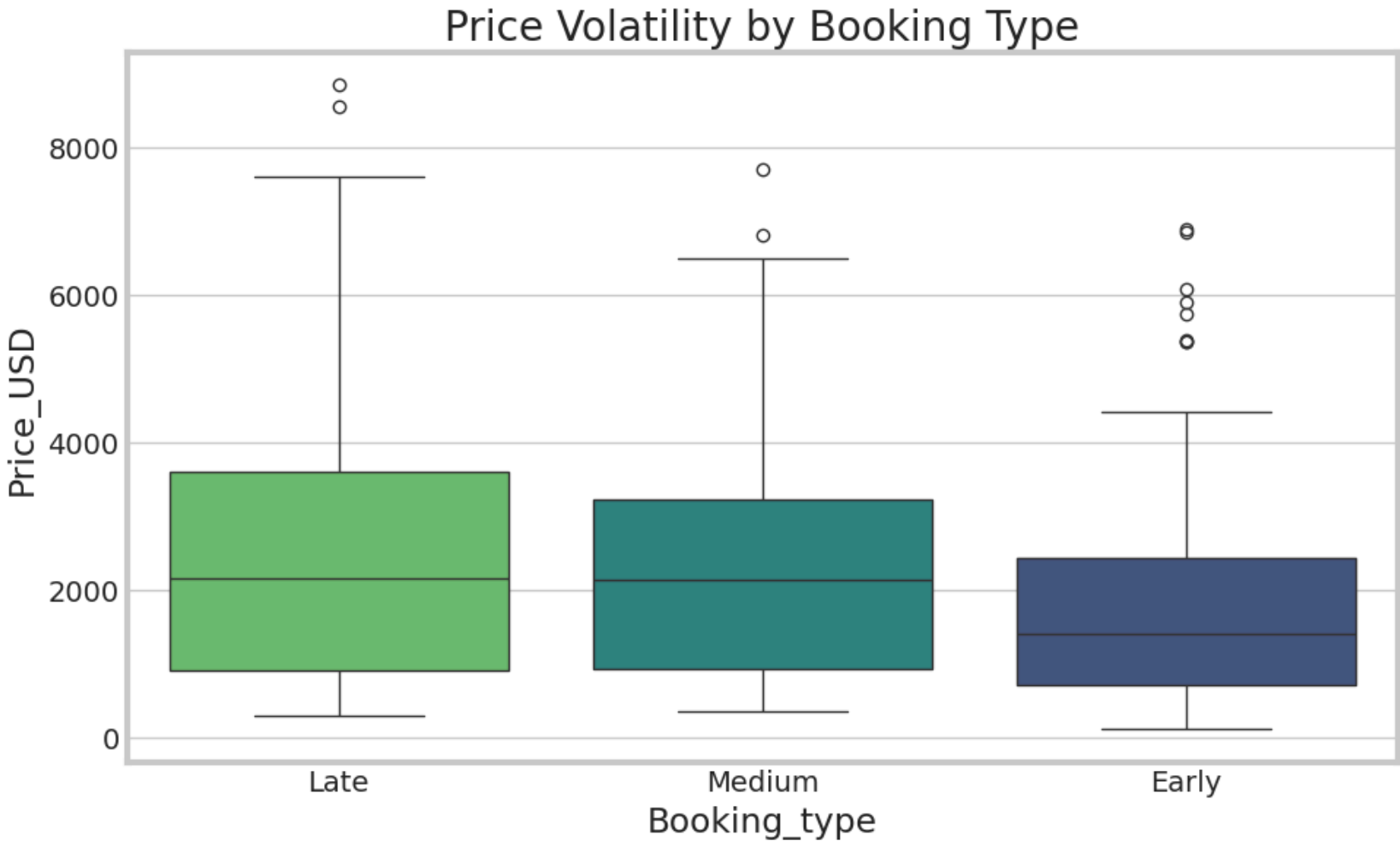
Next steps:

Generate code with STD

New interactive sheet

plt.figure(figsize=(10,6))
sns.boxplot(x="Booking_type", y="Price_USD",data=df,palette="viridis_r") # # Early / Medium / Late


```
plt.title("Price Volatility by Booking Type")
plt.show()
```



If last-minute STDDEV > early STDDEV

Prices are more volatile (unstable, risky)

“Last-minute tickets ke price fix nahi hote — wo unpredictable hote hain.”

✅ 1. Late (Last-minute bookings) Box sabse wide hai 🖱️ high spread Whiskers bhi long hain 🖱️ range zyada hai Outliers (2000+ tak) 🖱️ extreme prices

🖱️ Conclusion: ✅ Highest volatility ✅ Prices unpredictable hain

✅ 2. Medium bookings Box moderate size ka hai Thoda variation hai but controlled

🖱️ Conclusion: ✅ Moderate volatility

✅ 3. Early bookings

Box sabse chhota hai 🖱️ tight range Prices zyada clustered hain Outliers kam hain

🖱️ Conclusion: ✅ Lowest volatility ✅ Prices stable hain

Which airline has the most consistent pricing? 🖱️ Low standard deviation

```
df.columns
```

```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',
      'Days_Before_Departure', 'Price_USD', 'Distance_Range', 'Route',
      'Booking_type', 'price_per_km', 'Booking_Channel', 'Departure_Range'],
      dtype='object')
```

```
airlines_std = df.groupby("Airline")["Price_USD"].std().round(2).reset_index()
airlines_std = airlines_std.sort_values(by = "Price_USD")
airlines_std
```

	Airline	Price_USD	
2	Etihad	1153.54	
3	Flynas	1416.63	
6	Saudia	1443.45	
7	Turkish Airlines	1639.43	
0	British Airways	1692.07	
1	Emirates	1700.64	
4	Lufthansa	1971.05	
5	Qatar Airways	2036.43	

Next steps:

[Generate code with airlines_std](#)

[New interactive sheet](#)

```
airlines_std.head(1)
```

	Airline	Price_USD	
2	Etihad	1153.54	

Next steps:

Generate code with airlines_std

New interactive sheet

Which route has the highest price variation?

df.columns

```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',  
      'Days_Before_Departure', 'Price_USD', 'Distance_Range', 'Route',  
      'Booking_type', 'price_per_km', 'Booking_Channel', 'Departure_Range'],  
      dtype='object')
```

```
route_var = df.groupby("Route")["Price_USD"].std().reset_index().rename(columns={"Price_USD":"highest_price_variation"})  
route_var = route_var.sort_values(by="highest_price_variation", ascending=False)  
route_var
```

	Route	highest_price_variation	
47	London→Mumbai	4633.493960	
58	Mumbai→Riyadh	4564.112643	
60	New York→Doha	4476.777885	
15	Doha→New York	4193.751324	
46	London→Jeddah	3340.146160	
...	...	...	
61	New York→Dubai	NaN	
72	Paris→Jeddah	NaN	
74	Paris→Mumbai	NaN	
78	Riyadh→Doha	NaN	
81	Riyadh→Jeddah	NaN	

85 rows × 2 columns

Next steps:

Generate code with route_var

New interactive sheet

route_var["highest_price_variation"].std()

1053.5068281126655

route_var["highest_price_variation"].mean()

np.float64(1484.97885691578)

theresold = route_var["highest_price_variation"].mean() + route_var["highest_price_variation"].std()

theresold

np.float64(2538.4856850284455)

```
route_var = route_var[route_var["highest_price_variation"]>theresold]  
route_var
```

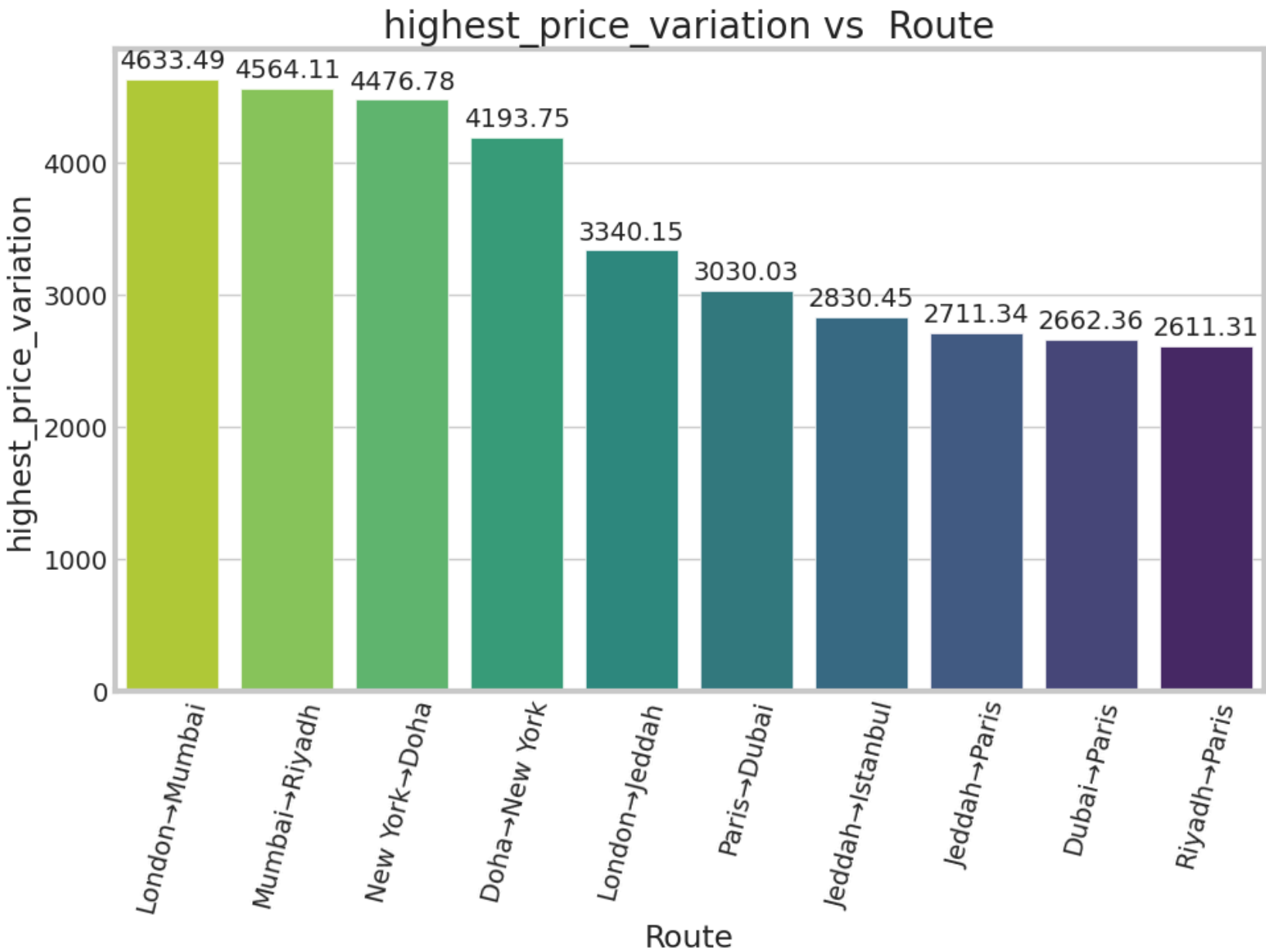
	Route	highest_price_variation	
47	London→Mumbai	4633.493960	
58	Mumbai→Riyadh	4564.112643	
60	New York→Doha	4476.777885	
15	Doha→New York	4193.751324	
46	London→Jeddah	3340.146160	
70	Paris→Dubai	3030.025592	
38	Jeddah→Istanbul	2830.453647	
40	Jeddah→Paris	2711.340213	
24	Dubai→Paris	2662.363097	
84	Riyadh→Paris	2611.309952	

Next steps:

Generate code with route_var

New interactive sheet

```
plt.figure(figsize=(10,6))  
sns.barplot(x="Route", y="highest_price_variation",data=route_var,palette="viridis_r")  
for c in plt.gca().containers:  
    plt.gca().bar_label(c,padding=3)  
plt.title("highest_price_variation vs Route")  
plt.xticks(rotation = 76)  
plt.show()
```



Are longer routes always more expensive per km? 🏹 Normalize using price_per_km

“Longer routes are not always more expensive per km. After normalizing using price_per_km

df.columns

```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',  
      'Days_Before_Departure', 'Price_USD', 'Distance_Range', 'Route',  
      'Booking_type', 'price_per_km', 'Booking_Channel', 'Departure_Range'],  
      dtype='object')
```

```
routes = df.groupby("Distance_Range")["price_per_km"].min().reset_index()  
routes
```

	Distance_Range	price_per_km	
0	0-500	1.220309	
1	500-1000	1.143636	
2	1000-2000	1.363010	
3	2000-5000	1.334189	
4	5000-10000	1.280226	
5	10000+	1.259843	

Next steps:

[Generate code with routes](#)

[New interactive sheet](#)

```
df[["price_per_km"]].corr()
```

	price_per_km	
price_per_km	1.0	

Which airline is best for long-distance travel (lowest price per km)?

df.columns

```
Index(['Ticket_ID', 'Airline', 'Origin', 'Destination', 'Distance_km', 'Class',  
      'Days_Before_Departure', 'Price_USD', 'Distance_Range', 'Route',  
      'Booking_type', 'price_per_km', 'Booking_Channel', 'Departure_Range'],  
      dtype='object')
```

```
df.head()
```

	Ticket_ID	Airline	Origin	Destination	Distance_km	Class	Days_Before_Departure	Price_USD	Distance_Range	Route
0	1	Saudia	Mumbai	Paris	2643	Economy	28	555.30	2000-5000	Mumbai→Paris
1	2	Etihad	Istanbul	Jeddah	8266	Business	98	2070.27	5000-10000	Istanbul→Jeddah
2	3	British Airways	Dubai	Riyadh	2698	First	21	1883.11	2000-5000	Dubai→Riyadh
3	4	Emirates	London	Istanbul	7956	First	24	5349.65	5000-10000	London→Istanbul

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

```
df['Distance_Range'].unique()

['2000-5000', '5000-10000', '1000-2000', '10000+', '0-500', '500-1000']
Categories (6, object): ['0-500' < '500-1000' < '1000-2000' < '2000-5000' < '5000-10000' < '10000+']
```

```
long_dis = df.groupby("Airline")["price_per_km"].mean().reset_index().rename(columns={"price_per_km":"longest_dis_price_per_km"})
```

	Airline	longest_dis_price/km
0	British Airways	3.639479
1	Emirates	4.191980
2	Etihad	3.840805
3	Flynas	4.125753
4	Lufthansa	3.838152
5	Qatar Airways	4.786068
6	Saudia	3.503547
7	Turkish Airlines	3.792349

Next steps:

[Generate code with long_dis](#)

[New interactive sheet](#)

```
plt.figure(figsize=(10,6))
sns.barplot(x="Airline", y="longest_dis_price/km",data=long_dis,palette="viridis_r")
for c in plt.gca().containers:
    plt.gca().bar_label(c,padding=3)
plt.title("longest_dis_price/km vs Airline")
plt.xticks(rotation = 76)
plt.show()
```

